

Advancing Intrusion Detection in V2X Networks: A Comprehensive Survey on Machine Learning, Federated Learning, and Edge AI for V2X Security

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Abstract—The security of Vehicle-to-Everything (V2X) networks is fundamental to the realization of next-generation intelligent transportation systems. However, the dynamic nature of V2X environments introduces critical challenges in ensuring robust Intrusion Detection Systems (IDS), particularly concerning false alarm rates, adversarial attacks, computational complexity, and real-world deployment constraints. Traditional centralized machine learning-based IDS suffer from high computation costs, privacy risks, bandwidth constraints, and scalability limitations, making them impractical for real-time, distributed vehicular networks. To address these gaps, this paper provides a comprehensive and structured survey of IDS methodologies in V2X security, focusing on Federated Learning (FL) and Edge AI for privacy-preserving and scalable IDS solutions. Unlike prior works, we systematically analyze and benchmark intrusion detection datasets, highlighting limitations in detecting zero-day attacks and exploring the need for hybrid datasets that integrate real-world vehicular data with adversarial attack scenarios. Furthermore, we investigate the adversarial robustness of ML-based IDS, analyzing AI-based evasion techniques, data poisoning threats, and misbehavior detection challenges. A key novelty of this work lies in the detailed examination of computational complexities in IDS deployment, including sensor fusion methods, noise reduction techniques, and false alarm mitigation strategies, which are often overlooked in previous surveys. We also explore deep learning-based IDS, providing a comparative evaluation of simulated versus real-world performance. Additionally, we present an in-depth discussion on post-quantum cryptographic techniques and blockchain integra-

tion for enhancing security in Federated Learning-based IDS. This survey bridges the gap between theoretical IDS models and real-world V2X deployment, addressing key constraints such as energy efficiency, communication overhead, and scalability in resource-constrained vehicular networks. By studying the state-of-the-art methodologies, identifying critical research gaps, and proposing practical advancements, this paper serves as a definitive resource for researchers and industry professionals, guiding the development of robust, adaptive, and privacy-preserving IDS solutions for next-generation autonomous and connected vehicles (CAVs).

Index Terms—V2X security, intrusion detection systems (IDS), machine learning, federated learning (FL), edge AI, false alarm reduction, adversarial threats, distributed sensor fusion, blockchain, post-quantum cryptography, hybrid IDS, autonomous vehicles, 5G/6G security.

I. INTRODUCTION

VEHICLE-TO-EVERYTHING (V2X) networks facilitate seamless communication among vehicles, infrastructure, and pedestrians, forming the backbone of intelligent transportation systems and the autonomous vehicle industry. These networks are essential for enhancing road safety, optimizing traffic management, and improving overall driving efficiency. However, securing V2X networks remains a formidable challenge due to their highly dynamic nature, decentralized architecture, and stringent real-time requirements. Effective security mechanisms are critical to ensuring message integrity, identity authentication, network availability, and protection against unauthorized access, all while preserving user privacy [1], [2], [3].

Intrusion Detection Systems (IDS) play a pivotal role in safeguarding V2X communication by continuously monitoring network traffic to identify and mitigate security threats [4], [5], [6]. Traditional IDS approaches, while effective in conventional network environments, face significant limitations when applied to V2X networks due to the high mobility of vehicles, low-latency communication demands, and the complexity of distributed attack surfaces. Therefore, the adoption of advanced AI and Machine Learning (ML)-based IDS has become increasingly crucial in V2X security research [7], [8].

Recent advancements in ML and Artificial Intelligence (AI) have significantly improved the ability to detect and respond to security threats in V2X networks. ML-based IDS

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enhance intrusion detection through anomaly detection, predictive analytics, automated response mechanisms, real-time traffic monitoring, and behavioral analysis. By learning normal communication patterns, ML models can identify deviations indicative of cyber threats, enabling proactive detection and mitigation. ML techniques such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), Deep Learning models, and Federated Learning (FL) have demonstrated promising results in enhancing the robustness of V2X IDS [9].

The increasing complexity and scale of V2X networks necessitate AI/ML-based IDS due to several reasons:

- **Scalability and Adaptability:** ML models can handle vast amounts of real-time vehicular data and adapt to dynamic network conditions more efficiently than traditional rule-based detection mechanisms [10].
- **Early Threat Detection:** ML-based IDS enables proactive identification of security threats by learning from historical attack patterns and detecting anomalies in vehicular communication.
- **Automated Threat Mitigation:** AI/ML models can automatically classify and respond to cyber threats, reducing response time and minimizing network disruptions.
- **Handling Encrypted Traffic:** Unlike signature-based IDS, ML-based approaches can detect malicious activities in encrypted network traffic without requiring payload inspection.
- **Federated Learning and Privacy Preservation:** Federated Learning (FL) allows decentralized training of IDS models across multiple vehicles while preserving data privacy, making it an essential technique for V2X security [11], [12], [13].

A. Research Methodology and Objectives of This Survey

In this survey, we present a comprehensive and systematic study on intrusion and misbehavior detection in V2X networks, focusing on AI/ML-based security solutions. Our methodology ensures a structured, objective-based approach to gathering and analyzing state-of-the-art research in this field. This section elaborates on our research objectives, methodology, dataset analysis, and comparative evaluation to ensure clarity and reproducibility.

1) *Research Objectives:* This study aims to analyze, evaluate, and compare machine learning (ML)-based Intrusion Detection Systems (IDS) and Misbehavior Detection Systems (MDS) in V2X networks. The key objectives of this survey include:

- **Comprehensive Investigation of V2X Security Threats**
 - Examine traditional and adversarial attacks in V2X networks.
 - Identify unique cybersecurity challenges in connected and autonomous vehicles (CAVs).
 - Explore emerging attack vectors in 5G-V2X and Beyond (6G-V2X).
- **Survey and Categorization of AI/ML-Based Intrusion Detection Approaches**
 - Analyze conventional machine learning models (e.g., SVM, RF, MLP, CNN, and LSTM) used in V2X IDS/MDS.

- Investigate the effectiveness of deep learning models for anomaly detection in vehicular networks.
- Identify the role of generative models such as GANs, BiGANs, and Variational Autoencoders (VAEs) in detecting zero-day attacks.

- **Study the Impact of Federated Learning (FL) and Edge AI in V2X Security**

- Discuss how privacy-preserving techniques such as federated learning enhance IDS deployment.
- Evaluate the trade-offs between centralized, distributed, and edge-based IDS architectures.
- Explore real-world feasibility of FL-based security models for autonomous vehicles and IoV (Internet of Vehicles).

- **Dataset Evaluation and Comparative Analysis**

- Identify and compare benchmark datasets used for V2X security research.
- Analyze attack diversity, real-world applicability, and dataset attributes.
- Study how synthetic vs. real-world datasets impact model generalizability.

- **Identify Challenges in Real-World Deployment of AI-Based IDS**

- Investigate the computational complexity of ML models for large-scale V2X IDS deployment.
- Address challenges in sensor fusion, redundant observer-based IDS, and edge-device resource constraints.
- Evaluate how network latency, communication overhead, and adversarial perturbations affect model performance.

- **Propose Future Directions for Secure V2X Communications**

- Provide recommendations for hybrid approaches (e.g., AI + blockchain for enhanced security).
- Suggest standardized datasets and evaluation benchmarks for future research.
- Explore the role of quantum cryptography, homomorphic encryption, and explainable AI (XAI) in V2X cybersecurity.

2) *Research Methodology:* To achieve the above objectives, we conducted a structured multi-phase research methodology consisting of:

3) *Systematic Literature Review (SLR) Approach:* We followed a structured literature review methodology, ensuring thorough and unbiased coverage of existing research.

Data Sources:

- We retrieved research papers from Google Scholar, Scopus, IEEE Xplore, SpringerLink, and ACM Digital Library.
- Our search focused on high-impact journals, conferences, and preprint servers (e.g., arXiv, Springer, Elsevier, Wiley, MDPI).

Keyword-Based Search Strategy:

- We used a combination of keywords and Boolean operators to ensure a broad yet focused search:
 - (“Machine Learning” OR “Deep Learning”) AND (“Intrusion Detection” OR “Misbehavior Detection”) AND (“V2X Security” OR “Connected Vehicles”).

Paper Selection Process:

- 1) **Initial Screening** → Collected 300+ research papers related to V2X security & AI-based IDS/MDS.
- 2) **Relevance Filtering** → Focused on papers specifically addressing AI/ML techniques for V2X IDS.
- 3) **Full-Text Analysis** → Analyzed methodologies, datasets, and findings of top 150 relevant studies.
- 4) **Categorization & Comparison** → Organized papers into attack detection techniques, dataset studies, and IDS deployment challenges.

Benchmark Surveys Considered:

- We compared our survey against existing benchmark studies and identified research gaps.
 - Our survey differentiates itself by including federated learning, adversarial attack studies, dataset comparisons, and real-world deployment challenges.
- 4) *Comparative Analysis of V2X Datasets:* A major component of our methodology was the detailed evaluation of existing V2X security datasets.

Dataset Selection Criteria:

- We selected benchmark datasets based on the following criteria:
 - Attack coverage (e.g., DDoS, DoS, Man-in-the-Middle, Spoofing, Replay, Adversarial Attacks).
 - Real-world applicability (i.e., collected from real vehicles vs. simulated environments).
 - Feature representation (i.e., CAN traffic, GPS, timestamps, sensor fusion).
 - Suitability for ML training (i.e., balanced vs. imbalanced classes, labeled vs. unlabeled data).

Key Datasets Evaluated:

- CICIDS 2017, TON_IoT, Car-Hacking Datasets, VeReMi, SYNDOS, NSL-KDD, KDD'99, UNSW-NB15, and custom V2X datasets.
- We compared datasets in terms of attack diversity, real vs. synthetic traffic, and feature granularity.

Findings and Limitations:

- Many datasets lack real-world vehicular attack data, limiting their applicability to V2X-specific security challenges.
 - Zero-day attacks are poorly represented, necessitating generative approaches (GANs) for synthetic attack generation.
 - High imbalance in intrusion datasets leads to bias in ML model predictions, requiring SMOTE-based oversampling techniques.
- 5) *Experimental Considerations for AI-Based IDS:* To ensure real-world applicability, we evaluated:
- Computational Complexity of AI models for real-time V2X IDS deployment.
 - Trade-offs between centralized vs. federated IDS architectures for connected vehicles.

Performance Metrics for IDS Evaluation:

- Detection Rate, False Alarm Rate (FAR), Accuracy, F1-score, Precision, Recall, AUC-ROC, Latency, and Communication Overhead.

Comparison of AI Models in Terms of:

- Lightweight models for resource-constrained vehicles.

- Scalability of IDS models in large V2X environments.
- Integration of IDS with SDN, blockchain, and cloud computing.

B. Identification of Research Gaps

Despite significant progress in AI/ML-based IDS for V2X security, existing survey studies exhibit several limitations that hinder their applicability in real-world deployment. Through an in-depth review of prior surveys, the following critical research gaps have been identified:

- **Real-World Deployment Challenges:** Existing surveys focus on theoretical ML models, overlooking constraints like computing resources, energy efficiency, and network latency in V2X environments.
- **Insufficient IDS Dataset Benchmarking:** Many studies fail to compare generalized IDS datasets with V2X-specific ones in terms of attack coverage and real-world applicability.
- **Limited Discussion on Adversarial Attacks:** The increasing sophistication of cyber threats, particularly AI-based evasion techniques, remains underexplored in IDS research.
- **Federated Learning & Edge AI in V2X:** Despite FL's potential for privacy-preserving IDS, existing studies lack information on scalability, security risks, and feasibility of deployment.
- **IDS Deployment on Resource-Constrained Devices:** Key challenges like energy consumption, communication overhead, and computational costs remain largely unaddressed.
- **5G/6G-V2X Security Analysis:** There is a lack of dedicated surveys evaluating how next-generation networks impact IDS performance and emerging security threats.
- **Sensor Fusion & Computational Complexities:** Prior work overlooks issues like sensor noise, false alarms, and computational overhead. We explore noise mitigation and optimization strategies for real-world IDS deployment.

C. Enhancements and Novel Contributions

To strengthen the novelty and impact of this work, we have significantly expanded the scope of our survey to address the key research challenges identified above. The following contributions distinguish this study from previous surveys:

1) *Evaluation of Real-World Deployment Challenges:* We analyze the practical challenges of deploying ML-based IDS in resource-constrained V2X environments, considering:

- Computational limitations and energy consumption constraints that impact IDS feasibility in vehicular networks.
- Communication overhead and real-time performance requirements, which are critical for deploying IDS solutions in dynamic and bandwidth-sensitive V2X environments.

2) *Comprehensive Dataset Benchmarking and Zero-Day Attack Detection:* Our survey systematically compares and categorizes existing IDS datasets, distinguishing between generalized IDS datasets and V2X-specific datasets by:

- Evaluating dataset attributes, attack coverage, and suitability for real-world deployment.
- Proposing hybrid dataset recommendations that integrate real-world and synthetic adversarial attacks to improve zero-day attack detection.

3) *Modern Threats and Adversarial Attacks in V2X Security*: our study significantly expands the discussion on adversarial ML attacks that exploit vulnerabilities in AI-based intrusion detection models. We examine:

- AI-powered evasion techniques, data poisoning, and adaptive adversarial attacks targeting IDS.
- Defensive mechanisms such as adversarial training, robust ML architectures, and differential privacy-based protection techniques to mitigate adversarial threats.

4) *Federated Learning for Privacy-Preserving IDS*:

- We study the impact of FL-based IDS on privacy, decentralization, scalability, and real-world V2X implementation.
- We explore the Security risks in FL-based IDS, including model inversion attacks, and gradient leakage, along with defensive strategies.

5) *Computational Complexity and Feasibility of IDS in V2X*:

- We provide a detailed analysis of the computational overhead of ML-based IDS, focusing on: Real-time processing constraints, energy efficiency, and network latency issues.

6) *Noise Reduction and False Alarm Rate Mitigation*:

- We introduce Distributed Sensor Fusion Techniques for improving signal quality and reducing IDS false alarms.
- We introduce a comparative analysis of feature engineering methods, denoising algorithms, and techniques for enhancing detection accuracy.

7) *Comparative Study of IDS Performance in Simulated Vs. Real-World Environments*:

- Evaluate the performance of CNNs for anomaly detection, LSTMs for sequential pattern recognition, GANs for attack scenario generation, and Transformer-based architectures for scalable intrusion detection.
- Conduct a comparative analysis of IDS performance in simulation-based vs. real-world environments, identifying performance gaps, and proposing realistic validation strategies.

8) *Blockchain and Post-Quantum Security for Federated Learning-Based IDS*: To strengthen the security of FL-based IDS, we introduce discussions on:

- We analyze blockchain-enhanced IDS frameworks to ensure secure model aggregation and data integrity in FL-based V2X security architectures.
- We discuss quantum-safe encryption techniques such as lattice-based cryptography, hash-based signatures, and quantum-resistant key exchange protocols, bridging cryptographic advancements with vehicular cybersecurity.

By addressing these key challenges and expanding discussions on emerging security paradigms, we believe that our work provides a comprehensive, high-impact contribution to the intelligent transportation and vehicular security research community.

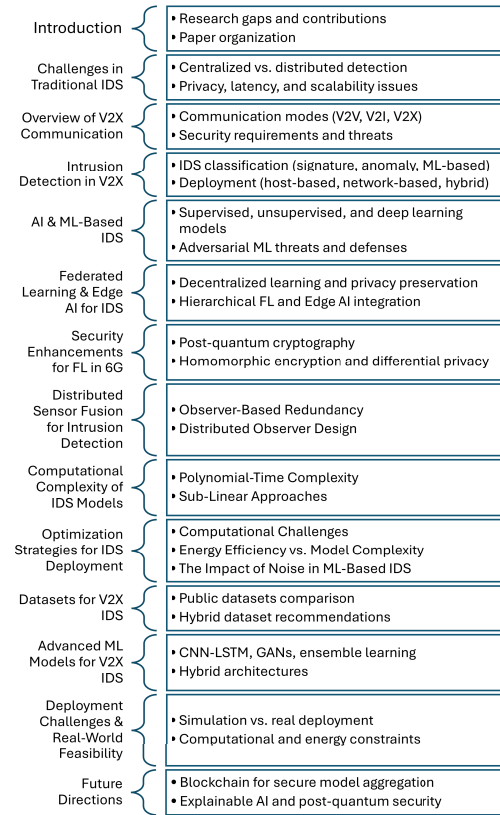


Fig. 1. Structure and organization of the paper.

D. Paper Structure and Organization

This paper provides a comprehensive survey on Intrusion Detection Systems (IDS) in V2X networks, covering traditional and emerging security approaches, dataset limitations, adversarial threats, and future research directions. The paper is organized as follows:

Section I introduces the research problem, highlights gaps in existing surveys, and presents the contributions of this study. **Section II** reviews traditional misbehavior detection techniques in V2X networks and their limitations, emphasizing the need for machine learning-based IDS. **Section III** critically examines existing machine learning-based IDS surveys for V2X, discussing their limitations and identifying research gaps. **Section IV** provides an overview of V2X communication fundamentals, including its modes, components, and standard protocols. **Section V** categorizes traditional attack models in V2X networks and reviews existing mitigation strategies. **Section VI** explores AI-based and adversarial attacks targeting V2X security, including simulation techniques and countermeasures. **Section VII** presents a taxonomy of IDS methods in V2X networks, classifying them based on detection approaches, deployment strategies, and network architectures. **Section VIII** provides an in-depth discussion of machine learning-based IDS, categorizing various ML approaches such as supervised, unsupervised, reinforcement, and deep learning techniques. **Section IX** reviews publicly available datasets for IDS in V2X, evaluating their limitations, and proposing recommendations for hybrid and more realistic datasets. **Section X**

TABLE I
LIST OF ABBREVIATIONS

Abbreviation	Full Form	Abbreviation	Full Form
5G-V2X	Fifth Generation Vehicle-to-Everything	6G-V2X	Sixth Generation Vehicle-to-Everything
AE	Autoencoder	AI	Artificial Intelligence
ANN	Artificial Neural Network	AP	Access Point
APTs	Advanced Persistent Threats	B5G	Beyond Fifth Generation
BiGAN	Bidirectional Generative Adversarial Network	BMP	Bearer Migration Poisoning
BSM	Basic Safety Message	CAN	Controller Area Network
CAV	Connected and Autonomous Vehicles	CNN	Convolutional Neural Network
CNN-LSTM	Convolutional Neural Network - Long Short-Term Memory	CSI	Channel State Information
DDoS	Distributed Denial-of-Service	DL	Deep Learning
DNN	Deep Neural Network	DoS	Denial-of-Service
DRL	Deep Reinforcement Learning	DSRC	Dedicated Short-Range Communications
EVA	Emergency Vehicle Alert	FBSM	First Basic Safety Message
FGSM	Fast Gradient Sign Method	FL	Federated Learning
FS	Feature Selection	GAN	Generative Adversarial Network
GANomaly	Generative Adversarial Network for Anomaly Detection	GPS	Global Positioning System
HIDS	Host-Based Intrusion Detection System	IDS	Intrusion Detection System
IoT	Internet of Things	IoV	Internet of Vehicles
IRS	Intelligent Reflecting Surface	KDD	Knowledge Discovery in Databases
LSTM	Long Short-Term Memory	MEC	Mobile Edge Computing
ML	Machine Learning	MLP	Multi-Layer Perceptron
MBSM	Majority-Basic Safety Message	MDP	Markov Decision Process
NIDS	Network Intrusion Detection System	NLOS	Non-Line-of-Sight
OBD	On-Board Diagnostics	OBU	On-Board Unit
PGD	Projected Gradient Descent	PIA	Policy Infiltration Attack
QoS	Quality of Service	RL	Reinforcement Learning
RSU	Roadside Unit	RSSI	Received Signal Strength Indicator
SCMS	Security Credential Management System	SMOTE	Synthetic Minority Over-Sampling Technique
SPaT	Signal Phase and Timing	SVM	Support Vector Machine
TCP	Transmission Control Protocol	UDP	User Datagram Protocol
UML	Unified Modeling Language	VANET	Vehicular Ad-Hoc Network
V2I	Vehicle-to-Infrastructure	V2N	Vehicle-to-Network
V2P	Vehicle-to-Pedestrian	V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything	VAE	Variational Autoencoder
VPN	Virtual Private Network	WBSM	Weighted-Basic Safety Message
XAI	Explainable Artificial Intelligence		

discusses the role of Federated Learning (FL) and Edge AI in developing privacy-preserving and decentralized IDS solutions for V2X networks. **Section XI** explores advanced deep learning architectures, including LSTM and GAN-based intrusion detection methods. **Section XII** addresses challenges in real-world deployment of ML-based IDS, focusing on simulation-based validation and practical implementation issues. **Section XIII** outlines future research directions, including blockchain-enhanced security, post-quantum cryptography, and the integration of FL with 6G-V2X networks. **Section XIV** concludes the paper by summarizing key findings and emphasizing the importance of privacy-preserving, scalable, and adaptive IDS frameworks for V2X security.

Figure 1 illustrates the structure and organization of the paper, providing a visual representation of the hierarchical flow of sections and subsections. As shown in Table I, commonly used abbreviations in V2X security research have been compiled based on previous studies.

II. TRADITIONAL MISBEHAVIOR DETECTION APPROACHES IN V2X NETWORKS AND LIMITATIONS

Ensuring the reliability and security of data transmission in Vehicle-to-Everything (V2X) communication systems is paramount. Misbehavior detection plays a crucial role in maintaining data accuracy, particularly in Basic Safety Messages (BSMs). Several studies have explored conventional

misbehavior detection techniques that do not rely on machine learning algorithms.

A. Prior Detection Approaches

Various traditional approaches have been proposed to identify misbehavior in V2X networks without leveraging machine learning techniques. These methods primarily rely on physical layer plausibility checks, statistical analysis, and rule-based anomaly detection.

So et al. [14] proposed a misbehavior detection technique utilizing the Received Signal Strength Indicator (RSSI) for essential safety messages (BSMs). Their study introduced three plausibility tests at the physical layer: First-BSM (FBSM), Majority-BSM (MBSM), and Weighted-BSM (WBSM), achieving a precision of 95.91% and a detection rate of 83.73% when tested on the upgraded VeReMi dataset. This work demonstrated that physical layer approaches outperform application layer plausibility checks in identifying V2X misbehavior.

Bißmeyer et al. [15] suggested a scalable centralized misbehavior detection system that enables any node to report an event while a central authority makes the final decision. However, a critical limitation of this approach is its reliance on the assumption that most network nodes behave honestly.

Sun et al. [16] introduced a technique that employs Doppler speed and the angle of arrival to Explain trustworthiness. This method utilizes chi-squared tests and Kalman filtering to evaluate the reliability of other vehicles by comparing them with a highly trustworthy node within a defined perimeter. However, its effectiveness is reduced in scenarios involving slow-moving traffic, where distinguishing between misbehavior and legitimate variations becomes challenging.

Yao et al. [17] proposed a technique to detect Sybil attacks based on the statistical properties of RSSI measurements. Vehicles with similar RSSI readings are grouped as potential Sybil attackers, suggesting that they originate from the same source. However, this method struggles against adversaries generating ghost vehicles at different locations.

Chen et al. [18] demonstrated the vulnerability of Intelligent Signalized Intersection (I-SIG) systems by showing how attackers can create traffic congestion using spoofed BSM payloads to manipulate traffic control signals. Their study highlights the weaknesses of application-layer plausibility checks.

Several studies have explored authentication and intrusion detection mechanisms for vehicular networks. Zhang and Zhu [19] proposed a distributed privacy-preserving intrusion detection system for Vehicular Ad-hoc Networks (VANETs). Mahmood et al. [20] addressed security concerns in VANET systems, while Irshad et al. [21] introduced a security key authentication framework to enhance vehicular access control.

Faisal and Zaidi [22] designed a detection system for Sybil attacks based on geographical coordinate verification. Similarly, Mahmood et al. [23] developed an anonymous identity-based key agreement protocol aimed at improving security in smart grid-integrated vehicular systems.

Mirsky et al. [24] introduced an autoencoder-based anomaly detection framework utilizing an ensemble structural algorithm to identify unauthorized users. Sharma et al. [25] proposed a certificateless authentication mechanism for V2X communication, reinforcing data integrity and confidentiality.

Additional approaches include Nayyar et al.'s [26] hybrid data model for intrusion detection in VANETs and Naqvi et al.'s [27] intrusion detection system (IDS) designed to counteract malicious vehicle behavior. Yao et al. [28] presented a geographic-based Sybil attack detection method, while Firl et al. [29] introduced MARV-X, a framework for validating vehicle maneuvers. However, its efficacy depends on the assumption that surrounding vehicles cooperate, making it vulnerable to adversarial scenarios.

Other studies, such as Guo et al. [30] and [31], proposed sensor-based anomaly detection approaches. However, these methods heavily rely on sensor data consistency, making them susceptible to data manipulation and Sybil attacks [32]. Several cooperative misbehavior detection models assume an honest majority [33], rendering them ineffective against collusion attacks. Abdelaziz et al. [34] introduced a signal arrival angle-based authentication mechanism, but this technique remains vulnerable to position-offset attacks.

Nguyen et al. [35] devised a signal-based relative positioning approach to verify the authenticity of V2V shared

data. Naserian and Lewis [36] suggested a frequency-based mechanism that detects misbehavior by analyzing changes in message transmission frequencies, leveraging the Doppler effect.

B. Why We Need Machine Learning-Based Intrusion Detection Systems in V2X

Despite their foundational contributions, traditional misbehavior detection approaches face significant limitations in real-world V2X deployments:

- **Dependence on Ideal Cooperation:** Methods such as MARV-X [29] assume that surrounding vehicles will cooperate, making them vulnerable to adversarial behaviors.
- **Susceptibility to Data Manipulation:** Techniques relying on sensor consistency [30], [31] are prone to attacks that manipulate sensor inputs, particularly Sybil attacks [32].
- **Honest Majority Assumption:** Several cooperative misbehavior detection approaches [33] assume that most vehicles are honest, making them ineffective against collusion attacks.
- **Practical Vulnerabilities:** Signal arrival angle-based authentication mechanisms [34] can be compromised by adversarial position manipulation.
- **Limited Scope of Verification:** Some techniques, such as the signal-based positioning approach, focus on specific attacks, limiting their ability to detect a wide range of misbehaviors.
- **Dependence on External Inputs:** Frequency-based misbehavior detection methods require accurate environmental data, reducing their effectiveness in non-line-of-sight (NLOS) scenarios.
- **High False Positive Rates:** Many traditional methods generate excessive false positives, leading to unnecessary alerts and decreased trust in the system.
- **Scalability Challenges:** As V2X networks grow, conventional detection methods struggle to scale efficiently due to high computational and communication overhead.
- **Inability to Adapt to Evolving Threats:** Rule-based approaches lack the flexibility to detect emerging cyber threats, as attackers continuously refine their techniques.

The heterogeneous nature of V2X communication introduces additional security challenges, as different wireless communication technologies, such as Dedicated Short-Range Communications (DSRC) and 5G-V2X, operate under distinct protocols, requiring adaptable intrusion detection systems (IDS) to ensure cross-technology security. These challenges are illustrated in Figure 2. Furthermore, inconsistencies in traditional misbehavior detection systems create gaps in scalability, accuracy, and real-time threat response, as shown in Figure 3. This survey addresses these gaps by systematically analyzing ML methodologies, evaluating benchmark datasets, and exploring real-world applicability, ensuring a structured and up-to-date reference for securing next-generation V2X networks.

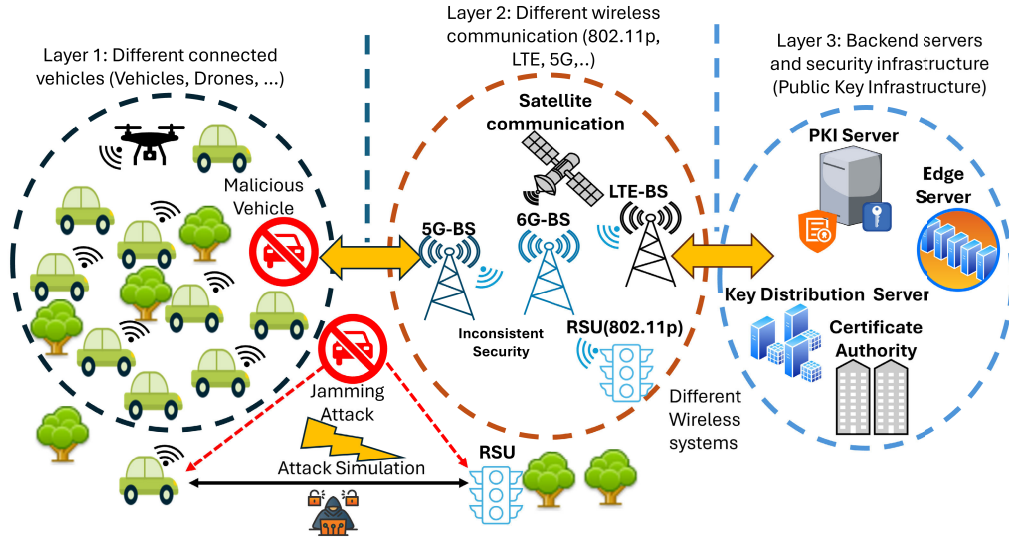


Fig. 2. Inconsistent security in V2X communication due to different wireless communications.

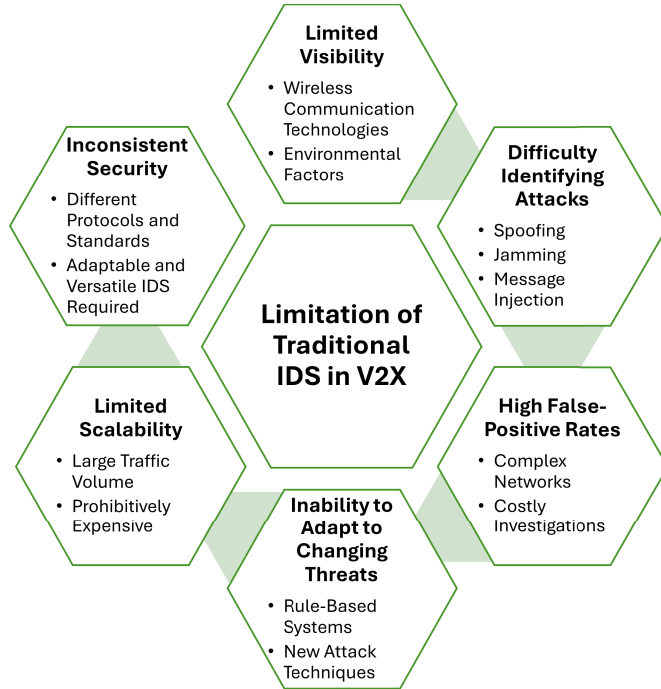


Fig. 3. Limitations of traditional misbehavior detection systems in V2X networks.

III. CRITICAL REVIEW OF EXISTING MACHINE LEARNING-BASED IDS SURVEYS FOR V2X: LIMITATIONS AND RESEARCH GAPS

ML has emerged as a powerful tool for addressing complex challenges across various domains, including vehicular networks. The wireless communication between vehicles and infrastructure exposes these networks to potential security threats, making ML an increasingly vital resource for identifying and mitigating risks in vehicle-to-everything (V2X) communication systems. This section provides an overview of existing surveys in the literature, highlighting their

contributions and limitations, while distinguishing our work from prior efforts.

A. Existing Machine Learning Based IDS Surveys

We conducted a thorough review of relevant survey studies, focusing on their exploration of security issues and the application of ML techniques in vehicular networks. Below, we summarize key findings from these studies and outline how our work diverges from theirs. Talpur et al. [37] conducted a comprehensive survey on the intersection of ML and vehicular networks, emphasizing security challenges and advanced ML-based solutions. While their work provides a detailed classification of security attacks and reviews ML techniques, it acknowledges limitations in privacy, trust, and scalability when implementing ML in vehicular networks. Our survey builds on this foundation by addressing these limitations and offering a more focused analysis of ML methodologies for V2X security. Sun et al. [38] explored cybersecurity in Connected and Autonomous Vehicles (CAVs), categorizing attacks into in-vehicle controller attacks, V2X attacks, and miscellaneous threats. Although their survey examines defense strategies and cybersecurity standards, it does not discuss the use of ML for security solutions. Our work fills this gap by providing a detailed analysis of ML-based approaches for securing V2X networks.

Dibaei et al. [39] investigated attacks and defenses in intelligent connected vehicles, classifying threats into in-vehicle and V2X communication attacks. While they discuss cryptographic techniques and network security measures, their study lacks a thorough evaluation of relevant datasets and real-time performance metrics. Our survey addresses these shortcomings by incorporating diverse benchmark datasets and discussing practical implementation challenges.

Ying et al. [40] provided a comprehensive overview of V2X communication security, focusing on historical developments and prevalent risks. However, their survey does not thoroughly

explore AI/ML-powered intrusion detection mechanisms or evaluate datasets for developing effective security solutions. Our work emphasizes recent advancements in ML techniques and their application in V2X networks. Gultarte et al. [41] focused on the market significance of V2X security, including investments, research, and patents. While their survey offers valuable insights, it lacks detailed discussions on specific attacks, mitigation methods, and AI approaches. Our study complements their work by providing an in-depth analysis of ML methodologies and dataset benchmarks. Solaas et al. [42] systematically reviewed anomaly detection in CAVs, analyzing AI algorithms such as LSTM, CNN, and autoencoders. However, their review primarily focuses on the CAN protocol and lacks specifics on AI methodologies and dataset benchmarks. Our survey expands on this by covering a broader range of V2X communication protocols and evaluating diverse datasets.

Boualouache et al. [43] investigated ML-based Misbehavior Detection Systems (MDS) for 5G and beyond vehicular networks. While their survey categorizes ML methodologies, it lacks a comprehensive analysis of practical implementation challenges. Our work addresses this gap by discussing real-time application scenarios and performance evaluation.

Sakiz et al. [44] examined security risks and detection methods in VANETs and IoV, focusing on conventional intrusion detection techniques. However, their survey does not cover recent advancements in ML-based solutions. Our study highlights the latest ML approaches for enhancing V2X security. Nagarajan et al. [45] explored ML-based Intrusion Detection Systems (IDS) for connected autonomous vehicles, focusing on intra-vehicle communication via the CAN-BUS. Their study does not consider alternative V2X communication protocols or provide a thorough analysis of datasets. Our survey addresses these limitations by including a broader range of protocols and datasets. Sedar et al. [46] conducted a comprehensive examination of V2X cybersecurity mechanisms, categorizing attacks and defense mechanisms. While they explore the potential of AI/ML, their survey lacks a detailed analysis of specific ML methods for intrusion detection. Our work provides a more focused discussion on ML techniques and their practical applications.

Sharma et al. [47] investigated IDS and proactive security mechanisms in VANETs, highlighting unresolved issues and future research areas. However, their paper does not offer an in-depth analysis of ML methods or examine datasets for practical applications. Our survey addresses these gaps by providing a detailed evaluation of ML methodologies and benchmark datasets. Tang et al. [48] examined ML methods in vehicular networks, focusing on the 6G era. While their survey offers valuable insights, it lacks specifics on ML algorithms, validation methods, and dataset benchmarks. Our work complements their study by addressing these aspects in detail. Rajbahadur et al. [49] surveyed anomaly detection techniques for connected vehicle cybersecurity, providing statistical overviews without detailed explanations of methodologies. Our survey offers a more in-depth analysis of ML techniques and their practical applications. Alrehan et al. [50] reviewed ML methods for DDoS attacks in VANETs, focusing primarily on DDoS threats. Their survey lacks comprehensive

coverage of other VANET-specific attacks and ML techniques. Our work addresses this by providing a broader analysis of security threats and ML methodologies.

Gonçalves et al. [51] reviewed intelligent IDS for vehicular communication, emphasizing the significance of IDS solutions. However, their study lacks an in-depth analysis of practical implementation specifics. Our survey addresses this gap by discussing real-world performance validation. Asuquo et al. [52] examined privacy and security concerns in vehicular networks, focusing on cryptographic mechanisms and privacy-enhancing schemes. Their study does not address ML-based solutions, which our survey comprehensively covers.

Liang et al. [53] reviewed ML frameworks for high-mobility vehicular networks, focusing on network optimization. While they discuss unauthorized access detection, their paper does not study security threats or ML methodologies in depth. Our work provides a more focused analysis of ML techniques for V2X security.

Kuutti et al. [54] investigated deep learning in vehicle control systems, showcasing its potential for improving control systems. However, their survey does not address security issues related to deep learning in vehicular networks. Our study fills this gap by exploring the role of ML in enhancing V2X security. Hossain et al. [55] reviewed ML methods in cognitive radio-based vehicular networks (CR-VANETs), focusing on security threats within the CR environment. Our survey expands on this by addressing a broader range of security challenges in V2X networks.

Lu et al. [56] provided an overview of advancements in VANET security, privacy, and trust management. However, their survey does not consider the importance of ML algorithms or their practical implementation. Our work addresses this by emphasizing the role of ML in enhancing VANET security.

Sheikh et al. [57] analyzed security concerns in VANETs, focusing on the structure and challenges of VANETs without discussing the use of ML for security issues. Our survey complements their work by providing a detailed analysis of ML-based solutions.

Tong et al. [58] surveyed ML in vehicular communication networks, briefly touching upon ML-based methodologies for identifying and preventing malicious activities. Our study builds on their work by comprehensively analyzing ML techniques and their applications in V2X networks.

Farsimadan et al. [59] review security challenges in V2X communications, highlighting vulnerabilities in Vehicular Ad-hoc Networks (VANETs) and classifying threats based on architecture, models, and attack types. They discuss countermeasures such as cryptographic authentication, trust management, and intrusion detection. However, the study lacks a detailed examination of machine learning-based intrusion detection systems (IDS), adversarial AI threats, and dataset benchmarking. While IDS approaches are mentioned, their effectiveness in anomaly detection and resilience to adversarial attacks remain unaddressed.

Zhang et al. [60] provide a survey on V2X communication in Intelligent Connected Vehicles (ICVs),

analyzing pre-communication, during-communication, and post-communication challenges. They examine wireless technologies, security threats, datasets, and experimental platforms, emphasizing their impact on traffic efficiency and autonomous driving. However, the study has notable gaps: it does not deeply evaluate intrusion detection methods or compare their performance against different attacks. The dataset review is limited to cooperative perception, overlooking real-world challenges such as sensor fusion and privacy-preserving communication. Additionally, while experimental platforms are discussed, the lack of benchmarking and performance comparisons weakens practical applicability. The study also overlooks key emerging areas, including quantum security, 6G-enabled V2X, and federated learning, which are essential for future secure V2X systems. While previous surveys provide valuable insights into ML-based security for V2X communication, they often lack comprehensive evaluations of specific ML architectures, dataset benchmarking, real-world deployment challenges, and federated learning scalability. Many studies focus on theoretical models without addressing computational constraints, adversarial threats, and emerging AI-driven attacks. Additionally, limited coverage of key performance metrics such as accuracy, precision, recall, and false alarm rates makes it difficult to assess IDS effectiveness.

Our survey addresses these gaps by systematically analyzing ML techniques for intrusion and misbehavior detection in V2X networks. By integrating the latest advancements up to 2025, we provide a comprehensive, up-to-date perspective on emerging security threats and technological developments. Tables II and III summarize the key limitations in recent V2X security surveys.

IV. V2X COMMUNICATION FUNDAMENTAL AND COMPONENTS

The architecture of V2X networks is structured hierarchically, enabling communication between vehicles, network entities, and infrastructure components. This section provides an analysis of V2X communication modes, protocols, fundamental components, message structures, and security concerns.

A. V2X Communication Modes, Layers and Standards Protocols

V2X communication include multiple modes, facilitating interactions between different entities within the network [61]:

- **Vehicle-to-Vehicle (V2V):** Direct wireless communication among vehicles within the same communication range, ensuring real-time data exchange for collision avoidance and cooperative driving [62].
- **Vehicle-to-Infrastructure (V2I):** Interaction between vehicles and roadside units (RSUs), traffic signals, and urban infrastructure to enhance road safety and traffic management [63].
- **Vehicle-to-Network (V2N):** Establishing a connection between vehicles and cellular networks, enabling access to cloud-based services, remote updates, and enhanced connectivity [64].

- **Vehicle-to-Pedestrian (V2P):** Sharing real-time situational awareness between vehicles and pedestrians, improving safety in urban environments [65].

Figure 4 illustrates the different communication modes within the V2X network architecture.

V2X communication relies on various protocols designed to support high-reliability, low-latency interactions [35]:

- **IEEE 802.11p (WAVE Standards):** An extension of the IEEE 802.11 standard, operating at 5.9 GHz, providing dedicated short-range communication (DSRC) with low latency for safety applications [66].
- **LTE-V2X (Long-Term Evolution for V2X):** A cellular-based technology that offers improved coverage and scalability, supporting advanced V2X applications and enabling future integration with 5G networks [67].
- **5G-V2X:** Extends LTE-V2X by leveraging Ultra-Reliable Low-Latency Communication (URLLC) to support real-time mapping, autonomous driving, and cooperative maneuvering [68].
- **6G-V2X:** Future communication standard expected to enhance ubiquitous connectivity, real-time decision-making, AI integration, and edge computing for intelligent transportation systems [69].
- **IEEE 1609.x Standards:** A family of standards defining security mechanisms and communication architecture to ensure confidentiality, integrity, and authenticity of transmitted messages [70].

Figure 5 summarizes different V2X communication standards, while Figure 6 represents the V2X protocol stack.

The V2X communication stack is layered to ensure efficient message transmission [71]:

- **Application Layer:** Handles V2X safety applications, such as Basic Safety Messages (BSM), Signal Phase and Timing (SPaT), and Emergency Vehicle Alerts (EVA) [72].
- **Transport Layer:** Utilizes Transmission Control Protocol (TCP) for reliable communication and User Datagram Protocol (UDP) for low-latency data exchange [73].
- **Network Layer:** Supports IPv6-based routing and Named Data Networking (NDN), which focuses on content-based data retrieval [74].
- **Data Link Layer:** Includes IEEE 802.11p for short-range wireless communication and LTE-V2X for cellular-based connectivity [75].
- **Physical Layer:** Encompasses DSRC and 5G-V2X technologies, providing high-speed and low-latency communication [76].

B. V2X Fundamental Components and Basic Messages

The efficient functioning of V2X networks depends on key components [77]:

- **On-Board Units (OBUs):** Vehicle-mounted devices responsible for transmitting and receiving V2X messages [78].
- **Roadside Units (RSUs):** Fixed infrastructure components that facilitate communication between vehicles and network entities [79].

TABLE II
RECENT SURVEY PAPERS AND THEIR LIMITATIONS (PART 1)

Study	Year	Title	Focus Area	Key Contributions and Limitations
Farsimadan et al. [59]	2025	A Review on Security Challenges in V2X Communications Technology for VANETs	General Security Survey in V2X Networks	Broad overview of V2X security; lacks focus on ML-based security, adversarial AI, and real-world challenges
Xinyu Zhang et al. [60]	2025	Vehicle-to-Everything Communication in Intelligent Connected Vehicles: A Survey and Taxonomy	V2X Communication in ICVs	Comprehensive V2X communication phases; lacks ML-based security and federated learning insights
Ying et al. [40]	2024	A Literature Review on V2X Communications Security	V2X Communication Security	Covers cryptographic threats but not AI/ML or dataset evaluations
Gularte et al. [41]	2024	Exploring the Cybersecurity Landscape through SLR	V2X Cybersecurity	Focuses on market stats; lacks attack mitigation and ML techniques
Solaas et al. [42]	2024	Anomaly Detection in Connected and Autonomous Vehicles	Anomaly Detection in CAVs	Discusses anomaly detection but lacks datasets and attack-specific analysis
Boualouache & Engel [43]	2023	ML-Based Misbehavior Detection in 5G-V2X	Misbehavior Detection in 5G-V2X	Emphasizes ML techniques; lacks real-world validations and dataset benchmarking
Nagarajan et al. [45]	2023	IDS for Connected Autonomous Vehicles	IDS for Autonomous Vehicles	IDS for CAN-BUS; lacks insights into broader V2X protocols
Sedar et al. [46]	2023	V2X Cybersecurity Mechanisms and Research Paths	V2X Cybersecurity	Covers traditional mechanisms; lacks DL and real-world application focus
Talpur & Gurusamy [37]	2022	ML for Vehicular Network Security	ML in Vehicular Networks	Comprehensive ML survey; limited by trust, privacy, and scalability challenges
Sun et al. [38]	2022	Cybersecurity of Connected and Autonomous Vehicles	CAV Cybersecurity	High-level overview; minimal coverage of ML defense techniques
Dibaei et al. [39]	2020	Attacks and Defenses in Intelligent Vehicles	Attack Mitigation in V2X	Focus on cryptography; lacks AI/ML and real-time strategies
Kuutti et al. [54]	2020	Deep Learning Applications in AV Control	Deep Learning in AVs	Focuses on vehicle control; does not cover cybersecurity aspects
Hossain et al. [55]	2020	ML in Cognitive Radio-VANETs	ML in CR-VANET	Addresses cognitive radio; omits security threat discussions
Sheikh & Liang [57]	2019	VANET Security Services in Traffic Management	VANET Security	Architecture-level discussion; lacks ML techniques
Tong et al. [58]	2019	AI in Vehicle-to-Everything Communication	ML in V2X Networks	Discusses AI-based V2X; limited IDS analysis
Tang et al. [48]	2019	Secure 6G Vehicular Networks via ML	ML in 6G V2X	ML in future networks; lacks practical algorithmic validation

TABLE III
RECENT SURVEY PAPERS AND THEIR LIMITATIONS (PART 2)

Study	Year	Title	Focus Area	Key Contributions and Limitations
Alrehan & Alhaidari [50]	2019	Detecting DDoS Attacks in VANETs via ML	ML for DDoS Detection	Focus on DDoS only; lacks diversity of threat models
Gonçalves et al. [51]	2019	Intelligent IDS in VANETs	IDS for VANETs	Covers ML and IDS datasets; lacks deployment analysis
Asuquo et al. [52]	2018	Security and Privacy in Location-Based Services	Privacy in VANETs	Cryptography-focused; omits ML-based IDS techniques
Liang et al. [53]	2018	Toward Intelligent Vehicular Networks	ML in Vehicular Networks	Network optimization focus; lacks intrusion detection context
Sharma & Kaul [47]	2018	IDS and Honeypot-Based Security in VANETs	IDS in VANETs	Proposes honeypots; no ML or dataset benchmarking
Rajbahadur et al. [49]	2018	Anomaly Detection for CV Cybersecurity	Anomaly Detection in CVs	Proposes taxonomy; lacks implementation depth
Sakiz & Sen [44]	2017	Attacks and Detection in VANETs/IoV	VANET/IoV Security	Reviews detection mechanisms; lacks ML insights
Lu et al. [56]	2018	Security, Trust, and Privacy in Vehicular Networks	Security and Trust in VANETs	Discusses architectures; omits ML-based frameworks

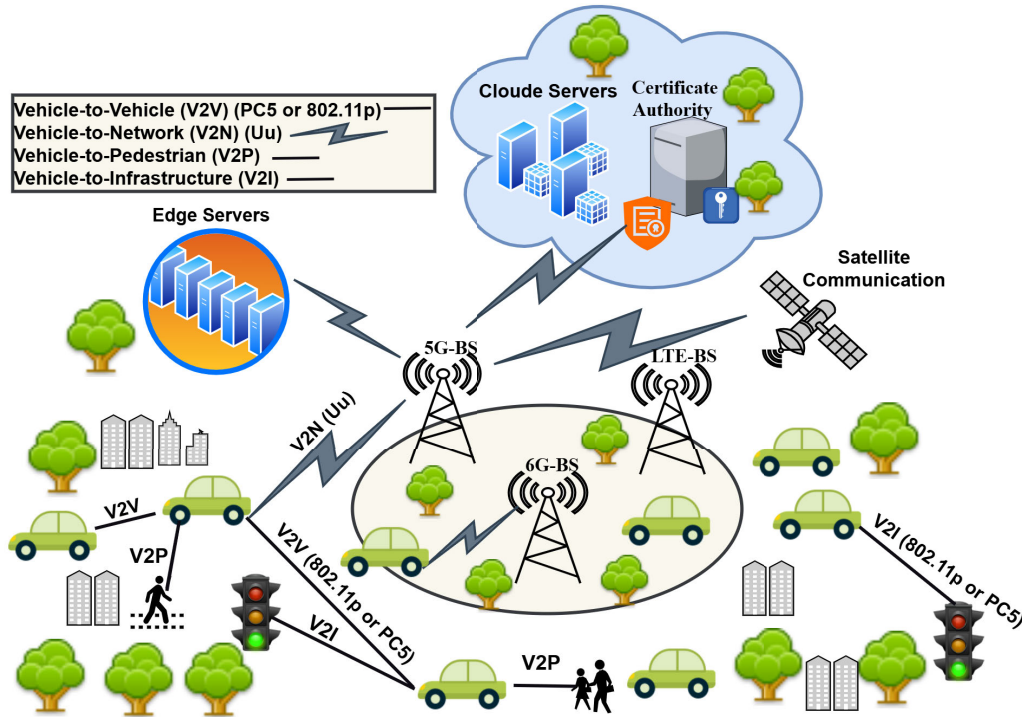


Fig. 4. V2X communication architecture.

- **Certificate Authorities (CAs):** Entities that issue digital certificates to authenticate communication participants [80].
- **Security Credential Management System (SCMS):** Manages the distribution, renewal, and revocation of

security certificates to maintain trust within the network [81].

V2X networks support various message types to enable seamless communication [82]:

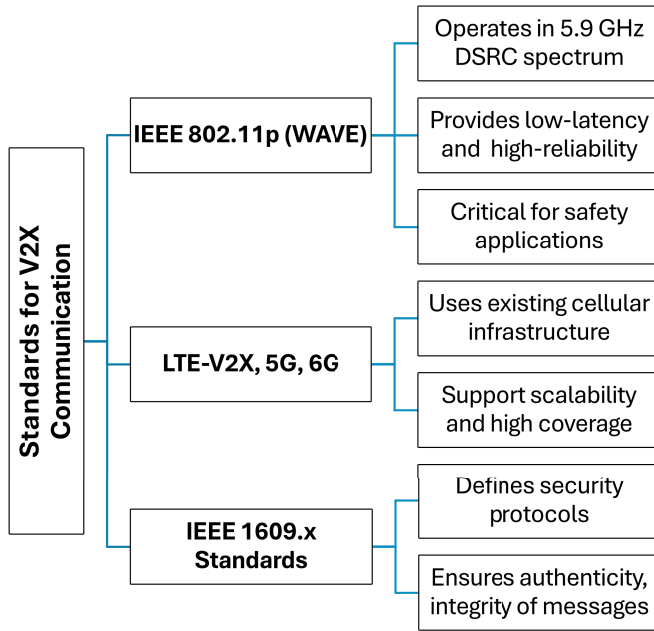


Fig. 5. V2X communication standards.

- **Basic Safety Messages (BSMs):** Provide critical information on vehicle position, speed, and acceleration [83].
- **Signal Phase and Timing (SPaT):** Transmit traffic signal status for improved intersection management [84].
- **Map Data Messages:** Offer detailed road topology, lane configurations, and speed limit data [85].
- **Emergency Vehicle Alerts (EVAs):** Notify nearby vehicles of emergency vehicle presence and intended routes [86].

Ensuring security in V2X networks is paramount to prevent unauthorized access and data breaches. The main security measures include:

- 1) **Message Authentication:** Cryptographic digital signatures verify message authenticity and prevent spoofing [87].
- 2) **Encryption:** Symmetric and asymmetric encryption techniques secure message confidentiality from eavesdropping threats [4].
- 3) **Certificate Management:** Certificate Authorities (CAs) issue and revoke digital certificates for network entities [88].
- 4) **Secure Key Management:** Proper key generation, distribution, and renewal protect cryptographic integrity [89].
- 5) **Privacy Preservation:** Pseudonymization techniques prevent vehicle tracking and unauthorized profiling.
- 6) **Intrusion Detection and Prevention:** IDS solutions detect and mitigate network anomalies to enhance cybersecurity.

Figure 7 illustrates the key security threats and mitigation strategies in V2X communication.

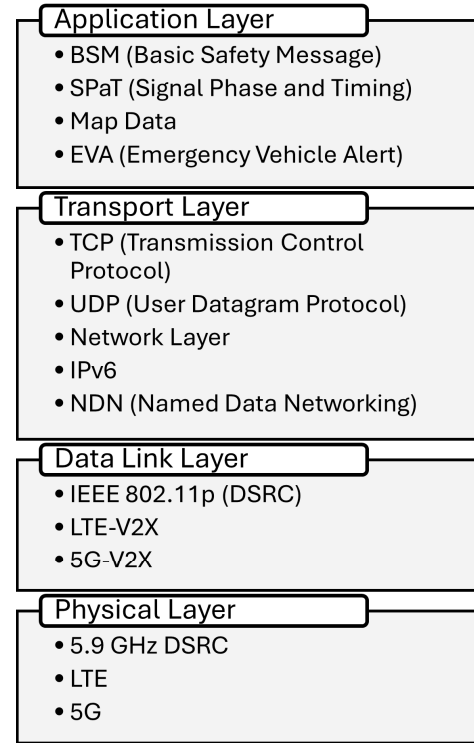


Fig. 6. V2X communication protocol stack.

V. TRADITIONAL ATTACKS IN V2X NETWORKS: ATTACK MODELS AND MITIGATION STRATEGIES

This section provides an overview of key vulnerabilities within V2X components, classifies potential traditional attack models, and discusses mitigation strategies.

A. Targets of Attacks

The interconnected nature of V2X communication systems relies on a range of sensors, controllers, and communication mechanisms. Any compromise to these components can lead to system malfunctions, unauthorized access, or data manipulation, ultimately affecting road safety. This subsection outlines the primary V2X components susceptible to cyber threats, supported by both theoretical and empirical attack models [63] [66] [86] [90].

1) **On-Board Diagnostic Port (OBD):** The On-Board Diagnostic (OBD) port serves as a critical interface for monitoring vehicle parameters such as emissions, speed, and engine status. Initially introduced in 1987 with OBD-I and later standardized as OBD-II in 1996, it provides real-time access to Electronic Control Unit (ECU) data and supports firmware updates. However, its accessibility poses a security risk, allowing adversaries to manipulate ECU data or inject malicious firmware. Effective mitigation strategies include enforcing strict access control policies and implementing secure firmware update mechanisms.

2) **Electronic Control Units (ECUs):** ECUs govern multiple vehicular subsystems, including engine management, braking systems, tire pressure monitoring, and inertial measurement units (IMUs). These units are responsible for critical functions

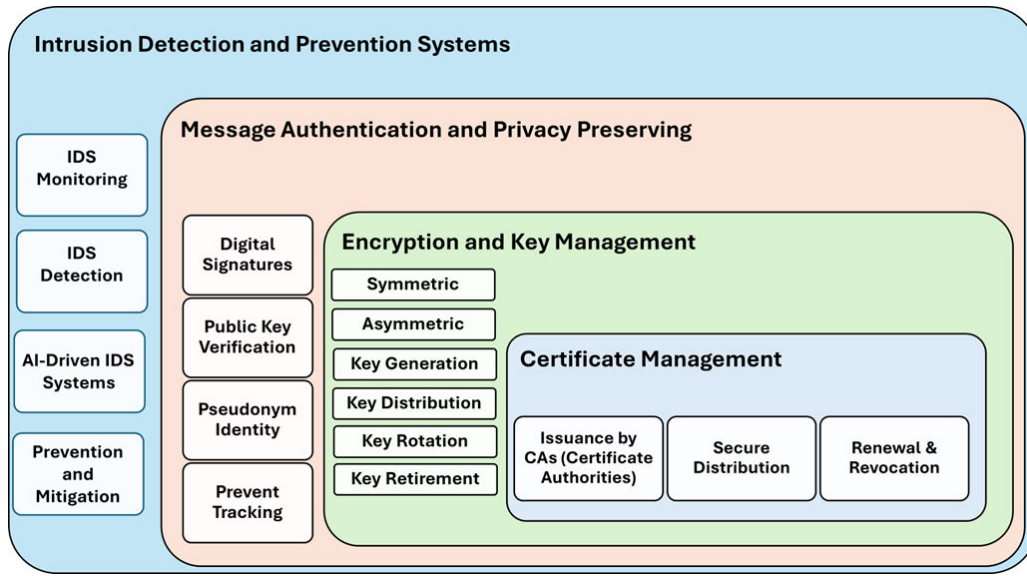


Fig. 7. Security consideration in V2X communication.

TABLE IV
VEHICLE COMPONENTS, VULNERABILITIES, AND MITIGATION TECHNIQUES

Component	Description	Vulnerabilities	Mitigation Techniques
On-board Diagnostic Port (OBD)	Gathers data on vehicle emissions, mileage, speed, and component info. OBD-II standard since 1996.	Real-time data access, ECU data alteration, and firmware updates.	Robust firmware update protocols and stringent access controls.
Electronic Control Units (ECUs)	Embedded control systems like brakes, engine, tire pressure, and IMUs.	Data manipulation, firmware attacks.	Strong encryption and advanced anomaly detection systems.
Controller Area Network (CAN)	Central network connecting ECUs uses a two-wire, half-duplex configuration for high-speed communication.	Lack of authentication fields, data interception, and injection attacks.	Network segmentation and secure communication protocols.
Sensors	LiDAR, Radar, GPS, and Cameras: Measure distances, detect objects and provide navigation and positioning data.	Jamming, spoofing, data manipulation attacks.	Multi-sensor data fusion, redundancy techniques.
Connection Mechanisms	V2V: Communication between vehicles. V2I: Communication between vehicles and road infrastructure.	Data interception, signal jamming, spoofing.	Robust communication protocols and advanced encryption techniques.

such as regulating air-fuel mixture, monitoring wheel-speed sensors, and ensuring real-time adjustments to braking pressure. Cyberattacks targeting ECUs often involve firmware manipulation or data injection, potentially leading to vehicle malfunctions. To safeguard ECUs, strong encryption mechanisms and advanced anomaly detection techniques should be implemented [91].

3) *Controller Area Network (CAN)*: The Controller Area Network (CAN) facilitates communication between ECUs within a vehicle through a two-wire, half-duplex configuration. Despite its efficiency, CAN lacks authentication mechanisms, making it vulnerable to data interception and injection attacks. Malicious actors can exploit this weakness to alter control messages or introduce fake commands. Countermeasures include network segmentation and the adoption of secure communication protocols to fortify the CAN bus against unauthorized intrusions.

4) *Sensors*: Modern vehicles incorporate a variety of sensors to support autonomous functionalities, including:

- **LiDAR**: Utilizes laser-based technology for distance measurement and object detection.
- **Radar**: Employs electromagnetic waves to identify objects and estimate their velocity.
- **GPS**: Relies on satellite signals to determine location and provide navigation assistance.
- **Cameras**: Capture 360-degree environmental data to enhance lane detection, obstacle avoidance, and traffic sign recognition.

These sensors are susceptible to jamming, spoofing, and data manipulation attacks. Effective countermeasures involve multi-sensor data fusion, redundancy techniques, and signal integrity verification. V2X communication relies on two primary communication mechanisms, each presenting unique security challenges. Both V2V and V2I communications are susceptible to data interception, signal jamming, and spoofing attacks. Implementing robust encryption standards and secure authentication protocols is essential to maintaining the integrity of these communication channels. Table IV outlines the associated attack vectors for different vehicle components.

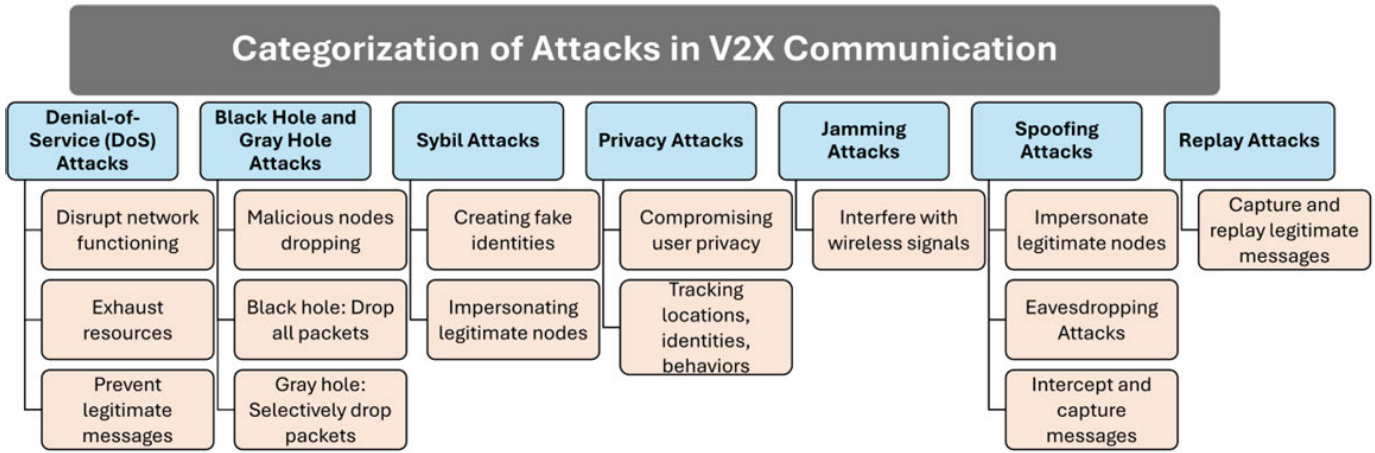


Fig. 8. Attacks categories in V2X communication.

B. Categories of Attack Models

V2X attacks can be classified based on the attacker's access level and primary objectives:

- **Remote-Access Attacks:** Conducted without physical access to the vehicle, these attacks typically involve signal interception, spoofing, or unauthorized data extraction. Examples include injecting false sensor data or manipulating wireless signals to deceive navigation systems.
- **Physical-Access Attacks:** Require direct access to the vehicle, often involving hardware manipulation, malicious firmware updates, or ECU reprogramming.

Cyberattacks on V2X communication networks are typically motivated by one of the following objectives:

- **Disrupt Operations:** Includes Denial-of-Service (DoS) attacks aimed at overwhelming vehicular networks, disabling essential functions, or disrupting navigation systems.
- **Gain Control:** Attackers attempt to take over vehicle control systems, allowing unauthorized manipulation of steering, acceleration, or braking.
- **Steal Information:** Exploits vulnerabilities to extract sensitive user data, which can be used for identity theft, industrial espionage, or illicit tracking.

Effective cybersecurity measures must be employed to protect V2X networks from evolving threats:

- **OBD Attacks:** Prevent unauthorized access to ECU data and vehicle operations by enforcing stringent access control mechanisms and secure firmware update protocols.
- **ECU Attacks:** Mitigate firmware tampering and data manipulation risks by deploying strong encryption techniques and real-time anomaly detection.
- **CAN Bus Attacks:** Address authentication weaknesses to prevent malicious data injection through secure network segmentation and communication filtering.
- **Sensor Attacks:** Counter spoofing and jamming attempts by integrating redundancy mechanisms, signal integrity verification, and multi-sensor data fusion techniques.

C. Traditional Attacks and Mitigation Strategies

This subsection classifies the major types of traditional attacks targeting V2X systems, highlighting their characteristics, potential impact, and mitigation strategies. A summary of these attacks is depicted in Figure 8.

1) *Denial-of-Service (DoS) Attacks:* DoS attacks aim to overwhelm network nodes, such as roadside units (RSUs) or onboard units (OBUs), with excessive data requests, rendering them incapable of processing legitimate information. These attacks can disrupt vehicle authentication, traffic updates, and security protocols. A more severe variation, distributed denial-of-service (DDoS) attacks, involves multiple compromised vehicles or bots sending coordinated malicious requests, making detection and mitigation more challenging [85]. Traditional mitigation strategies, such as blocking attacker IP addresses, are ineffective against DDoS due to the use of multiple source addresses. Advanced anomaly detection and traffic filtering mechanisms can enhance resilience against these attacks.

2) *Black-Hole Attacks:* In a black-hole attack, a compromised vehicle intercepts and drops all network packets instead of forwarding them to their intended destination, creating a "black hole" that disrupts communication. A variant of this attack, known as the grey-hole attack, selectively drops packets to avoid detection [4]. These attacks degrade network performance and obstruct vital data flow. Implementing sequence numbers in packet transmissions allows legitimate nodes to detect anomalies in message forwarding. Additionally, machine learning-based solutions, such as the Black Hole Attack Detection Algorithm (BADA), can help identify malicious nodes based on historical routing behaviors.

3) *Replay Attacks:* Replay attacks involve intercepting and retransmitting previously recorded legitimate messages to deceive network participants. Attackers can use this technique to bypass authentication mechanisms and gain unauthorized access to the network [84]. Such attacks are particularly concerning in vehicle authentication and secure communication protocols. Mitigation measures include implementing strong encryption, incorporating time stamps or nonces in message

authentication protocols, and employing Virtual Private Networks (VPNs) to secure data exchanges.

4) *Sybil Attacks*: Sybil attacks involve an adversary generating multiple false identities or impersonating multiple legitimate nodes to manipulate V2X network behavior. Attackers may create congestion by simulating high traffic density or misleading vehicles into rerouting unnecessarily. GPS spoofing is often used in conjunction with Sybil attacks to manipulate navigation data. Machine learning-based detection methods that analyze node behaviors, such as spatial inconsistencies and abnormal movement patterns, can help mitigate Sybil attacks. Cryptographic authentication techniques and certificate-based verification further enhance network integrity [71].

5) *Impersonation Attacks*: Impersonation attacks occur when an adversary masquerades as a legitimate RSU or vehicle, thereby gaining unauthorized access to network resources or sensitive information. Such attacks may grant the attacker privileged access to traffic updates, navigation data, or vehicle control functions [90]. Countermeasures include employing strong encryption for authentication, integrating location-based verification mechanisms, and utilizing clustering techniques to validate network entities.

6) *Malware Attacks*: Due to the dynamic nature of vehicle networks, software updates and data exchanges occur frequently. If these updates originate from untrusted sources, malware can infiltrate the vehicle's system, leading to data corruption, system failures, or unauthorized control over vehicle functionalities. Firewalls and intrusion detection systems (IDS) serve as the primary defense mechanisms against malware by filtering suspicious messages. Additionally, reputation-based verification systems can help ensure that only trusted updates are accepted [89].

7) *False Information Attacks*: Attackers can disseminate misleading data regarding road conditions, accidents, or congestion to manipulate traffic flow or cause safety hazards. For example, an adversary might transmit false congestion reports, diverting vehicles away from an optimal route. Conversely, attackers could suppress warnings about actual hazards, increasing the risk of collisions [84]. Reputation-based trust models that reward accurate reporting and penalize misleading information help mitigate this threat. Anomaly detection techniques and cross-validation with multiple data sources further enhance the reliability of shared information.

8) *Timing Attacks*: Timing attacks involve delaying or manipulating the transmission of time-sensitive messages, disrupting real-time applications such as collision avoidance and emergency braking. Delays in V2X communication can be catastrophic, particularly in autonomous driving scenarios, where decision-making depends on instantaneous data exchange [68]. Mitigation strategies include enforcing strict timing constraints, using synchronized clocks, and implementing delay monitoring techniques to detect intentional message delays.

9) *Privacy Attacks*: Privacy breaches in V2X networks occur when attackers track vehicle locations, identities, and behavioral patterns by intercepting communication signals.

Adversaries may exploit side-channel information to infer private user data, such as frequently visited locations or personal driving habits [87]. Privacy-preserving techniques, such as encryption, pseudonymization, and data obfuscation, play a crucial role in preventing unauthorized tracking.

10) *Jamming Attacks*: Jamming attacks interfere with wireless signals, preventing vehicles from transmitting or receiving critical safety messages. This type of attack can disrupt V2V and V2I communication, leading to loss of situational awareness and potential traffic incidents [4]. Machine learning-based IDS can help identify abnormal signal interference patterns and distinguish legitimate transmissions from malicious jamming attempts.

11) *Spoofing Attacks*: Spoofing attacks involve an adversary pretending to be a legitimate V2X node to manipulate traffic conditions, reroute vehicles, or introduce false safety alerts. These attacks undermine trust in V2X communication networks and can cause severe disruptions. IDS leveraging machine learning algorithms can detect spoofed nodes by analyzing behavioral inconsistencies and comparing them with trusted network entities.

12) *Eavesdropping Attacks*: Eavesdropping attacks involve unauthorized interception of V2X messages, allowing adversaries to access confidential vehicle information, such as user credentials or navigation data. Attackers may exploit intercepted data for identity theft, industrial espionage, or targeted attacks [69]. Implementing end-to-end encryption, secure key management, and anomaly-based IDS helps prevent eavesdropping threats.

Table V provides a summary of the traditional attack models in V2X communication.

VI. AI-BASED ATTACKS AND ADVERSARIAL THREATS IN V2X NETWORKS

The integration of AI in V2X communication introduces new vulnerabilities. AI-based attacks, particularly adversarial attacks, target the ML models used for network optimization, resource allocation, intrusion detection, and traffic prediction, compromising safety-critical applications such as collision avoidance, remote driving, and platooning. This section examines adversarial attack types, their impact on V2X communication systems, a review of state-of-the-art research, and mitigation strategies to secure vehicular networks.

A. Definition of AI-Based and Adversarial Attacks in V2X Networks

The primary objective of adversarial attacks in V2X networks is to reduce the reliability of AI models, disrupt resource allocation, and compromise safety-critical vehicular communications. This subsection provides a detailed exploration of adversarial attack simulations in V2X ML models and deep reinforcement learning (DRL)-based mitigation strategies, incorporating insights from Sharma et al. [92], Sedar et al. [93], and others. Adversarial attacks in V2X security exploit ML vulnerabilities to manipulate vehicular data and mislead misbehavior detection models. These attacks

TABLE V
TYPES OF TRADITIONAL ATTACKS IN V2X NETWORKS AND MITIGATION STRATEGIES

Attack Type	Attack Strategy	Mitigation Techniques
Denial-of-Service (DoS)	Overloading the network with excessive data to disrupt legitimate communication.	Blocking malicious IP addresses, network traffic monitoring, and anomaly detection.
Black-Hole Attack	Malicious nodes drop data packets instead of forwarding them, creating a communication void.	Implementing packet sequence numbers, anomaly-based routing mechanisms.
Replay Attack	Intercepting and retransmitting previously recorded messages to deceive network participants.	Strong encryption, time-stamp validation, VPN-based secured communication.
Sybil Attack	Generating multiple fraudulent identities to manipulate network behavior.	Cryptographic authentication, anomaly detection techniques, and trust-based validation.
Impersonation Attack	Malicious entities disguise themselves as legitimate nodes to gain unauthorized access.	Encryption-based authentication, location verification, clustering-based detection mechanisms.
Malware Injection	Infiltrating vehicle systems via untrusted updates, leading to data corruption or system failures.	Firewalls, signature-based intrusion detection, and reputation-based verification of updates.
Falsified Information Attack	Disseminating misleading traffic data to manipulate vehicle routing and create disruptions.	Reputation-based trust mechanisms, cross-validation with multiple data sources.
Timing Attack	Delaying critical safety messages to disrupt time-sensitive applications.	Enforcing fixed data transmission rates, real-time latency monitoring, and timestamp validation.
Privacy Breach	Unauthorized tracking of user locations, identities, and behavioral patterns.	Encryption, pseudonymization, and privacy-preserving cryptographic methods.
Jamming Attack	Disrupting wireless communication by emitting interference signals.	Machine learning-based intrusion detection systems (IDS) for signal pattern recognition and anomaly filtering.
Spoofing Attack	Falsifying identity to manipulate data exchange and network decisions.	Machine learning-based IDS for behavioral analysis and authentication validation.
Eavesdropping Attack	Unauthorized interception of sensitive messages to extract confidential data.	End-to-end encryption, secure key management, and anomaly-based IDS techniques.

can lead to traffic congestion, safety hazards, and compromised emergency response systems. With the increasing deployment of AI-based decision-making in V2X communication, attackers exploit vulnerabilities in deep learning models through various attack vectors [94], [95]:

- **Data Poisoning Attacks:** Injecting malicious data into training datasets, leading to incorrect traffic flow predictions or sensor fusion errors.
- **Evasion Attacks:** Modifying input signals (e.g., spoofed sensor data, altered CSI values) to mislead AI models into making incorrect decisions.
- **Model Extraction Attacks:** Reconstructing AI models through repeated queries to steal decision-making algorithms.

Adversarial attacks involve intentionally inputs that mislead AI-based V2X systems, affecting network security and efficiency. These attacks are categorized into:

- **Open-Box (Transparent) Attacks:** The attacker has full access to the AI model, allowing precise manipulation of network decisions.
- **Closed-Box (Opaque) Attacks:** The attacker has limited knowledge of the model but alters the AI's responses through repeated inputs.

B. Simulating Adversarial Attacks in V2X ML Models

in this section, we introduce some simulated adversarial attacks on V2X systems.

1) *Adversarial Attacks Targeting ML-Based Misbehavior Detection Models for CAVs:* Sharma et al. [92] present an experiment that follows a structured adversarial attack simulation to evaluate the effectiveness of current ML models.

Step 1: Generating Adversarial Data

- **Dataset Selection:** The study uses the VeReMi dataset, a benchmark dataset for VANET misbehavior detection, containing Basic Safety Messages (BSMs) with vehicle parameters such as speed, acceleration, and position.

- **Generating Adversarial Examples:** The study employs a traditional attack engine to create attack samples simulating:

- Location Spoofing Attacks (GPS manipulation)
- Sybil Attacks (Fake identities)
- Denial-of-Service (DoS) Attacks (Message flooding)

Step 2: Training the ML Model to Generate Hard-to-Detect Attacks

- **Supervised ML Training:** ML models, including K-Nearest Neighbor (KNN), Random Forest (RF), Logistic Regression (LR), and Long Short-Term Memory (LSTM), are trained using time-series data extracted from normal BSMs.
- **Adversarial Model Training:** A second ML model is trained to generate attack samples that mimic normal behavior, bypassing conventional misbehavior detection mechanisms.

Step 3: Evaluating Attack Effectiveness

- The generated adversarial attacks are tested against standard ML-based misbehavior detection systems.
- **Findings:** LSTM models demonstrate better resistance but still exhibit reduced detection performance against adversarial attacks.

2) *Adversarial Attacks on V2X Resource Allocation:* **Fast Gradient Sign Method (FGSM):** FGSM is a one-step adversarial attack that perturbs resource allocation parameters (e.g., channel state, transmission power) to force incorrect AI-based scheduling decisions, reducing [96], [97]:

TABLE VI
DRL PERFORMANCE AGAINST ADVERSARIAL ATTACKS

Attack Type	DRL Accuracy	LSTM Accuracy	MLP Accuracy
Label-Flipping (5%)	99.44%	73.80%	74.97%
Label-Flipping (10%)	99.31%	73.63%	75.43%
Policy Induction ($p = 0.2$)	91.58%	70.35%	70.28%
Policy Induction ($p = 1.0$)	50.68%	49.94%	61.48%

- Data transmission efficiency.
- Packet delivery rates.
- QoS performance for V2X applications.

Projected Gradient Descent (PGD): PGD [98] improves FGSM by applying iterative perturbations to mislead DRL-based V2X resource allocation models. It:

- Creates cumulative misallocations over multiple iterations.
- Significantly increases network congestion.
- Reduces V2X connectivity reliability.

Policy Infiltration Attack (PIA): PIA is a gradient-based adversarial attack that tricks DRL algorithms into misallocating network resources [99]. It:

- Reduces packet delivery rates by 77.74%.
- Increases latency by 150%.
- Disrupts ultra-reliable low-latency communication (uRLLC).

3) *Adversarial Attacks on Signal Processing & Physical Layer:* **Signal Manipulation Attacks:** Adversaries inject fake wireless signals that trick AI-based spectrum classifiers. These attacks:

- Mislead deep learning models in modulation classification.
- Trigger false alarms in intrusion detection systems (IDS).
- Compromise AI-powered interference mitigation.

Bearer Migration Poisoning (BMP): BMP exploits weaknesses in bearer migration procedures [100], causing:

- Traffic misrouting in V2X networks.
- Denial-of-Service (DoS) attacks on connected vehicles.
- Severe delays in collision avoidance systems.

Intelligent Reflecting Surface (IRS) Manipulation: Attackers control IRS configurations to redirect signals incorrectly, causing:

- Beamforming errors.
- Network disconnection in vehicular systems.
- Increased power consumption in smart vehicle networks.

C. Mitigation Strategies for Adversarial Attacks in V2X Networks

To counter adversarial threats, the following mitigation strategies can be implemented:

- **Adversarial Training:** ML models are trained with adversarial examples to improve robustness.
- **Defensive Distillation:** Distillation-based ML models improve resistance to adversarial perturbations.
- **Blockchain-Enabled Authentication:** Blockchain technology ensures tamper-proof vehicle authentication and prevents Sybil attacks.

- **Anomaly Detection with Deep Learning:** Autoencoders and anomaly detection models can identify subtle attack patterns.

- **Federated Learning for Secure Training:** Decentralized ML training prevents model poisoning attacks by ensuring no single entity controls all training data.

To counter adversarial ML threats, Sedar et al. [93] propose a DRL-based defense mechanism for vehicular communication systems. Their study focuses on defending against:

- **Label-Flipping Attacks:** Adversaries flip training data labels to mislead ML classifiers.
- **Policy Induction Attacks:** Adversaries inject perturbations into state observations, corrupting DRL models.

1) *DRL-Based Defense Framework:* The proposed DRL defense system operates at RSUs, classifying vehicular data and detecting misbehavior in V2X networks.

Step 1: Reinforcement Learning for Attack Detection

- **Defining the Environment:** The RSU receives real-time BSMs and maintains a time-series repository.
- **Training the DRL Agent:** A Markov Decision Process (MDP) is used for sequential decision-making. The Q-learning algorithm ensures adaptive attack detection.

Step 2: Evaluating DRL Performance Against Adversarial Attacks

Adversarial attacks pose a significant risk to V2X communication security, allowing attackers to manipulate vehicle data, evade intrusion detection, and disrupt traffic systems. Studies by Sharma et al. [92] and Sedar et al. [93] demonstrate that conventional ML-based security mechanisms fail against AI-adversarial attacks. The DRL-based detection framework offers a promising defense mechanism, achieving 99% detection accuracy for label-flipping attacks and maintaining moderate robustness under extreme adversarial conditions. Future research should focus on real-world adversarial datasets and adaptive security models to ensure resilient V2X networks. As shown in Table VI, Deep Reinforcement Learning (DRL) outperforms LSTM and MLP in handling adversarial attacks, achieving higher accuracy in label-flipping and policy induction scenarios.

Adversarial Training: Training AI models with adversarial examples improves defense against FGSM, PGD, and PIA attacks.

Secure Reinforcement Learning (RL) Models:

- Regularizing policy updates in DRL models reduces policy infiltration risks.
- Entropy-based reinforcement learning ensures secure V2X decision-making.

TABLE VII
TYPES OF ADVERSARIAL AND MODERN ATTACKS IN V2X NETWORKS AND MITIGATION STRATEGIES

Attack Type	Attack Strategy	Mitigation Techniques
Adversarial Machine Learning Attack	Manipulating input data to deceive AI/ML models used in misbehavior detection and intrusion detection systems (IDS).	Adversarial training, model hardening, and defensive distillation to improve AI model robustness.
Fast Gradient Sign Method (FGSM)	Single-step perturbation of AI model inputs to mislead resource allocation and traffic control decisions.	Adversarial training, gradient masking, and input sanitization techniques.
Projected Gradient Descent (PGD)	Iterative attack refining adversarial perturbations to fool deep learning models used in V2X communication.	Robust training with adversarial examples, model ensembling, and secure DRL policy updates.
Policy Infiltration Attack (PIA)	Injecting adversarial states into deep reinforcement learning (DRL)-based V2X resource management models.	Entropy-based policy regularization, secure Markov decision process (MDP) models, and multi-agent validation.
Model Extraction Attack	Stealing AI models by analyzing their responses to repeated queries, enabling reverse engineering of decision-making algorithms.	Secure federated learning, differential privacy, and zero-trust authentication for AI models.
Evasion Attack	Crafting deceptive input signals (e.g., altered CSI values, manipulated sensor data) to bypass AI-based intrusion detection.	Anomaly detection using AI-based IDS, robust ML training with adversarial scenarios.
Data Poisoning Attack	Injecting malicious samples into training datasets, misleading AI-based vehicular behavior predictions.	Secure data validation, blockchain-based data provenance, and SMOTE-based dataset balancing.
Bearer Migration Poisoning (BMP)	Manipulating bearer migration procedures to disrupt V2X network routing and service availability.	Secure channel authentication, encrypted bearer migration updates, and anomaly-based routing validation.
Intelligent Reflecting Surface (IRS) Manipulation	Modifying IRS phase shifts to cause beamforming errors and signal misdirection in vehicular networks.	Secure phase control algorithms, AI-based beamforming anomaly detection, and IRS authentication.
Jamming Attack with AI Enhancement	AI-based adaptive jamming that exploits reinforcement learning to optimize interference patterns.	Spread spectrum techniques, AI-powered anti-jamming strategies, and adaptive frequency hopping.
Spoofing Attack with GANs	Using Generative Adversarial Networks (GANs) to generate fake sensor data, misleading vehicular decision-making systems.	Blockchain-based message authentication, anomaly-based data validation, and cryptographic integrity checks.
DeepFake Attacks on V2X Perception	Generating synthetic images to deceive AI-based object recognition in autonomous vehicles.	AI-based DeepFake detection models, cross-validation with multiple sensor inputs, and secured perception frameworks.

AI-based Intrusion Detection Systems (IDS): Deploying adversarially trained IDS improves anomaly detection accuracy and prevents evasion attacks.

Secure Open Interfaces in V2X:

- End-to-end encryption for V2X communication prevents signal injection attacks.
- Zero-trust authentication for connected vehicles strengthens security [101].

Radio Positioning & Trusted Location Validation: Using multi-array antennas for UE positioning validation prevents fake CSI-based adversarial attacks.

Multi-Agent AI for Distributed Security:

- Distributed multi-agent learning enhances attack detection capabilities.
- Cross-validation among V2X nodes prevents DRL policy exploitation.

As summarized in Table VII, adversarial and modern attacks pose significant security challenges in V2X networks. Various countermeasures, including adversarial training, secure federated learning, and anomaly-based detection, have been proposed to mitigate these threats.

VII. INTRUSION DETECTION IN V2X NETWORKS: DETECTION METHODS, DEPLOYMENT STRATEGIES, AND SCOPE OF PROTECTION

This section discusses IDS classification based on detection methodology, deployment location, network architecture, and scope of detection, as depicted in Figure 9.

A. IDS Categorization Based on Detection Approach

Various IDS models have been proposed to detect cyber threats in V2X communication. These systems can be broadly classified into the following categories:

- **Signature-Based IDS:** This type of IDS detects known threats by comparing incoming network traffic against a database of predefined attack signatures. These signatures are derived from previous attack patterns and known vulnerabilities. When a match is identified, the IDS generates an alert or initiates mitigation measures. While signature-based IDSs are highly efficient in recognizing previously documented threats, their major limitation is the inability to detect novel or zero-day attacks, as they rely solely on predefined signatures [102].
- **Anomaly-Based IDS:** Unlike signature-based detection, anomaly-based IDSs identify intrusions by analyzing

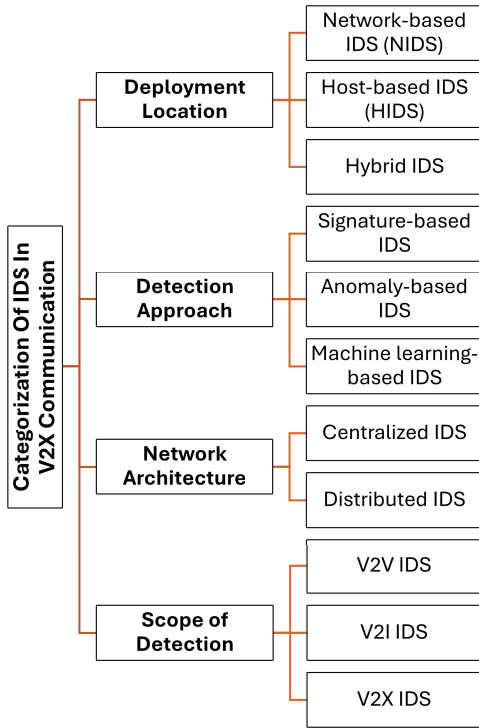


Fig. 9. IDS types in v2x.

deviations from normal network behavior. These systems establish a baseline of legitimate activity and flag any anomalies as potential threats. Statistical models and machine learning techniques are often employed to enhance detection accuracy. Although anomaly-based IDSs are more effective in detecting unknown attacks compared to signature-based IDSs, they tend to produce higher false positive rates due to their dependency on baseline deviations [103], [104].

- **Machine Learning-Based IDS:** This approach employs supervised or unsupervised machine learning algorithms to detect abnormal traffic patterns. By training models on historical network data, these IDSs can identify complex attack patterns and detect emerging threats. While machine learning-based IDSs offer significant advantages in detecting zero-day attacks, their performance depends on the availability of high-quality training datasets and continuous model updates to maintain accuracy.

B. IDS Categorization Based on Deployment Location

IDS deployment in V2X networks can be categorized based on whether they operate at the host level (individual vehicles) or across the network infrastructure.

- **Host-Based IDS (HIDS):** HIDS operates at the individual vehicle or infrastructure level, monitoring internal system activities such as system logs, file integrity, and process execution. By analyzing system behavior, HIDS can detect unauthorized access or malicious activities occurring within a specific node. However, one limitation is that HIDS may not be able to detect network-wide attacks occurring beyond the monitored device [105].

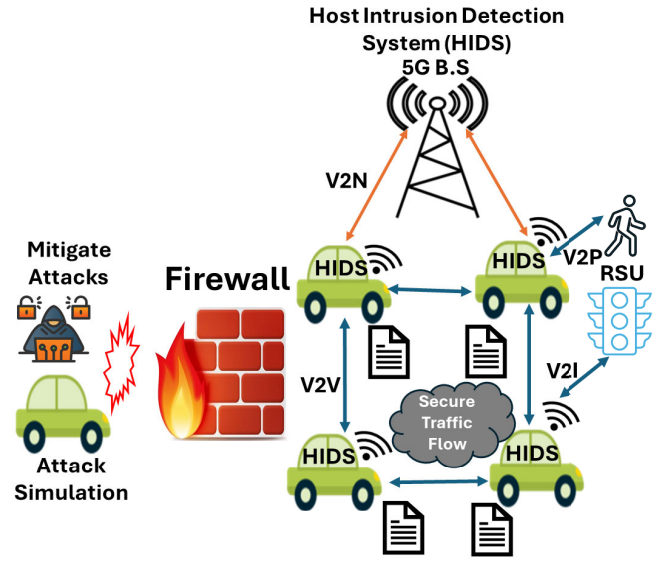


Fig. 10. Host-based IDS (HIDS) in V2X Networks.

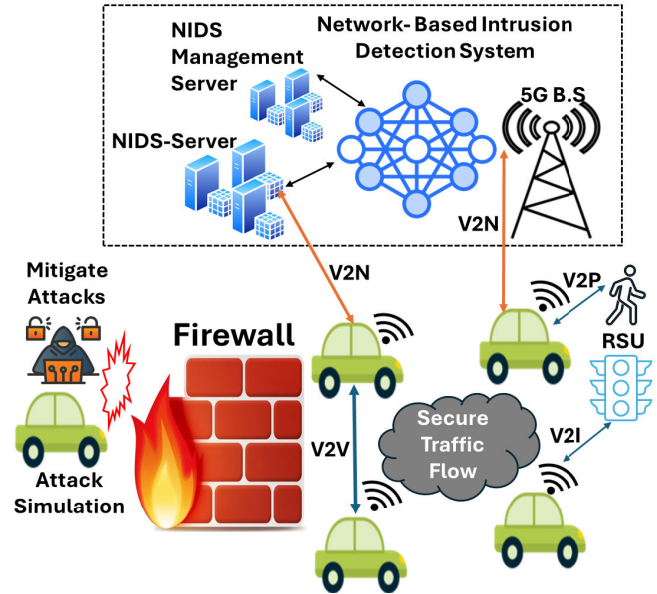


Fig. 11. Network-based IDS (NIDS) in V2X Networks.

Figure 10 illustrates the role of HIDS in V2X networks, where each vehicle autonomously monitors and analyzes network traffic to enhance security.

- **Network-Based IDS (NIDS):** Deployed at strategic network locations such as Roadside Units (RSUs), NIDS monitors real-time network traffic for malicious activity. These systems analyze communication patterns, packet payloads, and network anomalies to detect potential security threats. Since NIDS operates at network ingress and egress points, it is particularly effective in identifying large-scale attacks. Figure 11 demonstrates how NIDS functions in a V2X environment, with RSUs overseeing traffic flows and identifying network vulnerabilities.
- **Hybrid IDS:** Combining the capabilities of both HIDS and NIDS, hybrid IDSs provide a more holistic approach

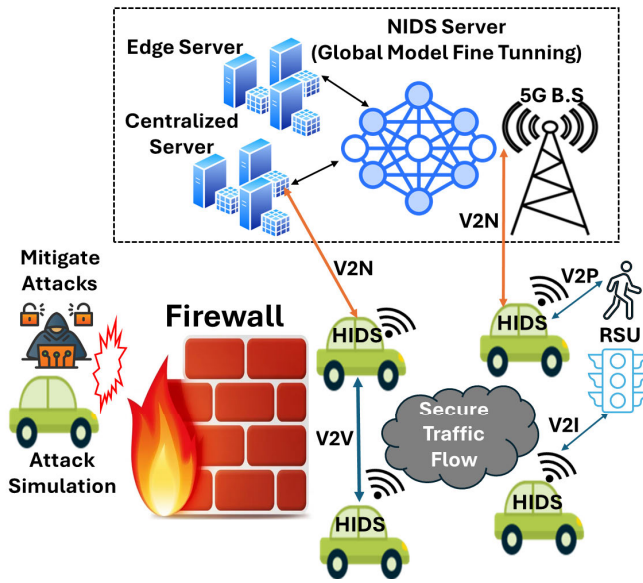


Fig. 12. Hybrid IDS in V2X Networks.

to intrusion detection. By integrating host-level monitoring with network-wide traffic analysis, hybrid IDSs can detect a broader range of security threats, including sophisticated multi-vector attacks. Figure 12 illustrates a hybrid IDS framework in V2X networks, demonstrating the synergy between decentralized (HIDS) and centralized (NIDS) detection mechanisms.

C. IDS Categorization Based on Network Architecture

Intrusion detection mechanisms in V2X networks can also be classified based on their architectural implementation:

- **Centralized IDS:** These systems operate from a central control point, such as a cloud-based security server or a core RSU, offering a unified view of network-wide threats. Centralized IDSs facilitate efficient monitoring and management but may be vulnerable to single-point failures and scalability limitations [93].
- **Distributed IDS:** Distributed IDSs are deployed across multiple nodes within the V2X network, enabling collaborative detection and mitigation of security threats. By distributing detection responsibilities, these systems enhance resilience against localized attacks. However, distributed IDSs may introduce increased complexity and higher false positive rates due to inconsistencies in data aggregation and correlation [106], [107], [108].

D. IDS Categorization Based on Scope of Detection

V2X networks encompass different communication paradigms, each presenting distinct security challenges. IDSs can be classified based on their scope of detection:

- **V2V IDS:** Designed to monitor and secure direct vehicle-to-vehicle (V2V) communication, these IDSs detect threats such as message spoofing, replay attacks, and unauthorized data manipulation.
- **V2I IDS:** Focused on securing interactions between vehicles and infrastructure components, V2I IDSs detect

false data injection, denial-of-service (DoS) attacks, and other security threats targeting critical infrastructure.

- **V2X IDS:** Providing a comprehensive security framework, V2X IDSs cover all communication modes within the V2X ecosystem. These systems detect multi-layered attacks, including malware propagation, coordinated cyberattacks, and large-scale data breaches affecting both vehicles and infrastructure [109]. Table VIII provides a comprehensive classification of IDSs based on their deployment location, detection techniques, network architecture, and scope of detection.

VIII. COMPREHENSIVE TAXONOMY OF MACHINE LEARNING-BASED INTRUSION DETECTION SYSTEMS FOR V2X SECURITY

ML-based IDS leverage data-based models to learn complex patterns and detect anomalies that may otherwise be overlooked. Unlike traditional IDS methods that rely on predefined rules and signatures.

A. Classification of Machine Learning Approaches

ML-based IDS enhance security by learning from network traffic patterns, continuously updating detection models, and autonomously responding to cyber threats. These models operate under different learning paradigms, including supervised, unsupervised, and reinforcement learning, each offering unique benefits for V2X security [110], [111] [112].

1) *Supervised Learning:* Supervised learning relies on labeled datasets to classify network traffic as benign or malicious. Once trained, these models effectively detect predefined attack patterns and generalize to unseen threats. Figure 13 illustrates the typical supervised learning framework in IDS.

Common Supervised Learning Algorithms:

- Logistic Regression
- Support Vector Machines (SVM)
- Random Forest
- k-Nearest Neighbors (KNN)
- Neural Networks
- Gradient Boosting Algorithms (XGBoost, AdaBoost, LightGBM)
- Ensemble Learning (Voting, Stacking)

Supervised learning is well-suited for detecting known attack types but requires high-quality labeled datasets, which may not always be available in dynamic V2X environments [113]. Supervised learning methods are illustrated in Figure 14.

2) *Unsupervised Learning:* Unsupervised learning detects anomalies without labeled data, making it ideal for identifying unknown or evolving threats. These models analyze deviations from normal network behavior to flag potential intrusions [114]. Figure 15 presents key unsupervised learning techniques for IDS.

Common Unsupervised Learning Algorithms:

- Gaussian Mixture Models (GMM)
- Hierarchical Clustering
- K-Means Clustering
- One-Class SVM
- Isolation Forest

TABLE VIII
IDS CATEGORIZATION BASED ON DEPLOYMENT LOCATION, DETECTION METHODS, SCOPE OF DETECTION, AND NETWORK ARCHITECTURE

Category	Subcategory	Description
Deployment Location	Network-Based IDS (NIDS)	Deployed at various network points to analyze real-time traffic and detect potential threats.
	Host-Based IDS (HIDS)	Operates on individual devices, monitoring system activity to detect unauthorized or malicious behavior.
	Hybrid IDS	Combines NIDS and HIDS techniques to enhance defense mechanisms against security threats and attacks.
Detection Approach	Signature-Based IDS	Detects known attacks by comparing incoming network traffic against a database of predefined attack signatures or patterns.
	Anomaly-Based IDS	Identifies unknown attacks by analyzing deviations from expected network behavior.
	Machine Learning-Based IDS	Employs machine learning algorithms to analyze network traffic and detect anomalous behavior patterns.
Network Architecture	Centralized IDS	Managed from a centralized location, offering a holistic view of network-wide security threats.
	Distributed IDS	Deployed across multiple network nodes, collaboratively detecting and mitigating threats.
Scope of Detection	V2V IDS	Focuses on detecting attacks and threats between connected vehicles.
	V2I IDS	Monitors attacks and vulnerabilities in communication between vehicles and infrastructure.
	V2X IDS	Covers the entire V2X ecosystem, detecting threats across connected vehicles, infrastructure, and other components of the V2X communication system.

TABLE IX
MACHINE LEARNING TECHNIQUES CATEGORIZATION

Category	Techniques	Description
Supervised Learning	Logistic Regression, Extra Tree, Random Forest, SVM, KNN, AdaBoost, Decision Stump, XGBoost, Ensemble Learning, LGBM, IBL, Ensemble Voting, Naive Bayes, Neural Networks	Classifying data into predefined classes or predicting values.
Unsupervised Learning	Gaussian Mixture Models, K-means, Hierarchical Clustering, Elliptic Envelope, Isolation Forest, One-class SVM, SST, LOF, PCA, Missing Value Ratio	Identifying patterns and reducing data dimensions.
Deep Learning	CNN, RNN, LSTM, GRUs, Autoencoders	Handling large datasets, image processing, sequence data.
Advanced ML Concepts	Reinforcement Learning, Federated Learning, Semi-Supervised Learning, Transfer Learning, GANs	Distributed learning, goal-oriented learning, leveraging existing knowledge.

- Local Outlier Factor (LOF)
- Autoencoders

While effective for zero-day attack detection, unsupervised models often require fine-tuning to minimize false positives.

3) *Reinforcement Learning*: Reinforcement Learning (RL) enhances IDS by dynamically optimizing security policies through trial-and-error learning. Unlike supervised models, RL continuously improves its attack response strategies based on reward-based feedback. This approach is particularly beneficial in adaptive threat detection and mitigation for V2X networks.

Applications of RL in IDS:

- Autonomous anomaly detection and adaptive response
- Optimized resource allocation for security mechanisms

- Adversarial learning to improve robustness against AI attacks

A diverse range of ML techniques can be employed to enhance IDS performance in V2X networks. Table IX categorizes key methods based on their learning paradigm.

4) *Deep Learning for IDS*: Deep Learning (DL), a subset of ML, utilizes multi-layer neural networks to analyze complex attack patterns in large-scale V2X datasets. DL-based IDS can identify intricate cyber threats by leveraging high-dimensional feature extraction and sequential analysis. A detailed comparison of deep learning-based IDS techniques is provided in Table X, highlighting their performance and associated challenges.

TABLE X
DIFFERENT DEEP LEARNING METHODS AND RELATED DATASETS WITHIN VARIOUS APPLICATION SCENARIOS

Deep Learning Method	Description	Datasets Used and Application Scenarios
LSTM (Long Short-Term Memory)	RNN type that learns long-term dependencies. Effective in anomaly detection in sequential data.	CIC-IDS2017, UNSW-NB15, VeReMi, ROAD: Detecting temporal patterns in network traffic, identifying sequential attack patterns.
GRU (Gated Recurrent Unit)	LSTM variant with fewer parameters, making it faster and computationally efficient.	CIC-IDS2017, CIC-IDS2018, UNSW-NB15: Like LSTM, used for sequence prediction and anomaly detection in network data.
CNN (Convolutional Neural Network)	Primarily used for image data but can be adapted for time series and other structured data.	CIC-IDS2017, UNSW-NB15, ToN-IoT: Feature extraction from network traffic data, identifying spatial patterns in data packets.
DNN (Deep Neural Network)	A neural network with multiple hidden layers that can model complex relationships in data.	KDD Cup 99, NSL-KDD, UNSW-NB15, CIC-IDS2018: General-purpose intrusion detection, network traffic classification into normal and various attack types.
Autoencoders	A neural network type used for unsupervised learning of efficient coding, effective for anomaly detection.	NSL-KDD, CIC-IDS2018, ToN-IoT: Identifying anomalies by learning a compressed representation of normal network traffic.
RNN (Recurrent Neural Network)	Designed for sequential data processing, capable of capturing temporal dependencies in time-series data.	CIC-IDS2017, VeReMi, ROAD: Analyzing sequences of network packets to detect intrusions.
GAN (Generative Adversarial Network)	Consists of a discriminator and a generator, useful for data augmentation and anomaly detection.	ToN-IoT, UNSW-NB15, KDD Cup 99: Generating synthetic attack data and identifying anomalies by learning normal data distribution.
Hybrid Models (e.g., LSTM-CNN)	Combines different deep learning architectures to leverage their strengths.	CIC-IDS2017, UNSW-NB15, VeReMi: Enhanced feature extraction and sequence learning, improved detection accuracy.
DBN (Deep Belief Network)	A deep neural network or generative graphical model comprising multiple layers of hidden variables.	KDD Cup 99, NSL-KDD: Unsupervised learning of complex representations, feature learning for intrusion detection.
RBM (Restricted Boltzmann Machine)	A generative stochastic artificial neural network that learns input probability distributions.	KDD Cup 99, NSL-KDD: Feature learning and dimensionality reduction, anomaly detection.
Transformer Networks	Utilizes self-attention mechanisms, effective in capturing long-range dependencies in data.	CIC-IDS2018, UNSW-NB15: Advanced sequence modeling, effective in analyzing complex network traffic patterns.
Denoising Autoencoder (DAE)	A type of autoencoder designed to remove noise from data.	UNSW-NB15, CIC-IDS2017: Improving the robustness of IDS models by removing irrelevant features and noise.
Variational Autoencoder (VAE)	A type of autoencoder that uses variational inference to learn the latent space.	ToN-IoT, NSL-KDD: Anomaly detection by modeling the distribution of normal data and identifying deviations.
AnoGAN	A GAN-based anomaly detection method.	KDD Cup 99, UNSW-NB15: Identifying anomalies by learning the distribution of normal network traffic and detecting deviations.
f-AnoGAN	An improved version of AnoGAN with better computational efficiency.	KDD Cup 99, UNSW-NB15: Faster anomaly detection with reduced computational overhead.
GANomaly	A GAN-based approach that uses two encoder networks for enhanced anomaly detection.	KDD Cup 99, UNSW-NB15: Improved performance in identifying anomalies by comparing the latent spaces of actual and reconstructed data.
Efficient GANs	GANs with optimized architecture for better performance.	KDD Cup 99, UNSW-NB15: Efficiently identifying anomalies with reduced training time and computational cost.

Common Deep Learning Architectures:

- **Convolutional Neural Networks (CNNs):** Effective for spatial pattern detection in network traffic.
- **Recurrent Neural Networks (RNNs):** Suitable for sequential data processing, including time-series intrusion detection.



Fig. 13. Typical supervised learning framework in IDS.

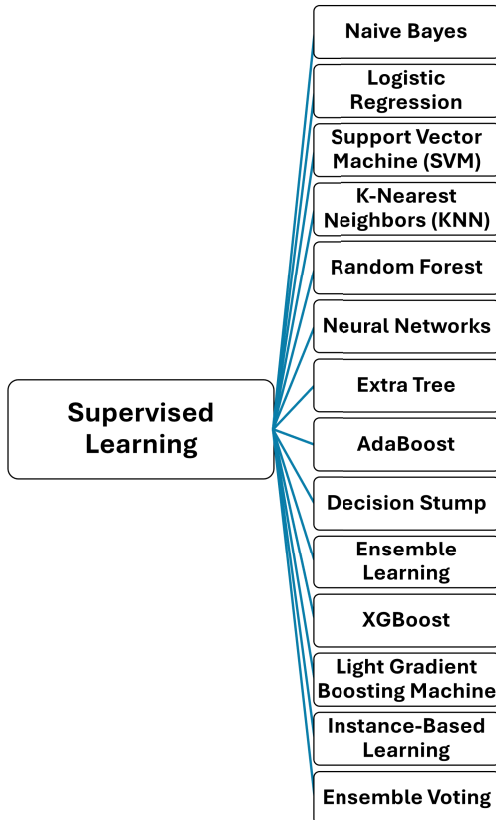


Fig. 14. Supervised learning common algorithms used in IDS-V2X.

The neural network architecture illustrated in Figure 16 represents a multi-layer perceptron (MLP) model with fully connected layers. The input layer processes raw data, which is subsequently transformed through three hidden layers comprising artificial neurons. Each hidden

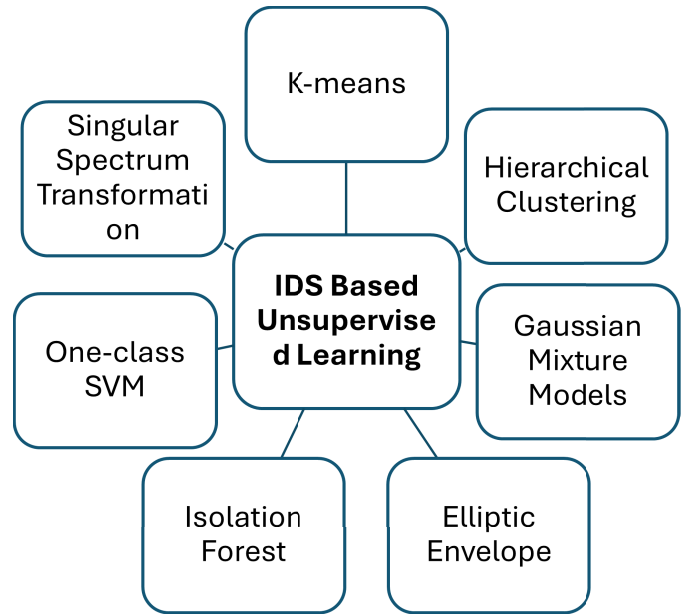


Fig. 15. Unsupervised learning common algorithms used in IDS-V2X.

layer applies nonlinear activation functions to extract hierarchical features. The output layer generates the final predictions. Such deep architectures are widely utilized in intrusion detection systems (IDS), image recognition, and natural language processing (NLP) due to their ability to capture intricate patterns within high-dimensional data.

- **Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs):** Handle long-range dependencies in attack sequences.
- **Autoencoders:** Identify anomalies by reconstructing normal traffic patterns.
- **Generative Adversarial Networks (GANs):** Generate synthetic attack scenarios to improve IDS robustness.

5) *Federated Learning and Edge AI for V2X Security:* Federated Learning (FL) enables decentralized IDS training across multiple V2X nodes without sharing raw data, preserving privacy while improving collaborative threat detection. Edge AI further enhances real-time intrusion detection by deploying ML models directly on vehicular devices, reducing latency and communication overhead.

B. Evaluation Metrics for Machine Learning-Based IDS

To explore the effectiveness of ML-based IDS in V2X networks, several key performance metrics are employed. These metrics help evaluate detection accuracy, false alarm rates, and overall system robustness. Additionally, hyperparameter tuning significantly impacts these metrics, as discussed later in this section.

- **Accuracy:** Measures the proportion of correctly identified instances, both benign and malicious.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

While accuracy is a widely used metric, it may be misleading in highly imbalanced datasets where benign traffic dominates.

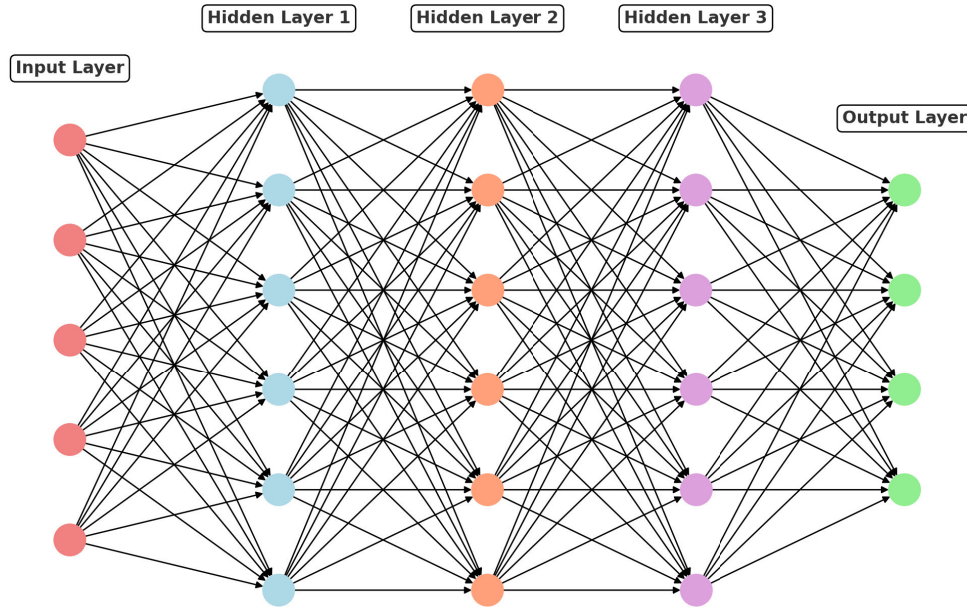


Fig. 16. Illustration of a Deep Neural Network (DNN) architecture consisting of an input layer, multiple hidden layers, and an output layer. Each neuron in a layer is fully connected to all neurons in the next layer.

- **Precision:** Determines the proportion of detected attacks that are actually malicious.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

High precision is crucial in IDS to reduce false alarms. A precision value close to 1.0 indicates fewer false positives, which is essential in real-time V2X security applications.

- **Recall (Sensitivity):** Evaluates the ability of the IDS to correctly detect all actual attacks.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

A low recall value suggests that many attacks go undetected, making the system vulnerable to security threats.

- **F1-Score:** A harmonic mean of precision and recall, balancing false positives and false negatives.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

This metric is particularly useful when dealing with imbalanced datasets, ensuring a balance between detection rate and false alarms.

- **True Positive Rate (TPR):** Measures the proportion of correctly detected attacks.

$$TPR = \frac{TP}{TP + FN} \quad (5)$$

TPR is directly linked to recall and is a critical metric for evaluating IDS detection capabilities.

- **True Negative Rate (TNR):** Explains the proportion of correctly identified benign traffic.

$$TNR = \frac{TN}{TN + FP} \quad (6)$$

A high TNR ensures minimal disruption to normal vehicular communication.

- **False Positive Rate (FPR):** Identifies the proportion of benign instances falsely classified as attacks.

$$FPR = \frac{FP}{FP + TN} \quad (7)$$

A high FPR leads to unnecessary security interventions, negatively impacting system efficiency.

- **False Negative Rate (FNR):** Captures the number of undetected attacks.

$$FNR = \frac{FN}{FN + TP} \quad (8)$$

A high FNR indicates that many real attacks are missed, making the system unreliable.

The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between sensitivity and specificity, with the Area Under the Curve (AUC) quantifying the classifier's effectiveness. AUC values range from 0 to 1, where higher values indicate better model performance in distinguishing between attack and benign instances.

1) **Hyperparameters and Their Impact on IDS Performance:** In addition to evaluation metrics, hyperparameters play a crucial role in optimizing the performance of ML-based IDS models. Proper tuning of these parameters directly influences the IDS's ability to detect attacks, minimize false alarms, and ensure real-time responsiveness.

- **Learning Rate (η):** Determines the step size for updating model weights during training.

- **Typical Range:** $0.0001 \leq \eta \leq 0.01$
- A high learning rate speeds up convergence but may lead to instability.
- A low learning rate improves stability but increases training time.

TABLE XI
EVALUATION METRICS FOR ML-BASED IDS IN V2X NETWORKS

Metric	Formula	Definition	Significance in V2X IDS
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	The proportion of correctly identified instances (both attacks and benign traffic).	Measures overall IDS effectiveness, but may be misleading in imbalanced datasets.
Precision	$\frac{TP}{TP+FP}$	The proportion of true attacks correctly classified out of all detected attacks.	High precision ensures fewer false alarms, crucial for reducing unnecessary security interventions.
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	The proportion of actual attacks correctly detected.	Essential for minimizing undetected threats and improving security reliability.
F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	The harmonic mean of precision and recall, balancing both metrics.	Ensures IDS effectiveness, particularly when dealing with imbalanced datasets.
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$	The proportion of benign traffic incorrectly classified as an attack.	Reducing FPR minimizes false alarms, ensuring normal operations in V2X networks.
False Negative Rate (FNR)	$\frac{FN}{FN+TP}$	The proportion of attacks that go undetected.	A low FNR is critical to avoid security threats that could impact vehicle safety.
True Positive Rate (TPR)	$\frac{TP}{TP+FN}$	The proportion of actual attacks that are correctly classified.	High TPR ensures strong attack detection capabilities in IDS.
True Negative Rate (TNR)	$\frac{TN}{TN+FP}$	The proportion of benign instances correctly classified.	Helps in correctly identifying normal traffic, reducing unnecessary defensive actions.
AUC-ROC (Area Under Curve - Receiver Operating Characteristic)	Measures classifier performance in distinguishing attacks from normal traffic.	Provides a comprehensive measure of IDS ability to separate benign and malicious activity.	Higher AUC values indicate better discrimination between attacks and normal traffic.
Detection Latency	Time taken to detect an attack after it occurs.	Measures IDS response time to an attack event.	Low latency is critical for real-time threat mitigation in V2X networks.

- **Batch Size:** Controls the number of samples processed before updating model parameters.
 - **Typical Range:** $32 \leq \text{Batch Size} \leq 512$
 - Larger batches lead to faster training but require more memory.
 - Smaller batches improve model generalization but slow down training.
- **Dropout Rate:** Introduces regularization to prevent overfitting in deep learning models.
 - **Typical Range:** $0.1 \leq \text{Dropout Rate} \leq 0.5$
 - A higher dropout rate enhances generalization but may hinder learning.
 - A lower dropout rate allows deeper learning but risks overfitting.
- **Number of Hidden Layers:** Determines the model's depth and capacity for feature learning.
 - **Typical Range:** $2 \leq \text{Hidden Layers} \leq 10$
 - More layers enhance feature extraction but increase computational complexity.
 - Fewer layers simplify the model but may limit detection accuracy.

Table XI presents the key evaluation metrics used in ML-based IDS for V2X networks, highlighting their significance in attack detection and prevention.

C. False Alarm Rate in ML-Based Intrusion Detection for V2X Security

False Alarm Rate (FAR) is a crucial metric in evaluating the effectiveness of intrusion detection systems (IDS) in vehicular networks. It measures the proportion of benign traffic misclassified as an attack, directly impacting the trustworthiness and efficiency of IDS models. A high FAR leads to unnecessary security interventions, potentially disrupting traffic flow and causing delays in decision-making. Conversely, an overly cautious system that avoids false positives may fail to detect actual threats, reducing its overall security effectiveness. This trade-off makes FAR a key consideration in designing and optimizing IDS for V2X networks.

Mathematically, FAR is defined as:

$$FAR = \frac{FP}{FP + TN} \quad (9)$$

where FP represents false positives (benign traffic incorrectly classified as an attack), and TN denotes true negatives (correctly classified benign traffic). Reducing FAR is particularly challenging in V2X environments due to dynamic network conditions, variations in vehicular density, and potential adversarial attacks that mimic benign behavior.

Several studies have addressed the issue of FAR in ML-based IDS. Jiménez et al. [237] reported a False Alarm Rate of 3% in their Autoencoder + LSTM-RNN framework, which was tested in synthetic autonomous vehicle network (AVN) attack scenarios. Despite achieving a detection rate

of 95%, the presence of false positives indicates that deep learning models, while highly accurate, still struggle with misclassifying benign anomalies. Similarly, Salazar et al. [246] examined a real-world anomaly detection system (VCADS) for V2X communications, where detection performance varied between datasets, suggesting that IDS models trained in controlled environments often generalize poorly to real-world traffic conditions, leading to inconsistencies in FAR.

The computational complexity of detection models further influences FAR. Kong et al. [250] demonstrated that deep learning-based IDS in V2V networks improved detection throughput (93%) but suffered from computational inefficiencies, likely contributing to higher false alarm rates in real-time deployment. The trade-off between accuracy and computational feasibility is critical, as reducing FAR often requires advanced model fine-tuning and feature engineering, which increases the processing burden on resource-constrained vehicular units.

Raja et al. [106] developed the ETADC framework based on reinforcement learning for anomaly detection in 6G-V2X networks. While the model demonstrated a 97% accuracy, it also exhibited a low false positive rate, indicating its effectiveness in reducing FAR. However, the study emphasized the need for real-world validation to Explain the IDS's adaptability to diverse attack scenarios.

Korba et al. [239] applied Federated Learning (FL) for zero-day attack detection in 5G-enabled V2X networks. The model achieved an impressive 99.99% accuracy and an F1-score of 99.99%, with a False Positive Rate of only 0.01%, making it one of the most reliable approaches in terms of minimizing FAR. The federated model effectively reduced communication overhead while maintaining low FAR, demonstrating the advantages of decentralized learning strategies.

To mitigate high FAR in ML-based IDS, researchers have explored various optimization techniques. Adaptive thresholding, which dynamically adjusts classification boundaries based on network conditions, has been effective in reducing false alarms. Hybrid IDS architectures, combining signature-based detection with behavioral anomaly analysis, have also shown promise in improving classification precision while maintaining a low FAR. Federated learning-based IDS, such as those examined by Korba et al. [239], leverage collaborative model training across multiple vehicular nodes to refine detection parameters without sharing raw data, effectively reducing false positives while preserving detection accuracy.

Another promising approach is real-time feature selection, where decision trees and random forests prioritize critical input variables, filtering out noisy or ambiguous features that contribute to false alarms. Additionally, context-aware intrusion detection, which integrates external factors such as location, traffic patterns, and vehicle types, has been proposed as a method to enhance classification accuracy by differentiating between anomalous yet benign behavior and genuine cyber threats. Future research should focus on balancing computational efficiency with FAR optimization by integrating lightweight deep learning architectures with edge AI solutions. The emergence of TinyML and adaptive federated learning

techniques offers potential for reducing model complexity while maintaining strong security performance. Additionally, explainable AI (XAI) could provide interpretability in IDS decision-making, ensuring that misclassifications leading to high FAR can be systematically corrected. Dynamic model adaptation, where IDS adjusts its computational resources based on real-time network constraints, may further enhance the scalability of IDS models in V2X security applications.

Addressing FAR in ML-based IDS is essential for ensuring reliable and practical intrusion detection in connected vehicular environments. As the complexity of cyber threats in V2X networks continues to evolve, improving the robustness of IDS models against false alarms while maintaining detection accuracy will be a critical area for future research and development.

Considerations in Reducing FAR

Reducing FAR in ML-based IDS for V2X security is crucial to avoid unnecessary interventions while maintaining strong threat detection capabilities. Several challenges contribute to higher FAR:

Data Imbalance in IDS Training: Many datasets contain significantly more benign traffic than attack instances. This imbalance can lead to models favoring benign classifications, thus increasing FAR.

Possible Solution: Techniques such as SMOTE (Synthetic Minority Over-sampling Technique) and class-weighted loss functions can help mitigate this issue.

Variability in Real-World Traffic Conditions: Unlike simulated environments, real-world networks experience fluctuating traffic loads, environmental noise, and diverse driving behaviors, which can introduce false positives.

Possible Solution: Real-world dataset augmentation and adaptive thresholding techniques to dynamically adjust IDS sensitivity based on network traffic patterns.

Generalization of IDS Across Different Attack Types: Many IDS models are trained on specific datasets and fail to generalize when exposed to new attack scenarios, leading to misclassifications.

Possible Solution: Federated Learning (FL) and continuous adversarial training can help models adapt to unseen attack patterns while minimizing FAR.

Trade-off Between Detection Sensitivity and False Positives: Increasing an IDS's sensitivity improves attack detection but also raises the risk of misclassifying benign traffic as malicious.

Possible Solution: Hybrid detection models, which combine signature-based and anomaly-based IDS techniques, can improve balance.

By addressing these challenges, future research can focus on refining ML-based IDS to achieve a more optimal balance between security effectiveness and operational efficiency in V2X networks.

Figure 17 illustrates the proposed data flow for a misbehavior detection system in V2X networks using ML-based techniques. This system integrates data collection from infrastructure and vehicle sensors, followed by data cleansing, transformation, and feature extraction. A continuous model training and feedback loop ensures adaptive learning and improves the reliability and accuracy of intrusion detection.

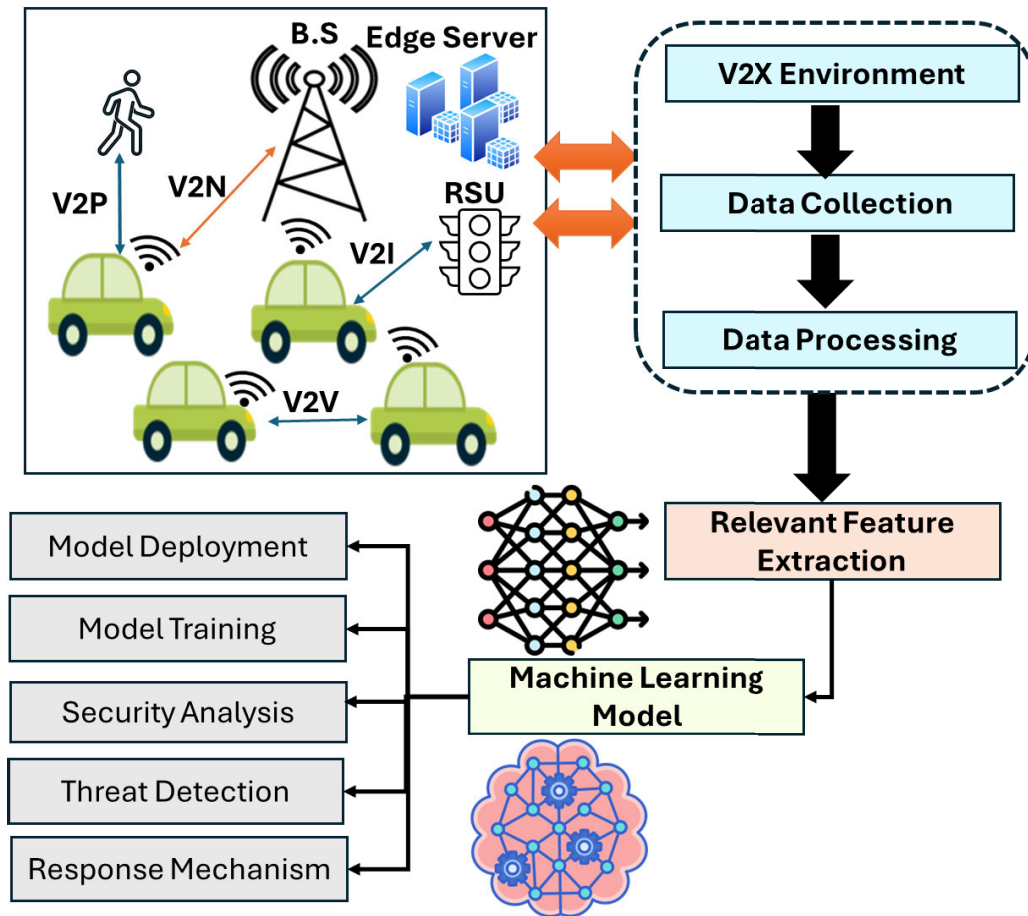


Fig. 17. The proposed data flow for a misbehavior detection system in V2X networks using ML-based techniques.

D. Conventional Machine Learning Approaches in V2X IDS

Zhang et al. [115] conducted an extensive study on ML-based intrusion detection techniques, categorizing them into traditional ML, ensemble learning, and deep learning approaches. Their study employed seven ML algorithms, including Naïve Bayes, decision trees, SVM, XGBoost, random forests, recurrent neural networks (RNN), and convolutional neural networks (CNN). The NSL-KDD and KDD Cup 99 datasets were used for evaluation, measuring accuracy, F1-score, and AUC. Results demonstrated that ensemble learning models outperformed traditional ML and deep learning models. Moreover, Naïve Bayes, despite its relatively poor performance on known data, showed potential for detecting zero-day attacks. Their findings also highlighted the critical role of hyperparameter tuning and model structure in deep learning IDS performance. Grover et al. [116] proposed an ML-based misbehavior classification model for vehicular networks. Their approach utilized single-class ML to distinguish between honest and dishonest nodes, followed by multi-class classification to identify specific misbehavior types. They evaluated six attack scenarios affecting V2V and V2R communications. Using features extracted from the NCTUns-5.0 simulator [117], their model achieved a 92% true-positive rate for single-class classification and 93% for multi-class classification, with random forest and J48 emerging as the most effective classifiers.

Li et al. [118] enhanced the random forest model for IDS in Internet of Vehicles (IoV) environments. Their proposed lightweight model demonstrated improved detection speed, accuracy, and lower false alarm rates, effectively protecting IoV networks from cyber threats.

Sarhan et al. [119] analyzed feature extraction techniques for ML-based IDS in IoT networks. They compared three feature extraction approaches—Linear Discriminant Analysis (LDA), Principal Component Analysis (PCA), and Autoencoders (AE)—against six ML models: CNN, Deep Feedforward Networks (DFF), RNN, Decision Trees (DT), Logistic Regression (LR), and Naïve Bayes (NB). Their evaluation on the ToN-IoT, UNSW-NB15, and CSE-CIC-IDS2018 datasets revealed that no single feature extraction method was optimal across all datasets, emphasizing the need for dataset-specific optimizations.

Yang et al. [120] introduced a hybrid IDS for the Internet of Vehicles, integrating both signature-based and anomaly-based detection using ML techniques. Their IDS operates at three levels: vehicle, edge, and cloud, where different detectors process diverse data streams. Signature-based IDS components detect known threats using Snort rules, while anomaly-based detection leverages random forest, k-NN, and SVM classifiers to identify zero-day attacks. Their IDS demonstrated high accuracy and low processing overhead when tested against the CAN-intrusion and CSE-CIC-IDS2018 datasets.

Sedjelmaci and Senouci [121] introduced a clustering-based IDS for VANETs that does not require specialized monitoring agents. Their framework detects malicious nodes at both local and global levels using Bayesian game theory and SVM learning, effectively identifying attacks such as wormholes, black holes, Sybil, selective forwarding, packet duplication, and resource exhaustion.

Li et al. [122] implemented an SVM-based anomaly detection model that incorporates behavioral and contextual data (velocity, GPS coordinates, and channel status) to identify malicious vehicles. Their distributed IDS approach aggregates data from neighboring nodes to enhance awareness of adversarial behavior while optimizing recall, accuracy, and communication efficiency.

Shams et al. [123] developed an SVM-based IDS for V2X networks. Their kernel-based model successfully distinguished between known and unknown threats but struggled in high-density vehicular environments.

Shenfield et al. [124] employed Artificial Neural Networks (ANNs) to create a non-signature-based IDS for shellcode detection. Their model achieved high accuracy with minimal false positives, but they noted a key limitation: ML-based IDS often lack explainability, making model decisions difficult to interpret.

Mane et al. [125] addressed explainability issues by proposing an Explainable AI (XAI) framework for IDS. They utilized feature importance analysis, Local Interpretable Model-agnostic Explanations (LIME), and counterfactual reasoning to provide visual and textual justifications for attack detection decisions.

Zhao et al. [126] explored the application of Deep Belief Networks (DBN) and Probabilistic Neural Networks (PNN) for intrusion detection in V2X networks. Their study evaluated the classification performance of these models in identifying cyber threats and improving the accuracy of intrusion detection systems (IDS).

Alsaedi et al. [127] developed a machine learning-based IDS for V2X security, leveraging the ToN-IoT dataset [128], which encompasses various cyber threats, including DoS/DDoS, TCP SYN flooding, UDP flooding, and HTTP flooding. Their research applied different ML techniques for binary and multiclass classification, employing Chi-square (Chi2) for feature selection and SMOTE for handling class imbalance. The results demonstrated that XGBoost outperformed other classifiers in terms of accuracy and recall.

Grover et al. [129] enhanced IDS accuracy through an ensemble learning approach, integrating multiple classifiers using majority voting. Their model was designed to detect vehicular misbehavior, such as position and identity spoofing and replay attacks. Using the WEKA toolset, they analyzed physical properties and message content, demonstrating that random forest and J48 classifiers provided superior performance.

Singh et al. [130] applied supervised learning to detect false position information in vehicular safety messages. They used the open-source VeReMi dataset [131], which simulates misbehavior attacks, including position falsification and traffic congestion. Their study compared Support Vector Machine

(SVM) and Logistic Regression (LR), finding that LR with a normalized feature set provided higher detection accuracy.

So et al. [132] evaluated machine learning models for location-spoofing misbehavior detection using the VeReMi dataset. Their findings indicated that ML models significantly outperformed traditional plausibility checks. They extracted feature vectors from location consistency and movement patterns, demonstrating that K-Nearest Neighbors (KNN) and SVM classifiers effectively detected spoofed locations.

E. IDS Based Advanced Machine Learning Approaches

Wahab et al. [133] proposed CEAP (Collection, Exchange, Analysis, and Propagation), an intelligent intrusion detection mechanism based on clustering protocols. Their approach leveraged SVM classification, with cluster heads acting as watchdogs to monitor packet transmission and classify behaviors. This method minimized computational overhead while ensuring effective IDS performance in infrastructure-free vehicular environments.

Gyawali et al. [134] introduced a misbehavior detection system (MDS) that integrates machine learning and reputation theory to mitigate internal attacks in V2X networks. Their model combined ML classification with Dempster-Shafer theory to compute reputation scores for vehicles. The system synchronized these scores with a Certificate Authority (CA) for misbehavior detection, achieving an F1-score of 98% and a false alert rate of 96% for positional attack detection.

Wang et al. [117] and Hawlader et al. [135] examined various supervised ML techniques, including KNN, decision trees, and LR, for detecting false alerts and position falsification in vehicular networks.

Sharma et al. [136] utilized an augmented feature set from the VeReMi dataset, incorporating information from successive Basic Safety Messages (BSMs) to enhance the detection of position falsification attacks. Their approach outperformed prior ML-based misbehavior detection methods. Similarly, Ercan et al. [137] leveraged position-related features to detect decentralized position falsification attacks.

Yang et al. [138] proposed a supervised learning classifier capable of distinguishing between three different types of Sybil attacks and benign vehicles.

Tariq et al. [139] introduced a transfer learning (TL)-based IDS designed for securing the Controller Area Network (CAN). Their model, based on convolutional LSTM, achieved detection accuracy ranging from 88% to 95% for known and novel cyber threats using real-world vehicular data.

Tait et al. [140] conducted a comparative analysis of machine learning-based IDS techniques using the CICIDS2017 dataset, which includes multiple attack types such as PortScan, DoS, Web Attacks, Botnet, Infiltration, Heartbleed, and DDoS. Their study [141] demonstrated that multiclass classification techniques significantly enhanced intrusion detection accuracy by improving attack differentiation.

Almi'ani et al. [142] proposed a nonlinear intrusion detection framework using self-organizing maps. Their model employed cluster analysis to detect and classify network

attacks, demonstrating its effectiveness in various V2X environments.

Leandros et al. [143] developed a distributed IDS (DIDS) for vehicular networks, analyzing the impact of RSU placement, intruder velocity, and vehicle density on detection accuracy and response time. Their approach combined one-class SVM (K-OCSVM) with K-means clustering, efficiently identifying anomalies and relaying critical alerts to a security center for further processing.

Table XII summarizes the key findings and measured IDS-based machine learning performance metrics across different datasets for V2X security.

IX. A COMPREHENSIVE REVIEW OF DATASETS FOR V2X IDS: REAL, SIMULATED, COMPARATIVE ANALYSIS, LIMITATIONS AND RECOMMENDATIONS

Datasets play a critical role in evaluating IDS by providing standardized benchmarks for enhancing detection capabilities against cyber threats. However, acquiring real-world vehicular network traffic datasets remains challenging due to privacy and security concerns. This section categorizes datasets based on their origin (real-world or synthetic), attack diversity, and relevance to V2X networks.

A. Vehicular-Specific Datasets for V2X Security

Datasets for vehicular networks are essential for evaluating IDS in connected and autonomous vehicles. These datasets focus on misbehavior detection, message falsification, and GPS spoofing, common threats in VANETs.

- **ROAD Dataset** The ROAD dataset is a real-world dataset containing both normal and abnormal vehicular communication data collected from road segments in Germany. It encompasses multiple attack scenarios, including jamming, replaying, spoofing, and message falsification. Additionally, it provides critical vehicular attributes such as GPS coordinates, speed profiles, timestamps, and message types. The dataset is publicly available for research purposes [144].
- **VDDD Dataset:** The Vehicular Data for Detection and Defense (VDDD) dataset is a synthetic dataset generated using the SUMO (Simulation of Urban Mobility) software. It simulates both normal and abnormal traffic patterns, incorporating a range of cyberattacks, including Denial-of-Service (DoS), Sybil attacks, and blackhole attacks, which involve selective packet dropping. The dataset includes attributes such as vehicle IDs, positions, velocities, and accelerations. The authors provide a Python-based script for dataset generation [145].
- **VeReMi Dataset:** The VeReMi dataset is a hybrid dataset that integrates real-world data with synthetic data generated using the Vehicles in Network Simulation (Veins) framework. It includes attack scenarios such as DoS, identity spoofing (impersonation), and message falsification. The dataset provides key vehicle attributes, including positions, IDs, velocities, and accelerations [146]. VeReMi, which stands for “Vehicle Reference Misbehavior,” is specifically designed to evaluate misbehavior

detection mechanisms in VANETs. Generated using LuST and VEINS, the dataset consists of Basic Safety Message (BSM) logs collected via Dedicated Short-Range Communications (DSRC). The primary objectives of VeReMi are to (1) reduce computational overhead associated with frequent VEINS simulations and (2) establish a baseline for deploying misbehavior detection mechanisms at an urban scale. The dataset is structured into different density levels, five attack types, and three attacker densities. Each simulation log contains message logs for every receiving vehicle and a corresponding ground truth file. Researchers can access the dataset via the GitHub repository, which utilizes Git Large File Storage (LFS) for efficient data handling.

- **VDOS-LRS Dataset:** The VDOS-LRS dataset is another synthetic dataset that simulates vehicular traffic under attack conditions using the OMNeT++/Veins framework. It captures attack scenarios, including DoS, greyhole attacks (selectively dropping packets), and blackhole attacks (dropping all packets). Similar to the VDDD dataset, VDOS-LRS provides vehicle IDs, positions, velocities, and acceleration data. A Python-based script is available for dataset generation [147].

B. General Security Datasets Used for V2X IDS Evaluation

V2X security extends beyond vehicular misbehavior, requiring evaluation against broader cyber threats. Several traditional network security datasets are adapted for V2X IDS training, offering diverse attack types.

- **NSL-KDD Dataset:** The NSL-KDD dataset is an improved version of KDD Cup 99, addressing issues such as redundancy, imbalance, and a lack of novel attack instances. It excludes duplicate records and provides a more balanced distribution of training and testing samples. Nevertheless, it still suffers from limitations related to irrelevant features and outdated attacks, making it less suitable for modern V2X IDS applications [148], [149].
- **ToN-IoT Dataset:** The ToN-IoT dataset is a large-scale dataset collected from heterogeneous Internet of Things (IoT) environments. It incorporates a variety of attack types, such as HTTP flooding, DoS, Distributed Denial-of-Service (DDoS), TCP-SYN flooding, and UDP flooding. The dataset sources its data from IoT and Industrial IoT (IIoT) telemetry sensors, operating systems (Windows 7, Windows 10, Ubuntu 14, Ubuntu 18), and network traffic logs. ToN-IoT was developed at the Cyber Range and IoT Labs at the University of New South Wales (UNSW) Canberra, where a testbed network integrating Windows, Linux, and Kali operating systems was deployed. The dataset captures attack behaviors targeting IoT gateways, web applications, and networked computing systems. Due to its extensive coverage, ToN-IoT is widely used for evaluating AI-based cybersecurity applications, including IDS, threat intelligence systems, malware detection, and fraud prevention [150].
- **CIC-IDS2017 and CIC-IDS2018 Datasets:** The CIC-IDS2017 and CIC-IDS2018 datasets were developed by

TABLE XII
KEY FINDINGS AND MEASURED PERFORMANCE METRICS OF IDS-BASED MACHINE LEARNING FOR V2X COMMUNICATION

Study	Approach	Dataset/Results	Key Findings
Zhang et al. [115]	Comparative research on ML methods	Datasets: KDD CUP99, NSL-KDD; Algorithms: Decision Trees, Naive Bayes, SVM, Random Forest, XGBoost, CNN, RNN	Accuracy: Ensemble learning : 95%, F1 score: Ensemble learning : 0.9, AUC: Ensemble learning : 0.95
Grover et al. [116]	Single-class and multi-class ML	Dataset: NCTUns-5.0; Algorithms: Random Forest, J48	True-positive rate: Single-class: 92%, Multi-class: 93%
Li et al. [118]	Improved Random Forest	Lightweight model; Enhanced detection speed and accuracy	Detection speed: Improved by 30%, Accuracy: 98%, False alarm rate: Reduced by 20%
Sarhan et al. [119]	Feature extraction & ML models	Datasets: UNSW-NB15, ToN-IoT, CSE-CIC-IDS2018; Algorithms: PCA, AE, LDA, DFF, CNN, RNN, DT, LR, NB	Accuracy: PCA + CNN: 97%, AE + RNN: 95%, LDA + DT: 93%
Yang et al. [120]	Hybrid IDS	Datasets: CAN-intrusion, CSE-CIC-IDS2018; Algorithms: Random Forest, SVM, KNN	Accuracy: 96%, Processing time: Reduced by 40%
Sedjelmaci & Senouci [121]	Clustering scheme & SVM	VANETs; Attacks: wormholes, black holes, Sybil, selective forwarding, packet duplication, resource exhaustion	Detection accuracy: 94%, False positive rate: 5%
Li et al. [122]	SVM for anomaly detection	Behavioral and contextual data	Precision: 95%, Recall: 93%, Communication overhead: Reduced by 25%
Shams et al. [123]	Kernel-based SVM	IDS for V2X; Limitations in densely populated networks	Detection accuracy: 92%; Effective in sparsely populated networks, less effective in dense networks
Shenfield et al. [124]	ANN-based IDS	Non-signature-based IDS for malicious shellcodes	Accuracy: 98%, False positive rate: 2%
Mane et al. [125]	Explainable AI Framework	Techniques: Feature importance analysis, LIME, counterfactual explanations	Trust improvement: 20%, Explainability: Enhanced by 30%
Zhao et al. [126]	DBN and PNN	Classification models for V2X intrusion detection	Classification accuracy: DBN: 95%, PNN: 93%
Alsaedi et al. [127]	ML-based IDS	Dataset: ToN-IoT; Techniques: Chi-square, SMOTE	Accuracy: 98%, Recall: 97%
Grover et al. [129]	Ensemble-based ML	WEKA toolset; Algorithms: Random Forest, J-48	Classification accuracy: Random Forest: 95%, J-48: 94%
Singh et al. [130]	SVM and LR	Dataset: VeReMi; Attacks: position falsification, unexpected stops, traffic congestion	Detection accuracy: LR with normalized features: 97%, SVM: 95%
So et al. [132]	KNN and SVM	Dataset: VeReMi; Detects location-spoofing misbehavior	Detection accuracy: KNN: 96%, SVM: 95%
Boulalouache et al. [43]	Survey of ML-based misbehavior detection	Various datasets surveyed	Survey-based insights
Almi'ani et al. [142]	Non-linear IDS using cluster count	Various V2X scenarios	Detection accuracy: 94%, Clustering effectiveness: Improved by 20%
Leandros et al. [143]	Distributed IDS (DIDS) with K-OCSVM	Investigated effects of RSU placement, intruder velocity, vehicle density	Accuracy: 92%, Response time: Reduced by 30%
Tariq et al. [139]	TL-based IDS	Convolution LSTM-based models; Accuracy: 88%-95%	Detection accuracy: 88%-95%, Model effectiveness: Enhanced by 15%
Tait et al. [140]	Advanced IDS methods	Dataset: CICIDS2017; Types: DoS, PortScan, Botnet, Web Attacks, etc.	Detection accuracy: 94%, Performance comparison: Enhanced by 10%

the Canadian Institute for Cybersecurity (CIC) to provide realistic intrusion detection scenarios. These datasets include a variety of cyberattacks, such as brute force, DoS, DDoS, web-based attacks, and network infiltration. The datasets incorporate network traffic logs, system events, and statistical flow-based features extracted using CICFlowMeter [151].

- **KDD Cup 99 Dataset:** Originally introduced for the Third International Knowledge Discovery and Data Mining (KDD) competition, the KDD Cup 99 dataset comprises network traffic data simulated in a military network environment. It contains 41 features and five classes, including four categories of attacks: DoS, Probe, Remote-to-Local (R2L), and User-to-Root (U2R). However, the dataset is considered outdated due to redundant data, imbalanced distributions, and obsolete attack scenarios [152].
- **X-IIoTID Dataset:** The X-IIoTID dataset is designed specifically for Industrial IoT (IIoT) systems. It addresses challenges related to IIoT security, including protocol heterogeneity and incompatibility. The dataset comprises host resources, network traffic, system logs, and security alerts, making it highly suitable for IDS evaluation in IIoT environments [153].
- **UNSW-NB15 Dataset:** The UNSW-NB15 dataset offers a more contemporary intrusion detection benchmark. It includes nine attack types, such as fuzzers, backdoors, DoS, exploits, reconnaissance, shellcode, and worms. Developed by the Australian Centre for Cyber Security (ACCS), the dataset contains 49 features (43 continuous and six categorical) and provides an up-to-date labeled dataset for evaluating IDS under realistic conditions [154]. Table XIII provides an overview of commonly used IDS datasets, detailing their characteristics, number of features, and accuracy metrics. These datasets include both real-world and synthetic sources, ensuring a diverse evaluation environment for intrusion detection techniques. Table XIV compares various datasets based on their attributes, attack coverage, and applicability for detecting zero-day attacks in V2X security. This comparison highlights the importance of using datasets that closely mimic real-world vehicular network environments to ensure robustness against evolving cyber threats.

C. Comparative Evaluation of Datasets for Zero-Day Attack Detection in V2X Security

While several datasets exist for intrusion detection research, not all are suitable for detecting previously unseen zero-day attacks. Evaluating IDS effectiveness against such attacks requires datasets with diverse attack scenarios, realistic network traffic, and up-to-date cyber threat coverage. This section evaluates key datasets—VeReMi, CICIDS2017, ToN-IoT, UNSW-NB15, and ROAD—based on their attack diversity, attributes, attack coverage, real-world applicability, adaptability to unknown threats, and V2X specificity.

- **VeReMi Dataset:** Designed for Vehicular Ad Hoc Networks (VANETs), VeReMi focuses on misbehav-

ior detection, including message falsification, identity spoofing, and position manipulation. While valuable for V2X-specific scenarios, it lacks diversity in general cyberattacks, limiting its applicability for evaluating IDS against zero-day threats.

- **CICIDS2017 Dataset:** This dataset encompasses a wide range of cyber threats, such as brute force attacks, Distributed Denial of Service (DDoS), botnets, and web-based attacks. Its broad attack diversity makes it suitable for Explaining IDS performance against zero-day attacks in traditional IT and IoT systems. However, it does not include V2X-specific attack scenarios like GPS spoofing or VANET-based Sybil attacks.
- **ToN-IoT Dataset:** Comprising modern IoT and Industrial IoT (IIoT) attacks, ToN-IoT includes DDoS, TCP-SYN flooding, and IoT-specific malware. It serves as a good testbed for zero-day detection models in edge computing environments but offers limited coverage of V2X misbehavior detection.
- **UNSW-NB15 Dataset:** Providing a contemporary intrusion detection benchmark, UNSW-NB15 covers modern attack patterns such as fuzzers, backdoors, exploits, and DoS. While useful for evaluating zero-day attacks, it lacks V2X-specific features necessary for comprehensive V2X security assessments.
- **ROAD Dataset:** A real-world dataset featuring jamming, replaying, and spoofing attacks, ROAD is beneficial for vehicular cyberattack research. However, its focus on known attacks in vehicular environments limits its effectiveness in Explaining IDS performance against zero-day threats.

D. VeReMi Vs. CICIDS2017: Evaluating Zero-Day Attack Detection

Among the most frequently used datasets in V2X security and general cybersecurity are **VeReMi** and **CICIDS2017**. However, their suitability for detecting zero-day attacks differs significantly.

1) **VeReMi Dataset:** The **VeReMi dataset** was designed for VANET misbehavior detection, with a focus on:

- Message falsification, identity spoofing, and position manipulation in vehicular networks.
- Simulation of misbehavior in V2X communication but lacking modern network-based attack diversity.
- No coverage of malware, ransomware, or large-scale DDoS attacks.
- Limited zero-day attack applicability, as it primarily captures predefined VANET attack types.

Despite its utility for V2X security, VeReMi is not well-suited for evaluating IDS performance against modern cybersecurity threats that extend beyond vehicular misbehavior.

2) **CICIDS2017 Dataset:** The **CICIDS2017 dataset** provides a broader scope of cybersecurity threats, making it more applicable for zero-day attack detection:

- Includes brute force attacks, DDoS, botnets, and web-based attacks.

TABLE XIII
DIFFERENT DATASETS, CHARACTERISTICS, FEATURES, AND ACCURACY METRICS

Dataset Name	Description	Characteristics	# Features	Accuracy Metrics
ROAD	Real-world dataset with normal and abnormal traffic data from a road segment in Germany.	Includes jamming, replaying, and spoofing attacks; provides GPS coordinates, speed profiles, timestamps.	72	99.57% (Precision, Recall, F1-score)
VDDD	Synthetic dataset simulating normal and abnormal traffic using SUMO software.	Includes DoS, Sybil, and blackhole attacks; provides vehicle IDs, positions, velocities, accelerations.	68	94.7% (Accuracy, Precision, Recall)
VeReMi	Hybrid dataset combining real-world data with synthetic data from the Veins framework.	Includes DoS, impersonation, and falsification attacks; provides vehicle IDs, positions, velocities, accelerations.	60	97.37% (Accuracy, Precision, Recall, F1-score)
VDOS-LRS	Synthetic dataset simulating normal and abnormal traffic using OMNeT++/Veins framework.	Includes DoS, grey hole, and blackhole attacks; provides vehicle IDs, positions, velocities, accelerations.	55	91.3% (Precision, Recall, F1-score)
KDD Cup 99	Classic dataset with network traffic data from a military network environment.	Contains network traffic data with various intrusions; 41 features and 5 classes (normal and 4 types of attacks).	41	92.5% (Accuracy, Precision, Recall, F1-score)
NSL-KDD	Revised version of KDD Cup 99 addressing redundancy, imbalance, and novelty issues.	Addresses issues of KDD Cup 99; includes selected records from the original dataset.	41	93.45% (Accuracy, Precision, Recall, F1-score)
UNSW-NB15	Modern dataset with normal activities and contemporary attack behaviors.	Includes 9 types of attacks; 49 features (43 continuous, 6 categorical).	49	94.9% (Precision, Recall, F1-score)
CIC-IDS2017	New dataset with benign traffic mixed with multiple recent attack types.	Includes 14 types of attacks; 79 features (76 continuous, three categorical).	79	97.12% (Accuracy, Precision, Recall, F1-score)
CIC-IDS2018	Comprehensive dataset for evaluating intrusion detection systems.	Includes various attack scenarios; 80 features extracted using CICFlowMeter-V3.	80	96.5% (Accuracy, Precision, Recall, F1-score)
X-IIoTID	Intrusion dataset designed for Industrial Internet of Things (IIoT) systems.	Contains network traffic, host resources, logs, and alerts; 68 features, including normal and attack labels.	68	95.87% (Accuracy, Precision, Recall, F1-score)
ToN-IoT	Realistic dataset capturing various data sources from a heterogeneous IoT network.	Includes DoS/DDoS, TCP SYN flooding, UDP flooding, HTTP flooding; sources from IoT sensors, OS data, network traffic data.	83	95.12% (Accuracy, Precision, Recall, F1-score)

- Captures modern network-based attack behavior, suitable for training machine learning models against unseen threats.
- Lacks vehicular-specific attack scenarios such as GPS spoofing, cooperative misbehavior, or VANET-based Sybil attacks.
- Primarily applicable for evaluating zero-day attack detection in IT and IoT security, but lacks V2X adaptability.

Thus, while **VeReMi** is designed for VANET security, **CICIDS2017** offers broader attack diversity, making it more applicable for detecting zero-day attacks in general cybersecurity. However, neither dataset fully supports the detection of unknown threats in V2X environments.

The analysis shows that no single dataset fully supports zero-day attack detection in V2X networks. For instance:

- **VeReMi** is specialized for VANET security but lacks general cybersecurity attack diversity, limiting its applicability for zero-day attack detection in broader V2X environments.
- **CICIDS2017** offers a broad range of attack scenarios, making it suitable for zero-day attack detection in traditional IT and IoT systems, but it does not address V2X-specific threats.
- **ToN-IoT** and **UNSW-NB15** provide diverse attack patterns applicable to IoT and general network security, respectively, but they lack V2X-specific features.
- **ROAD** provides real-world vehicular communication data but focuses on known attack types, reducing its effectiveness for detecting novel attacks.

TABLE XIV
COMPARATIVE EVALUATION OF DATASETS FOR ZERO-DAY ATTACK DETECTION IN V2X SECURITY

Dataset	Attributes	Attack Coverage	Real-World Data?	Zero-Day Suitability	Real-World Applicability
VeReMi	Hybrid dataset (real-world + synthetic), structured into density levels and attacker densities.	DoS, Impersonation, Message Falsification (VANET-specific).	No	Low	High applicability for VANETs, but limited generalizability to other V2X environments.
CICIDS2017	Realistic network traffic dataset with modern attack coverage.	Brute Force, DoS, DDoS, Web Attacks, Infiltration.	No	High	High real-world applicability for general IDS, suitable for zero-day attack detection.
ROAD	Real-world dataset with vehicular attack scenarios.	Jamming, Replay, Spoofing, Message Falsification.	Yes	Low	High applicability for V2X-specific IDS, limited to vehicular attack scenarios.
ToN-IoT	IoT and IIoT-specific dataset with diverse sensor logs and network traffic.	DoS/DDoS, TCP SYN Flooding, UDP Flooding, HTTP Flooding.	Yes	Moderate	High applicability for IoT and IIoT environments, limited relevance to V2X security.
UNSW-NB15	Modern dataset capturing contemporary cyberattack behaviors.	Fuzzers, Backdoors, Exploits, DoS, Reconnaissance, Shellcode, Worms.	No	Moderate	High real-world applicability for general IDS, good coverage of modern threats.
KDD Cup 99	One of the oldest IDS datasets with predefined attack types.	DoS, Probe, R2L, U2R (Imbalanced attack coverage).	No	Low	Low real-world applicability due to outdated attacks, unsuitable for zero-day threats.
NSL-KDD	Improved version of KDD Cup 99 with more realistic attack distributions.	DoS, R2L, U2R, Probe.	No	Moderate	Limited applicability for zero-day attack detection, better than KDD Cup 99 but still insufficient.
CICIDS2018	Extension of CICIDS2017 with additional attack types and updated traffic.	Brute Force, DDoS, Botnets, Web Attacks.	No	High	High real-world applicability for general IDS, suitable for zero-day attack detection.
VDDD	Simulated vehicular dataset generated using SUMO.	DoS, Sybil Attacks, Blackhole Attacks.	No	Low	High applicability for VANETs, but limited generalizability to broader V2X environments.
VDOS-LRS	OMNeT++ and Veins-based synthetic dataset.	DoS, Greyhole, Blackhole Attacks.	No	Low	High applicability for VANETs, but limited in detecting emerging threats.
X-IIoTID	Designed for Industrial IoT (IIoT) with host logs and system events.	IoT-specific malware and various network threats.	Yes	Moderate	High applicability for IIoT environments, but limited relevance to V2X-specific security.

A hybrid approach is recommended to enhance IDS evaluation for zero-day attacks in V2X security. Combining elements from datasets like CICIDS2017 (network-based attacks), ToN-IoT (IoT-based anomalies), and VeReMi (V2X misbehavior detection) can create a more comprehensive evaluation framework. This approach addresses the limitations of individual datasets and provides a more realistic

testbed for zero-day attack detection in vehicular environments.

E. Limitations and Challenges of Datasets in V2X Intrusion Detection

The effectiveness of Intrusion Detection Systems (IDS) in Vehicle-to-Everything (V2X) networks is highly dependent on

the availability of realistic, diverse, and standardized datasets. However, existing datasets face several limitations that hinder the development of robust IDS models capable of detecting a broad range of cyber threats in vehicular environments.

A comprehensive study by Banafshehvaragh and Rahmani [155] analyzed research publications from 2018 to 2022 in Elsevier, IEEE, and Springer, Explaining the datasets and simulators used in intrusion, anomaly, and attack detection for smart vehicles. Their findings revealed several critical issues:

- **Limited Attack Diversity:** Many datasets focus on specific attack types, failing to capture the breadth necessary to evaluate IDS performance against a wide range of threats, including zero-day attacks. The study found that 27.58% of research works relied solely on static attack signatures rather than leveraging datasets for training ML models. Furthermore, while 15.38% of datasets were collected from Controller Area Networks (CAN) [156], making them more realistic, they are not fully representative of V2V communication, limiting their applicability to broader vehicular network security.
- **Reliance on Synthetic Data:** A significant proportion of IDS research depends on simulator-generated data, which does not fully capture the real-world complexities of vehicular networks. The study highlighted that 26.91% of research works used non-vehicular datasets, including NSL-KDD (11.53%), UNSW-NB15 (7.69%), and CIC-IDS2017 (7.69%), all of which were originally designed for general network security rather than vehicular intrusion detection. While some studies employed more relevant datasets, such as CAN-intrusion [156], Car-Hacking [157], ROAD, and NAIST CAN, their ability to reflect the dynamic and heterogeneous nature of V2X communication remains limited.
- **Lack of Standardization:** Variations in data collection methodologies, attack scenarios, and feature sets across different datasets create inconsistencies that hinder the development of universally applicable IDS solutions. This lack of benchmarking frameworks results in ML models that perform well on specific datasets but fail to generalize across different vehicular environments. Additionally, feature extraction tools like Tranalyzer [158], commonly used to process network traffic data from .pcap files, introduce further variability in dataset structure and preprocessing methods, making cross-dataset comparison and model transferability challenging.
- **Bias in IDS Evaluation:** Many IDS models report near 100% classification accuracy, yet this is often due to dataset biases rather than true generalizability. This issue arises because network simulators typically implement predefined attack behaviors rather than capturing real-world adversarial conditions. As a result, ML models trained on synthetic datasets may fail to detect complex or evolving attack patterns when deployed in real vehicular networks.
- **Challenges in Real-World Data Collection:** A major limitation in IDS research is the scarcity of large-scale, real-world vehicular network data. Due to privacy

concerns, security risks, and infrastructure constraints, only a few studies have successfully collected real-time vehicular IDS datasets. This data scarcity affects the ability to train, validate, and benchmark ML models under realistic conditions, further limiting their deployment feasibility [159].

- **Impact on Safety-Critical Applications:** The importance of high-quality datasets extends beyond security; it also impacts vehicle safety and operational reliability. In platooning applications, for example, Quality of Service (QoS) metrics require communication latencies between 10 and 500 milliseconds to ensure synchronization among autonomous vehicles [160]. Studies have demonstrated that a 1-second delay in platoon leader information transmission can lead to vehicle crashes due to loss of coordination [161], underscoring the need for low-latency, high-availability datasets that accurately reflect real-time vehicular interactions.

F. Recommendation of Robust and Realistic Datasets

Addressing these challenges requires the development of hybrid datasets that integrate real-world V2X traffic data with diverse attack scenarios, ensuring better generalizability and resilience in IDS models. Future research should focus on:

- Merging real-world vehicular data with synthetic attack scenarios to capture both authentic communication patterns and adversarial behaviors.
- Establishing standardized feature sets and benchmarking frameworks to enable consistent evaluation of IDS models across different datasets.
- Enhancing real-time data collection mechanisms through collaborations between academia, industry, and standardization bodies, fostering access to large-scale, labeled vehicular datasets.
- Ensuring dataset adaptability for emerging V2X applications, particularly in 5G and 6G-enabled vehicular networks, where low-latency and high-reliability communication are critical.

X. DECENTRALIZED AND PRIVACY-PRESERVING INTRUSION DETECTION: FEDERATED LEARNING AND EDGE AI FOR V2X SECURITY

Traditional machine learning-based IDS rely on centralized data processing, which leads to challenges such as privacy risks, bandwidth constraints, and high computational overhead. These limitations hinder scalability and real-time adaptability in highly dynamic V2X environments. Traditional IDS, which rely on centralized architectures, face several critical limitations when applied to V2X environments.

A. Challenges of Centralized IDS in V2X

Centralized IDS solutions depend on cloud-based or core network servers to process and analyze vehicular data for detecting cyber threats. While effective in traditional networks, they are not well-suited for V2X due to the following challenges:

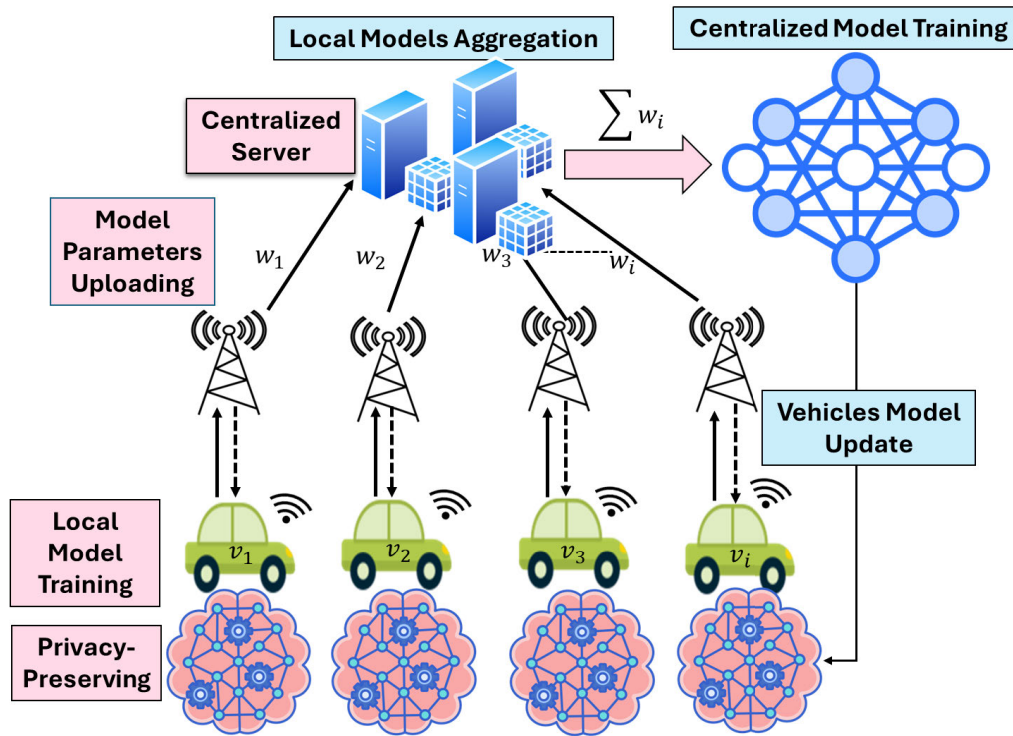


Fig. 18. Misbehaviour detection in V2X networks using federated learning method.

- **Privacy Concerns:** Centralized IDS require continuous data transmission from vehicles to remote servers, raising significant privacy risks and potential exposure to adversarial attacks.
- **High Latency:** The time taken to detect and respond to threats increases due to the communication overhead between vehicles and centralized IDS, making them unsuitable for real-time security applications.
- **Bandwidth Constraints:** The massive volume of data generated by connected vehicles places excessive demands on network bandwidth, leading to congestion and delays.
- **Single Point of Failure:** Centralized architectures introduce vulnerabilities where the failure or compromise of a central IDS server can disrupt the entire security framework.
- **Scalability Issues:** As V2X networks expand with the integration of 5G and 6G technologies, centralized IDS become increasingly impractical due to the need for extensive computational and storage resources.

To overcome these challenges, there is a growing need for decentralized IDS frameworks that can operate efficiently in dynamic V2X environments while preserving user privacy and reducing communication overhead. Decentralized approaches ensure that security mechanisms are distributed across vehicles, roadside units (RSUs), and edge computing nodes, thereby minimizing latency and improving system resilience.

B. Federated Learning and Edge AI as Solutions

Federated Learning (FL) and Edge Artificial Intelligence (Edge AI) have emerged as promising solutions for

privacy-preserving and real-time intrusion detection in V2X networks:

- **Federated Learning (FL):** A decentralized machine learning approach where vehicles and RSUs collaboratively train a global intrusion detection model without sharing raw data, thus ensuring privacy and reducing bandwidth usage.
- **Edge AI:** Deploys lightweight AI models directly on vehicles and edge devices, enabling real-time threat detection without relying on cloud-based processing.

By integrating FL and Edge AI, V2X networks can enhance security while addressing the limitations of traditional IDS. These approaches enable distributed anomaly detection, adaptive threat mitigation, and privacy-preserving security analytics, making them crucial for next-generation vehicular networks.

Federated Learning (FL) is a decentralized machine learning approach that allows multiple clients (e.g., vehicles, roadside units) to collaboratively train a global model without sharing raw data [162]. Instead of centralizing sensitive data, FL aggregates locally trained models to improve accuracy while preserving privacy. This technique is particularly beneficial for V2X communication, where real-time security and privacy constraints prevent the transmission of raw vehicular data as shown in Figure 18.

Federated Learning can be classified into three main types:

- **Horizontal Federated Learning (HFL):** All participating clients have similar data distributions but different samples, making it suitable for homogeneous V2X networks.
- **Vertical Federated Learning (VFL):** Clients share different features of the same data, commonly used in

TABLE XV
COMPARISON OF CENTRALIZED, EDGE AI, AND FEDERATED IDS FOR V2X SECURITY

Feature	Centralized IDS	Edge AI IDS	Federated IDS
Data Privacy	Low (Data shared with a cloud server)	High (Processed on-device)	Very High (Only model updates shared)
Detection Latency	High (Cloud processing delay)	Very Low (On-device processing)	Low (Localized aggregation)
Scalability	Limited (Requires extensive cloud infrastructure)	Moderate (Deployable on edge devices)	High (Distributed model training)
Resilience to Attacks	Low (Single-point failure risk)	Moderate (Edge models can be compromised)	High (Decentralized aggregation reduces attack impact)
Computational Efficiency	Low (Requires centralized GPU resources)	Moderate (Limited by edge device capacity)	High (Collaborative learning optimizes resource usage)

heterogeneous V2X environments, such as collaborations between vehicles and smart infrastructure.

- **Federated Transfer Learning (FTL):** Applied when data distributions and features are different, allowing knowledge transfer across diverse vehicular datasets.

FL-based Intrusion Detection Systems (IDS) for vehicular networks employ various aggregation models to balance privacy, accuracy, and communication efficiency:

- **Federated Averaging (FedAvg):** A standard FL technique where local models are aggregated at a central server, minimizing communication overhead while maintaining accuracy [162].
- **Federated Stochastic Gradient Descent (FedSGD):** Instead of sharing full model parameters, clients transmit gradient updates, reducing bandwidth consumption.
- **Federated Proximal (FedProx):** An improved variant of FedAvg designed to stabilize training in heterogeneous V2X environments by adjusting local updates [163].
- **Hierarchical Federated Learning (HFL):** Extends FL by introducing multi-level model aggregation (e.g., at the vehicle, RSU, and cloud levels), improving scalability and efficiency in large-scale V2X networks.

FL significantly differs from centralized IDS in terms of privacy, efficiency, and scalability. Table XV highlights the advantages of FL-based IDS over centralized and edge-based approaches.

FL has been widely recognized as a privacy-preserving machine learning approach in V2X security applications. Several cryptographic techniques enhance privacy in FL-based IDS:

- **Homomorphic Encryption (HE):** Allows computations on encrypted model updates, preventing adversarial model inversion attacks.
- **Differential Privacy (DP):** Introduces controlled noise to prevent leakage of sensitive vehicular movement patterns.
- **Secure Multi-Party Computation (SMPC):** Ensures secure aggregation of model updates without relying on a central server.

C. Use Cases of FL in V2X Security

Several studies have demonstrated the effectiveness of FL for privacy-preserving IDS in V2X networks: H²-Fed by Song et al. [164], This study introduces a hierarchical federated learning framework for Cooperative ITS (C-ITS), where

roadside units (RSUs) act as intermediate aggregation nodes to reduce communication overhead and improve security. The system ensures scalable, privacy-preserving intrusion detection, especially in non-IID vehicular environments [164].

Li et al. [163] addresses system heterogeneity in FL-based IDS, stabilizing training for vehicular networks where participating nodes have varying computational resources. The study highlights a 22% improvement in convergence rates compared to FedAvg, making it more suitable for real-time V2X networks.

FL-DECO-BC by Narkedimilli et al. [165] integrates blockchain with federated learning for privacy and security enhancement in VANETs. By employing decentralized oracles, the system ensures secure data provenance tracking and trustworthy collaboration among vehicles while mitigating potential adversarial attacks in FL deployments.

Selamnia et al. [9] implemented a federated learning-based IDS for cellular V2X (C-V2X) networks. Their approach distributed detection tasks across multiple nodes while preserving data privacy. Their results indicated that federated learning significantly reduced network resource consumption while maintaining high detection accuracy.

Alekszejenko et al. [166] addresses privacy concerns in federated learning for vehicular networks by incorporating trusted V2V communication to balance data distribution and learning efficiency. Their proposed privacy-aware FL framework significantly reduces eavesdropping risks at aggregation servers while enhancing model generalization across AV fleets.

Federated Learning (FL) is a decentralized machine learning paradigm where multiple devices, such as vehicles and roadside units (RSUs), collaboratively train a global model without sharing raw data. This methodology is particularly well-suited for V2X networks due to its ability to enhance privacy, reduce communication overhead, and improve security without requiring centralized data aggregation [167]. By ensuring that data remains on local devices, FL mitigates risks associated with data breaches, unauthorized access, and large-scale cyberattacks [168].

In V2X and IoT environments, FL has proven effective in strengthening intrusion detection capabilities. Li et al. [169] introduced an FL-based network intrusion detection system (IDS) that significantly reduced communication costs and improved model accuracy, a crucial consideration for high-speed, low-latency vehicular networks. Similarly,

Huang et al. [170] proposed an Execution & Evaluation dual-network FL framework, allowing for personalized intrusion detection based on device-specific characteristics, a feature essential for the heterogeneity of V2X and IoT security applications.

The application of FL in Software-Defined Networking (SDN) has also been explored to enable privacy-preserving intrusion detection mechanisms in vehicular networks. Raza et al. [171] implemented FL-based intrusion detection in SDN-based architectures, reducing the risk of sensitive network data exposure while maintaining high detection accuracy. These studies highlight FL's capacity to adapt to diverse and distributed vehicular ecosystems, ensuring security across connected and autonomous vehicles (CAVs) without compromising data privacy.

A study by Chen et al. [172] introduced VAN-IDS, a hybrid intrusion detection system for Vehicular Ad-Hoc Networks (VANETs) that leverages both packet analysis and physical feature analysis. The system combines Bi-LSTM for temporal packet analysis and LightGBM for vehicle physical attributes, integrating their outputs using the Dempster-Shafer theory. Results from experiments conducted on the F2MD simulation environment demonstrated over 98.5% accuracy in detecting DoS and Sybil attacks, surpassing existing VANET IDS solutions. To further enhance privacy and efficiency, the study proposed VAN-FED-IDS, incorporating federated learning to expedite model training and updates while safeguarding local data privacy. The framework employs roadside units (RSUs) as edge computing nodes for local training, while cloud services aggregate models for collaborative learning. Using open-source FL frameworks like Flower and FedTree, the study evaluated communication overhead and training duration in both single-machine simulation and multi-machine deployments, demonstrating that FL effectively maintains detection accuracy while enhancing privacy.

Similarly, Huang et al. [173] introduced FED-IoV, an FL-based intrusion detection system for the Internet of Vehicles (IoV), addressing high computational demands and privacy concerns. The framework transforms vehicular communication traffic into images and employs MobileNet-Tiny for efficient feature extraction, enabling deployment on resource-constrained devices. Evaluations on the CAN-Intrusion and CICIDS2017 datasets demonstrated 98.51% and 97.74% accuracy, respectively, highlighting the system's ability to handle imbalanced datasets effectively. Additionally, FED-IoV maintained a prediction latency of under 10 milliseconds on Raspberry Pi 4 (8GB), achieving superior real-time performance compared to conventional methods. The study underscores the practicality of FL-based IDS in enhancing IoV security while preserving data privacy.

Renda et al. [174] explore the integration of **Federated Learning (FL)** and **Explainable Artificial Intelligence (XAI)** to enhance security and automation in **6G-enabled vehicle networking**. While FL has been widely applied using *stochastic gradient descent (SGD)* for neural networks, its use in inherently explainable models remains underexplored.

The study highlights **two key advantages** of this approach:

- **XAI improves user trust** by making in-network AI decisions more transparent.
- **FL enhances security and privacy** by preventing direct data sharing across vehicular networks.

The authors propose a framework for online training using live vehicular data, demonstrating its potential in V2X environments. They emphasize how FL-XAI models can provide decentralized, lightweight, and communication-efficient intelligence. Additionally, the paper discusses standardization perspectives and the importance of QoE-driven AI predictions for network operations.

D. Distributed Sensor Fusion for Intrusion Detection in V2X Networks

Intrusion detection in V2X networks has traditionally relied on centralized architectures, where data collected from vehicular sensors is aggregated at a central cloud or edge server for anomaly detection. However, this approach introduces several challenges, including **privacy risks**, **latency issues**, and **vulnerability to single points of failure**. To address these limitations, **distributed sensor fusion techniques** have emerged as a promising alternative, enabling vehicles and roadside units (RSUs) to collaboratively analyze security threats while minimizing data transmission overhead.

Distributed sensor fusion allows multiple agents (e.g., vehicles, RSUs, infrastructure nodes) to share **processed insights** rather than raw sensor data, leading to improved decision-making efficiency. One key approach in this domain is **consensus-based data fusion**, where each node in a vehicular network refines its local threat assessment by integrating **partial observations** from its neighboring nodes. [175] introduced a **multi-agent consensus-based approach** for detecting and mitigating biasing attacks, demonstrating its robustness in distributed sensor networks. Such techniques can be directly applied to **V2X intrusion detection**, where decentralized consensus mechanisms enhance the resilience of misbehavior detection in vehicular networks.

1) Advantages of Distributed Sensor Fusion in V2X IDS:

- **Reduced Data Transmission Overhead:** Unlike centralized IDS, where all data is sent to a remote cloud, **distributed fusion** enables vehicles to share only necessary insights, reducing communication bandwidth usage.
- **Improved Fault Tolerance:** By **aggregating multiple viewpoints**, the system can filter out erroneous or compromised sensor inputs, mitigating adversarial attacks.
- **Scalability:** Decentralized processing enables **scalable security analytics**, essential for future **5G and 6G-enabled V2X** networks.

2) Implementation in V2X Security: Recent research has explored how **multi-agent collaborative detection** can strengthen **intrusion detection and bias mitigation** in V2X environments. For example, [176] proposed a **two-layer vehicle-edge IDS**, where **edge computing nodes** play a key role in **distributed anomaly detection and attack mitigation**.

Autonomous Vehicles (AVs) rely on multi-sensor fusion to enhance situational awareness and improve decision-making in real-time by integrating data from various sources such

as LiDAR, radar, GPS, and cameras. However, the complexity of these fusion architectures raises concerns regarding transparency, interpretability, and accountability, which are critical factors in ensuring trustworthiness and safety in AV systems [177].

Recent studies highlight the necessity of incorporating Explainable AI (XAI) techniques within multi-sensor fusion frameworks to make AV decision-making processes more interpretable [177]. While multi-sensor fusion technologies significantly improve perception accuracy, their inherent closed-box nature makes it challenging to diagnose system failures and biases. As noted by Panduru and Walsh [177], a key research focus should be on balancing high-performance real-time fusion methods with explainability constraints to ensure transparency without compromising safety and computational efficiency.

Incorporating XAI-based sensor fusion strategies allows for better debugging of AV models, enhancing their interpretability for engineers, policymakers, and end-users alike. This approach also aligns with regulatory efforts aimed at ensuring that AVs meet safety and ethical standards in real-world deployments.

3) *Future Considerations:* Despite its advantages, implementing **distributed sensor fusion** in V2X security poses **challenges**, such as the need for **robust trust mechanisms** to prevent malicious data manipulation. Future work should focus on:

- **Hybrid distributed-federated IDS models** that combine **local anomaly detection** with **global model updates**.
- **Privacy-preserving consensus algorithms** to prevent **information leakage in sensor fusion**.
- **Integration with blockchain-based trust frameworks**, ensuring data integrity in multi-agent V2X IDS setups.

E. Observer-Based Redundancy in Federated Learning for Secure V2X Networks

In Federated Learning (FL)-based intrusion detection, vehicles collaboratively train IDS models without sharing raw data. However, FL-based security mechanisms face challenges such as data inconsistencies, poisoning attacks, and sensor failures, leading to reduced detection accuracy. To address these challenges, observer-based redundancy mechanisms can enhance FL by enabling vehicles to infer system state even in the presence of missing or unreliable data. Ensuring robust intrusion detection in Federated Learning (FL)-based IDS for V2X networks requires fault-tolerant mechanisms that can mitigate data loss and sensor failures. Observer-based redundant estimation techniques have emerged as promising solutions for maintaining estimation accuracy and system observability despite adversarial conditions or hardware failures.

1) *Distributed Observer Design for CAV Platooning:* A critical challenge in V2X networks is ensuring reliable estimation and fault detection in connected and autonomous vehicle (CAV) platoons. Doostmohammadian and Rabiee [178] propose a distributed observer-based approach to track vehicle dynamics in a platoon using a linear-matrix-inequality (LMI)-based estimation strategy. This method enhances network survivability by utilizing redundant observer designs that

maintain estimation accuracy under node or link failures. The study highlights that strong-connectivity in vehicular communication networks ensures observability and stability of state estimation, even in adverse conditions.

2) *Redundant Observer-Based Intrusion Detection:* Observer-based techniques can also enhance IDS robustness in V2X networks by providing redundant state estimation in FL models. Zabihi et al. [179] introduce a hybrid model-data vehicle sensor and actuator fault detection framework that integrates model-based estimation with data-based techniques to improve anomaly detection accuracy. By leveraging observer-based filtering, their approach mitigates sensor noise and data loss, making it particularly useful for distributed FL-based IDS in dynamic vehicular environments.

3) *Securing Distributed Filtering in Federated Learning:* Federated learning introduces unique security challenges, such as poisoning attacks and unreliable data aggregation. He et al. [180] address these concerns by proposing a secure distributed filtering approach that enhances resilience against sensor attacks. Their methodology ensures that faulty sensor readings do not propagate through the FL model, preserving the integrity of intrusion detection decisions. Integrating such filtering mechanisms within FL-based IDS frameworks can improve detection accuracy and robustness in V2X networks.

4) *Future Research Directions:* Future research should focus on integrating observer-based fault-tolerant estimation within FL frameworks for V2X security. Potential directions include:

- **Resilient Federated Learning:** Implementing redundant observer-based techniques in FL aggregation models to handle unreliable vehicular data.
- **Secure Model Updates:** Enhancing FL robustness by filtering out adversarial contributions through redundant state estimation.
- **Adaptive Network Topology:** Designing dynamically reconfigurable vehicular networks that maintain strong connectivity and observability under node or link failures.
- **Lightweight Redundant Estimation:** Developing computationally efficient observer-based FL models that can operate on resource-constrained edge devices in vehicular networks.

F. Advancements in Distributed Algorithms for V2X Networks

The evolution of distributed algorithms has revolutionized computation, information processing, filtering, and control methods, making them indispensable for large-scale and dynamic applications. Unlike traditional centralized schemes, distributed solutions offer enhanced scalability, robustness to node failures, and adaptability, while also facilitating parallel processing. The decentralized nature of these algorithms ensures that the failure of a single node does not compromise the integrity of the entire system, thereby improving fault tolerance and reliability [181].

In the context of intelligent transportation systems (ITS), distributed algorithms have gained traction in addressing challenges related to connected and autonomous vehicles (CAVs). Several studies have explored various control and

estimation strategies, including centralized and distributed frameworks. For instance, hybrid model-data vehicle sensor and actuator fault detection systems have been proposed to enhance vehicular security and reliability. Observer-based control mechanisms, particularly in leader-follower vehicle platooning scenarios, have also been examined for improving coordination and safety. Furthermore, consensus-based strategies have demonstrated efficacy in ensuring string stability under heterogeneous time-varying delays in CAV platoons.

Recent advancements in distributed computing, including cloud and edge computing, have further fueled interest in decentralized algorithms, given their alignment with these architectures. Notably, distributed algorithms have been successfully implemented across various domains, including distributed fault detection, distributed machine learning and optimization, and distributed attack mitigation. Moreover, their application in vehicular networks extends to distributed routing, coverage control, and resource allocation.

The integration of distributed reinforcement learning (RL) has also been proposed as a novel approach for optimizing mixed-traffic environments consisting of both human and automated vehicles [181]. Distributed filtering techniques, such as Kalman filtering and particle filtering, have been widely employed to improve multi-target tracking in autonomous vehicle fleets [181]. Additionally, consensus-based distributed filtering has been effectively utilized in applications involving autonomous aerial vehicles (AAVs), ensuring enhanced tracking and state estimation under uncertain conditions.

Despite the significant advancements in distributed algorithms, several challenges remain, particularly concerning security and resilience. The susceptibility of multi-agent networks to adversarial attacks necessitates the development of robust countermeasures. Distributed intrusion detection and mitigation mechanisms have been introduced to enhance network resilience against cyber threats [179]. Future research should focus on hybrid distributed-federated models, privacy-preserving consensus mechanisms, and blockchain-based trust frameworks to ensure the integrity and security of V2X networks.

G. Edge AI for Real-Time Intrusion Detection

Traditional cloud-based IDS approaches suffer from high latency, bandwidth constraints, and centralized processing bottlenecks, making them unsuitable for real-time vehicular security applications. In contrast, edge computing facilitates distributed intelligence by processing security threats closer to the vehicles, reducing reliance on centralized cloud resources.

Advantages of Edge AI for ML-Based IDS in V2X Networks:

- **Low Latency & Real-Time Response:** Deploying ML models at edge nodes enables V2X networks to detect anomalies and cyber threats with minimal delay, which is crucial for safety-critical applications like autonomous driving. Edge-based IDS significantly reduces decision-making time compared to cloud-based solutions.
- **Bandwidth Optimization & Scalability:** Centralized cloud processing requires high data transmission costs, which

is impractical in bandwidth-limited V2X environments. Edge computing enables localized ML inference, allowing vehicles to process security threats without excessive network load.

- **Federated Learning for Distributed Security:** **Federated Learning (FL)** can be integrated with edge computing to create **collaborative IDS models** across multiple vehicles and roadside edge nodes. This decentralized approach ensures **privacy-preserving training**, where vehicles learn from local data **without exposing sensitive information** to centralized servers.
- **Adaptability to Dynamic V2X Environments:** Edge computing allows **continuous model updates** by leveraging data from **real-world V2X interactions**. This adaptability helps in **detecting new attack patterns** and enhancing IDS robustness over time.

Edge AI refers to deploying AI-powered security models directly on edge devices such as RSUs, onboard vehicle processors, and fog nodes, allowing real-time threat detection without reliance on cloud-based systems. Unlike centralized approaches, Edge AI significantly reduces latency, enhances real-time decision-making, and lowers bandwidth consumption, making it ideal for intrusion detection and anomaly detection in V2X networks [182]. A practical **Edge-Assisted Federated Intrusion Detection System (EF-IDS)** can be implemented to improve vehicular security. This system can:

- Utilize **edge computing nodes (RSUs, MEC servers)** to process vehicular security threats in **real-time**.
- Deploy **lightweight deep learning models** optimized for resource-constrained edge devices.
- Leverage **federated learning and continual training** to adapt to evolving cyberattacks in V2X environments.

A key advantage of Edge AI in vehicular security is its ability to reduce dependency on continuous cloud connectivity. Alladi et al. [183] proposed a Mobile Edge Computing (MEC)-based IDS, integrating deep learning models to analyze V2X threats in real time. Similarly, Samy et al. [182] designed a distributed attack detection framework, leveraging fog computing to process intrusion detection locally, thereby minimizing response time and improving system resilience.

Another crucial aspect of Edge AI is computational efficiency. Modern Edge AI architectures employ energy-efficient deep learning models, ensuring that vehicular devices and IoT sensors can perform real-time anomaly detection without excessive power consumption [184]. Bangui et al. [185] further explored a hybrid security mechanism, dynamically allocating computational resources across V2X networks, thereby optimizing performance in diverse vehicular environments.

Rihan et al. [186] propose Deep-VFog, an AI-enabled, fog-assisted architecture for V2X networks, integrating deep learning (DL), deep reinforcement learning (DRL), and federated learning (FL) with fog computing (FC) to enhance vehicular communication efficiency. The study highlights how AI and FC can improve traffic management, autonomous driving, resource allocation, and vehicular cloud services. Their results demonstrate that DL models optimize traffic flow predictions, DRL enhances resource allocation in AAV-assisted vehicular networks, and FL strengthens privacy-aware,

decentralized V2V communication. The authors also discuss open challenges, including security risks, FL implementation, AAV-aided networks, and SDN/NFV exploitation in future 5G-V2X networks.

Williams et al. [187] propose a hybrid AI architecture integrating edge computing with cloud-based intelligence to enhance security in Vehicle-to-Everything (V2X) communication networks. The framework employs lightweight AI models at the edge for real-time anomaly detection while utilizing federated learning in the cloud to aggregate threat intelligence while preserving data privacy. The proposed system was evaluated using a large-scale V2X simulation environment, testing resilience against cyber threats such as Denial-of-Service (DoS), spoofing, and data manipulation. Experimental results show that the hybrid approach reduces latency by 30%, improves anomaly detection accuracy by 10%, and enables faster real-time responses (22ms at the edge vs. 58ms in the cloud) [9], [187], [188]. Furthermore, blockchain integration ensures secure, immutable communication, preventing unauthorized data tampering [189], [190]. The architecture optimally distributes computation between edge devices and cloud-based models, ensuring adaptive security updates without overloading vehicle resources. By combining localized edge-based detection with cloud intelligence, the system provides a scalable, privacy-preserving, and resilient cybersecurity solution for intelligent transportation. Future enhancements include optimizing federated learning for large-scale deployments, integrating quantum-resistant cryptography, and refining deep learning models with reinforcement learning to strengthen threat mitigation [191], [192].

Khan et al. [193] proposed a robust, resilient, and reliable architecture for V2X communications by integrating mobile edge computing (MEC) and blockchain technology. Their study introduced a three-stage computing service that leverages cloud computing (CC), edge computing (EC), and fog computing (FC) nodes to optimize resource allocation and network adaptability. The proposed blockchain-based transaction service enhances reliability, ensuring secure data exchange. Their experimental results demonstrated an accurate car connectivity prediction of 98% and a 92% increase in data blocks using a micro-blockchain solution.

Rishiwal et al. explore secure V2X communication networks in smart cities, focusing on human-centric security and privacy challenges. They analyze security threats, including data privacy breaches, traffic manipulation, and malicious attacks, and propose a blockchain-powered framework to enhance the security and integrity of V2X networks. The study highlights authentication mechanisms, encryption protocols, and intrusion detection systems as critical components of secure V2X communication. They demonstrate blockchain's potential through a case study, showing its role in decentralizing trust, ensuring data integrity, and preventing unauthorized access [194].

Haji Mirzaee et al. propose a two-layer collaborative vehicle-edge Intrusion Detection System (IDS) for securing vehicular communications. The system leverages machine learning (ML) techniques for attack detection, integrating an on-vehicle IDS module with an IDS platform at a mobile edge

computing (MEC) server. Their approach enhances security in V2X communication by significantly reducing communication latency and energy consumption by up to 80% while maintaining high detection accuracy [176].

Ibn-Khedher et al. [195] explore the role of *Edge Computing* and *Network Function Virtualization (NFV)* in optimizing resource allocation and service offloading for next-generation *autonomous vehicle networks (AVNs)*. The study highlights the computational challenges of existing optimization methods, which struggle with large-scale networks due to their exponential time complexity.

To address this, the authors propose a Deep Reinforcement Learning (DRL)-based approach to enhance Autonomous Vehicle (AV) service offloading and optimize edge resources. Specifically, they introduce:

- **Optimal Virtual Edge Autopilot Placement (OVEAP) Algorithm:** A method using Integer Linear Programming (ILP) to efficiently allocate autopilot services.
- **Autopilot Placement Protocol:** Designed to support the OVEAP algorithm for better network function virtualization (VNF) placement and chaining.
- **DRL Model for IoAV Networks:** A learning-based framework that improves network service allocation and edge resource utilization.

Extensive simulations validate the effectiveness of the proposed methods, showing significant improvements in total edge server utilization, allocation time, and autopilot success rates compared to state-of-the-art solutions.

Albogami et al. (2025) [196] propose an Intelligent Deep Federated Learning Model for Enhancing Security (IDFLM-ES) to address privacy and security concerns in IoT-enabled edge computing environments. Traditional centralized machine learning methods pose risks to data confidentiality, necessitating decentralized approaches like Federated Learning (FL). However, implementing FL in heterogeneous IoT environments remains challenging.

To mitigate these issues, the authors introduce a Federated Hybrid Deep Belief Network (FHDBN) model, which leverages FL for analyzing time-series data from IoT edge devices. The proposed model incorporates:

- **Data Preprocessing:** A feature selection mechanism using Golden Jackal Optimization (GJO) to enhance data quality.
- **Federated Learning-based Model:** A distributed learning approach to improve convergence and generalization in decentralized databases.
- **Hyperparameter Optimization:** A Dung Beetle Optimizer (DBO) to fine-tune the FHDBN model for improved performance.

Experimental validation on benchmark datasets demonstrates the superiority of IDFLM-ES, achieving an accuracy of 98.24%, outperforming state-of-the-art models in intrusion detection for IoT networks. The findings suggest that integrating deep learning with federated architectures significantly enhances security in IoT-edge computing environments.

H. FL and Edge AI for Resource-Constrained V2X and IoT Security

The combination of Federated Learning and Edge AI provides a powerful framework for securing resource-constrained vehicular and IoT environments, particularly in 6G-enabled networks where bandwidth and computational power are critical constraints. Key advantages include:

- **Computational Efficiency:** FL minimizes centralized processing requirements, enabling in-vehicle security analytics without overloading communication networks [184].
- **Low Latency:** Edge AI allows on-device threat detection, significantly reducing latency and response time compared to cloud-based solutions [197].
- **Energy Efficiency:** AI models optimized for edge devices consume minimal power, ensuring long-term viability in battery-operated vehicular and IoT systems [198].
- **Privacy Preservation:** By processing security threats locally within vehicles and edge devices, FL eliminates the need for large-scale data transfers, reducing the risk of data breaches [168].

The integration of FL with blockchain-based security frameworks has further strengthened privacy protection in distributed vehicular networks. Bhavsar et al. [198] demonstrated that blockchain-enhanced FL models significantly improve tamper resistance and intrusion detection accuracy. Additionally, [113] introduced a blockchain-powered FL framework for V2X edge computing, providing anti-tampering protection and secure collaborative learning across distributed vehicular nodes.

I. Performance Comparisons and Real-World Implementations

Quantitative analyses highlight the superiority of FL-based IDS over centralized models in terms of latency, detection accuracy, and communication cost. Hasan et al. [184] and Zhou et al. [168] reported that FL-based IDS achieved:

- Lower communication overhead by transmitting only model updates instead of raw data.
- Higher detection accuracy, leveraging heterogeneous datasets across distributed vehicles.
- Reduced response time, as security threats are detected and mitigated at the edge rather than relying on cloud-based solutions.

Moreover, Taslimasa et al. [197] found that ImageFed, an FL-based security model, demonstrated greater robustness in detecting cyberattacks compared to traditional centralized IDS frameworks. Similarly, Korba et al. [167] reported that FL-based IDS in V2X networks achieved an impressive 99.99% detection accuracy, with a false positive rate of just 0.01%, significantly outperforming centralized approaches.

J. Future Research Directions in FL and Edge AI for V2X Security

As V2X networks continue evolving, integrating FL and Edge AI will be pivotal for developing scalable,

privacy-preserving, and real-time security solutions. Several emerging directions include:

- **FL with Reinforcement Learning (FL-RL):** Research by [117], [118] explores FL + RL for mitigating jamming attacks, crucial for securing autonomous vehicle communications.
- **Quantum Federated Learning (QFL):** [116] introduced a Quantum FL framework for securing consumer networks, highlighting its potential in 6G-V2X security.
- **Explainable AI in FL-based IDS:** [115] proposed an explainable FL system, ensuring transparency in intrusion detection models for vehicular networks.

Despite these advancements, challenges remain, including secure model aggregation, adversarial attack mitigation, and optimizing energy-efficient learning protocols. Future work should focus on hierarchical FL architectures, where multi-layered FL frameworks (vehicle-level, RSU-level, and cloud-level) collaborate to enhance security and computational efficiency in next-generation V2X networks [184].

By integrating decentralized, real-time, and privacy-enhancing AI techniques, FL and Edge AI hold the potential to redefine security frameworks in V2X networks, ensuring robust and adaptive protection against evolving cyber threats.

K. Challenges and Future Directions for Federated Learning in V2X and IoT Security Within Beyond 5G Networks

The deployment of Beyond 5G (B5G) and 6G networks introduces unprecedented challenges for Federated Learning (FL) in V2X and IoT security. The architectural evolution of 6G networks, characterized by ultra-dense networks (UDNs), pervasive AI, terahertz (THz) communications, and massive Machine-Type Communications (mMTC), presents new vulnerabilities that threaten the privacy and security of FL-based Intrusion Detection Systems (IDS) and anomaly detection mechanisms in vehicular environments. Given the rapid expansion of connected autonomous vehicles (CAVs) and the reliance on FL for decentralized intrusion detection, addressing these challenges is critical for ensuring data privacy, computational efficiency, and system resilience.

1) *Ultra-Dense Networks (UDNs) and Model Poisoning Risks:* The unprecedented device density in 6G UDNs significantly increases the attack surface for FL-based security models. Since FL relies on decentralized training across distributed edge devices, malicious entities can inject poisoned model updates, compromising the integrity of the global model. Data poisoning and Sybil attacks become more feasible in 6G-enabled smart vehicular networks, where adversaries can masquerade as multiple legitimate vehicles, disrupting the trustworthiness of the FL model aggregation process [168].

2) *Eavesdropping and Interference in Terahertz (THz) Communications:* The use of THz bands in 6G provides ultra-high-speed data transmission with minimal latency, making it highly suitable for real-time FL aggregation in vehicular networks. However, THz signals are susceptible to eavesdropping and jamming attacks, allowing adversaries to intercept and manipulate FL model updates transmitted between vehicles

and roadside units (RSUs). Secure communication mechanisms such as homomorphic encryption (HE) and differential privacy (DP) must be integrated into FL-based IDS to prevent adversarial inference on vehicular network traffic [182].

3) Model Inversion and Membership Inference Attacks:

One of the major risks in FL for V2X security is the potential for model inversion and membership inference attacks, where adversaries attempt to reconstruct private training data from model gradients. In FL, while raw data remains on local devices, gradient updates can still leak sensitive information, particularly in high-dimensional V2X datasets [184]. Techniques such as differential privacy, secure multiparty computation (SMC), and encrypted model aggregation must be deployed to mitigate these risks while maintaining model performance.

L. Enhancing FL Security in 6G-V2X Networks

1) Secure Aggregation and Homomorphic Encryption (HE):

One of the primary solutions for enhancing FL security in V2X communication is the integration of homomorphic encryption (HE) for secure model aggregation. HE enables FL participants to perform computations on encrypted data without revealing the underlying information, ensuring privacy-preserving model updates. This technique is particularly beneficial in vehicular edge computing, where sensitive driving patterns and network traffic data must remain confidential [173].

2) *Differential Privacy for FL in IoT-V2X Communication:* Differential Privacy (DP) is a fundamental approach for preventing adversarial inference attacks in FL-based V2X networks. By adding controlled noise to gradient updates before transmission, DP ensures that no individual data point can be inferred from the shared model parameters. FL has widely adopted this technique for connected autonomous vehicles (CAVs) and smart IoT security, demonstrating high resistance against model inversion attacks [171].

3) Quantum-Resilient FL for Next-Generation Networks:

As quantum computing advances, traditional cryptographic mechanisms used in FL-based IDS may become vulnerable to quantum-enabled attacks. Quantum Key Distribution (QKD) integrated with FL is being explored as a secure alternative for model transmission and aggregation to counteract this. By leveraging quantum cryptographic techniques, QKD ensures that FL updates in V2X networks remain secure against quantum adversaries [168]. Figure 19 illustrates a privacy-preserving architecture where federated learning is enhanced with post-quantum cryptographic techniques for secure communication and intrusion detection. Hardware security modules (PQ-HSM) are incorporated to accelerate encryption, decryption, and key storage operations, ensuring real-time security while reducing computational overhead.

M. Addressing Computational and Energy Constraints in FL for V2X and IoT

A critical challenge in deploying FL-based IDS in 6G-enabled V2X networks is the high computational and energy requirements of deep learning models. The following strategies

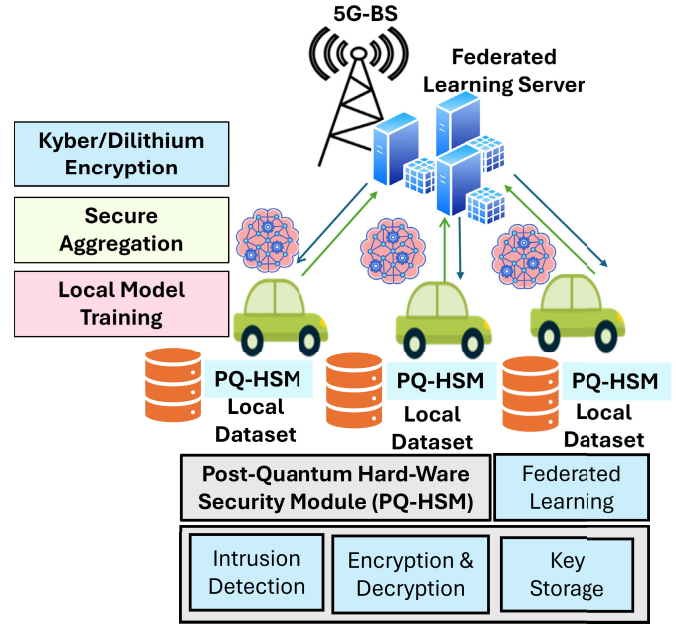


Fig. 19. An overview of integrating post-quantum cryptography (Kyber/Dilithium) and hardware security modules (PQ-HSM) to support secure model aggregation in federated learning for V2X security.

can be implemented to optimize FL for resource-constrained edge AI environments:

- **Lightweight FL Models for IoT and Connected Vehicles:** Optimized CNN-LSTM architectures must be adopted to reduce computational overhead while maintaining high intrusion detection accuracy [184].
- **Energy-Efficient Training Techniques:** Adaptive FL training reduces unnecessary model updates and minimizes energy consumption in battery-operated vehicular IoT systems. AI dynamic model pruning can be used to optimize FL model complexity based on real-time network conditions [168].

A comparative analysis of Federated Learning applications in V2X security is summarized in Table XVI and Table XVII, highlighting key findings, performance metrics, and specific scenarios where FL has been effectively implemented.

XI. ADVANCED DEEP LEARNING ARCHITECTURES FOR INTRUSION DETECTION IN V2X NETWORKS

Deep learning (DL) has emerged as a promising approach for designing intelligent and adaptive Intrusion Detection Systems (IDS) in V2X networks. Unlike traditional rule-based and machine learning-based IDS, deep learning models can autonomously learn complex patterns and detect anomalies with higher accuracy. This section provides an in-depth analysis of two advanced DL architectures—Long Short-Term Memory (LSTM) networks and Generative Adversarial Networks (GANs)—focusing on their architecture, advantages, and limitations.

A. Long Short-Term Memory (LSTM) Networks for IDS

LSTM networks are a special type of recurrent neural network (RNN) designed to handle long-term dependencies in

TABLE XVI
KEY FINDINGS AND PERFORMANCE METRICS OF FEDERATED LEARNING IN V2X SECURITY (PART 1)

Study	Key Findings	Metrics	Scenario
Song et al. [164]	Proposed H ² -Fed hierarchical FL with RSUs for cooperative ITS to reduce overhead	Scalability and privacy in non-IID vehicular data	Hierarchical FL for C-ITS with RSUs
Li et al. [163]	Tackled heterogeneity in FL-IDS and improved training stability	22% convergence improvement over FedAvg	FL training for heterogeneous V2X nodes
Narkedimilli et al. [165]	Introduced FL-DECO-BC, integrating blockchain to enhance FL security	Secure data provenance, reduced adversarial attacks	Blockchain-FL for VANET IDS
Selamnia et al. [9]	Implemented FL-based IDS for C-V2X ensuring privacy	Lower resource usage, high accuracy	FL for distributed IDS in C-V2X
Alekszejko et al. [166]	Proposed privacy-aware FL with trusted V2V communication	Reduced eavesdropping risk, better generalization	Privacy-preserving FL for AV fleets
Li et al. [169]	Developed FL-based IDS reducing costs while improving accuracy	15% drop in comm. overhead, better IDS detection	FL for high-speed, real-time V2X networks
Huang et al. [170]	Dual-network FL enabling device-specific IDS learning	Adaptive detection in heterogeneous networks	Personalized FL for IoV applications
Raza et al. [171]	Applied FL in SDN to ensure privacy in network data	High accuracy, privacy-preserving detection	FL integration with SDN for secure V2X
Chen et al. [172]	VAN-IDS: Bi-LSTM + LightGBM hybrid model in FL	Accuracy over 98.5% for DoS/Sybil detection	Hybrid IDS in VANETs with FL

sequential data. Given that vehicular network traffic exhibits temporal dependencies, LSTMs are particularly effective in analyzing time-series data for IDS.

1) *LSTM Architecture*: LSTM networks consist of multiple layers of memory cells, each comprising:

- **Forget Gate**: Determines what information from the previous state should be discarded.
- **Input Gate**: Updates the cell state with new information.
- **Cell State**: Maintains memory across time steps.
- **Output Gate**: Controls the next hidden state.

As shown in Figure 20, the Long Short-Term Memory (LSTM) model effectively captures sequential dependencies by regulating information flow through gating mechanisms. This structure enables it to retain long-term dependencies while mitigating vanishing gradient issues in recurrent neural networks (RNNs).

2) *Advantages of LSTM in IDS*:

- **Handles Sequential Data Efficiently**– Ideal for analyzing time-series network traffic data.
- **Detects Advanced Persistent Threats (APTs)**– Can identify stealthy attacks that evolve over time.
- **Robust to Noisy Inputs**– Learns relevant features even when training data contains inconsistencies.

3) *Limitations of LSTM in IDS*:

- **Computationally Intensive**– Requires substantial processing power and memory.
- **Training Complexity**– Longer training times compared to simpler ML models.
- **Vulnerability to Adversarial Attacks**– Susceptible to adversarial perturbations in network traffic data.

4) *Generative Adversarial Networks (GANs) for IDS*: GANs have gained attention in IDS due to their ability

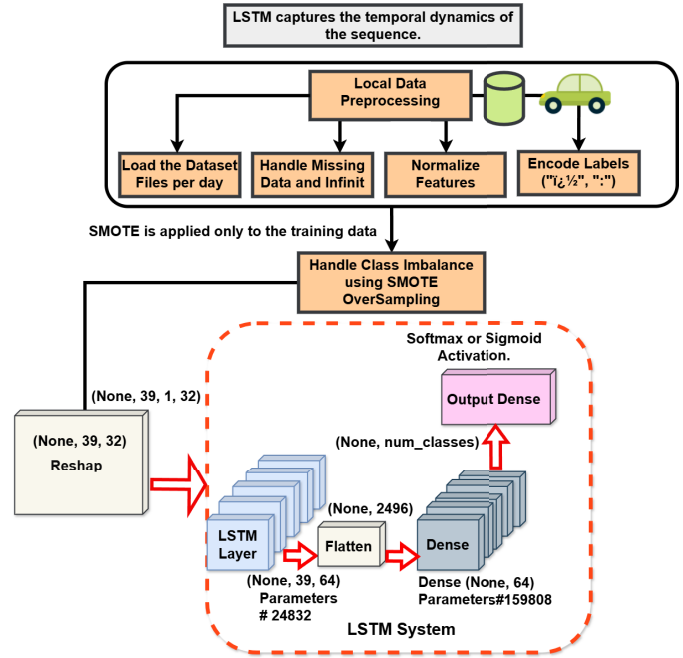


Fig. 20. LSTM-based IDS for V2X Networks.

to generate realistic synthetic attack traffic and improve anomaly detection performance. GANs consist of two neural networks—a Generator (G) and a Discriminator (D)—trained in an adversarial manner as shown in Figure 21.

B. GAN Architecture

- **Generator (G)**: Learns to create synthetic traffic samples resembling real network traffic.

TABLE XVII
KEY FINDINGS AND PERFORMANCE METRICS OF FEDERATED LEARNING IN V2X SECURITY (PART 2)

Study	Key Findings	Metrics	Scenario
Huang et al. [173]	Developed FED-IoV using MobileNet-Tiny for real-time detection	98.51% (CAN), 97.74% (CICIDS2017), latency <10ms	FL-IDS for IoV with fast detection
Renda et al. [174]	Merged XAI and FL to improve trust and transparency	Enhanced interpretability and decentralized security	FL-XAI for secure 6G-V2X networks
Chen et al. [172]	Proposed VAN-FED-IDS with edge-assisted FL aggregation	Reduced training duration, low overhead	RSU-edge-cloud FL in VANETs
Doostmohammadian et al. [178]	Observer-based redundancy for robust CAV platooning	Accurate state estimation despite failures	Observer-based FL for platoon stability
Zabihi et al. [179]	Fault-tolerant FL-IDS using hybrid model-data estimation	Sensor noise mitigation, anomaly accuracy improvement	Observer-enhanced FL-IDS in dynamic V2X
He et al. [180]	Secure distributed filtering for FL against sensor attacks	IDS robustness through attack-resilient filtering	FL model protection via secure aggregation
Taslimasa et al. [197]	Proposed ImageFed/FedCNN for CAN-bus IDS with FL	Resilient to cyberattacks, effective on IID data	FL-IDS models for CAN traffic security
Zhou et al. [168]	Introduced quantum-safe FL with post-quantum crypto	Quantum-resilient collaborative security	Post-quantum FL for secure V2X

- Discriminator (D): Differentiates between real and fake traffic samples.
- Both models iteratively improve through adversarial training.

1) Advantages of GANs in IDS:

- Enhances Anomaly Detection – Generates synthetic attack data to improve model generalization.
- Detects Zero-Day Attacks – Can identify novel attack patterns without explicit prior knowledge.
- Reduces Data Imbalance Issues – Addresses the problem of limited labeled attack samples in training datasets.

2) Limitations of GANs in IDS:

- Mode Collapse– The generator may produce a limited variety of attack patterns, reducing effectiveness.
- High Training Complexity – Requires extensive computational resources and careful hyperparameter tuning.
- Difficulty in Interpretability – Harder to analyze GAN decisions compared to rule-based or ML-based models.

3) *Comparison of LSTM and GAN for IDS:* Deep learning-based IDS provides powerful capabilities for securing V2X networks against cyber threats. LSTM networks excel in sequential traffic analysis, while GANs enhance anomaly detection by generating synthetic attack samples. However, computational cost, adversarial robustness, and interpretability must be addressed. Future research should focus on hybrid models that integrate LSTM and GAN architectures for enhanced IDS performance.

Deep learning is a powerful approach for classifying and detecting network intrusions. By leveraging multiple hidden layers, deep learning models can extract complex patterns and hierarchical features, enabling them to address challenging classification tasks more accurately than conventional machine learning methods [199]. Table XVIII compares LSTM and GAN-based intrusion detection systems in terms of learning

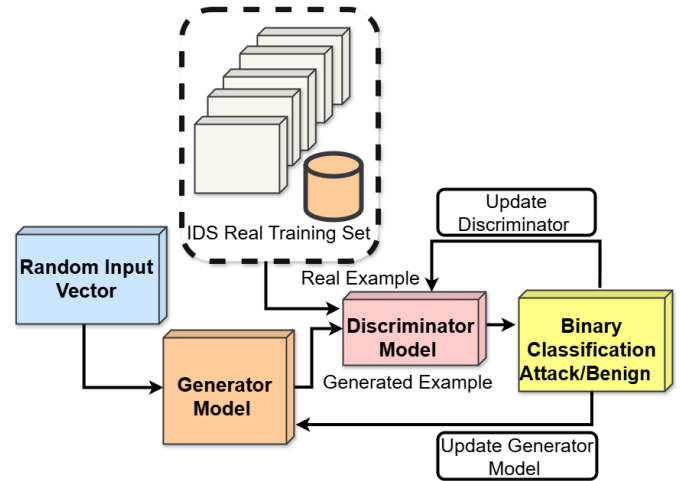


Fig. 21. GAN-based IDS for V2X Networks.

mechanisms, zero-day attack handling, interpretability, and computational cost.

C. Comparative Analysis of Deep Learning Models for IDS

Deep learning has been widely used in intrusion detection, particularly through autoencoders and Generative Adversarial Networks (GANs) [200], [201]. Autoencoder-based approaches rely on reconstruction algorithms to differentiate between normal and anomalous behavior. These models are typically trained on normal data samples, enabling them to learn the underlying data distribution and reconstruct inputs with high fidelity. Anomalous data points, which deviate significantly from the learned representation, exhibit higher reconstruction loss and are flagged as potential threats [202].

Several studies have refined these approaches to enhance intrusion detection performance. In [203], an analysis frame-

TABLE XVIII
COMPARISON OF LSTM AND GAN FOR INTRUSION DETECTION SYSTEMS

Feature	LSTM	GAN
Primary Use	Time-series anomaly detection	Synthetic attack generation and anomaly detection
Learning Mechanism	Sequential learning using memory cells	Adversarial learning between generator and discriminator
Handling Zero-Day Attacks	Limited ability to detect new attack types	Strong ability to generate new attack samples
Training Complexity	High (due to long sequences)	Very High (requires adversarial training)
Interpretability	Moderate (clear temporal dependencies)	Low (closed-box nature)
Computational Cost	High	Very High

work combined with a Denoising Autoencoder (DAE) was used to remove noise-related artifacts that could degrade classification accuracy. Similarly, Variational Autoencoders (VAE) incorporated probabilistic modeling techniques to capture latent feature distributions, improving anomaly detection by incorporating Gaussian priors in the reconstruction loss function.

GANs, originally introduced for computer vision applications, have been successfully adapted for IDS. AnoGAN, the first IDS model based on GANs, demonstrated promising results by generating realistic attack samples to improve classifier robustness. This was further refined in f-AnoGAN, which incorporated an encoder network to optimize the mapping of input data to latent representations, thereby reducing computational complexity [149]. Other advancements, such as GANomaly, enhanced anomaly detection by introducing dual encoder networks—one for learning the latent space of actual data and another for reconstructed data. Efficient GANs, inspired by Bidirectional GANs (BiGANs), demonstrated improved anomaly detection capabilities across benchmark datasets such as KDD Cup and image-based intrusion datasets [204]. A further refinement involved stabilizing model training by integrating autoencoder-style learning into BiGAN frameworks [92].

The effectiveness of deep learning-based IDS has been validated in multiple studies. Thakkar et al. [205] conducted a comprehensive survey on ML and DL techniques for intrusion detection in IoT environments. Their findings underscored the growing reliance on DL models for detecting sophisticated cyber threats. Aloqaily et al. [206] introduced a dataset derived from a simulated LAN of the U.S. Air Force over nine weeks, capturing various network intrusions and attack types. Their dataset, based on the KDD99 benchmark, provided valuable insights into intrusion detection model performance.

He et al. [207] proposed a cybersecurity framework incorporating a newly developed dataset, CAV-KDD, which builds on the existing KDD dataset to classify vulnerabilities specific to CAV systems. Their framework includes a Unified Modeling Language (UML)-based architecture inspired by the UK CAV cybersecurity framework, which systematically

evaluates attack vectors and countermeasures for secure CAV communications.

Thapa et al. [208] developed a novel deep learning-based Network Intrusion Detection System (NIDS) by evaluating ML and DL models across various datasets. Their study highlighted the limitations of static signature-based IDS and emphasized the need for adaptive security solutions. The proposed model, which integrated convolutional neural networks (CNN) with embedding layers and Long Short-Term Memory (LSTM) networks, achieved superior detection accuracy when tested on the CIC-IDS2017 dataset. Their findings demonstrated that hybrid deep learning architectures significantly outperformed traditional machine learning classifiers.

Deep transfer learning (DTL) has also been explored for IDS applications. Vu et al. [209] introduced a DTL framework that leverages knowledge transfer across multiple IoT device datasets. Their approach incorporated two Autoencoder (AE) models: AE1, trained in a supervised setting using labeled data from source domains, and AE2, which learned unsupervised representations from target datasets without labeled information. This method enabled IDS systems to generalize better to new attack scenarios without requiring extensive retraining.

Samy et al. [210] proposed a distributed IDS framework optimized for IoT environments. Their model leveraged DL techniques to achieve high detection rates for diverse cyber-attacks. By deploying detection mechanisms on fog nodes, they reduced the latency associated with centralized cloud-based IDS solutions. They introduced a robust distributed attack detection framework leveraging deep learning techniques to enhance intrusion detection in IoT environments. Their system was deployed on fog nodes, which offered proximity to edge devices, substantial computational capacity, and a distributed architecture, enabling real-time threat detection. Roopak et al. [211] addressed the challenge of high-dimensional network data by introducing a feature selection algorithm based on multiple optimization techniques. Their approach improved IDS performance in detecting Distributed Denial of Service (DDoS) attacks in IoT networks.

In addition to centralized and federated learning approaches, Taslimasa et al. [212] introduced ImageFed and FedCNN, two distributed intrusion detection models for V2X security. These approaches leveraged federated learning to detect attacks while preserving data privacy across connected vehicles. They developed two privacy-preserving distributed IDS solutions, ImageFed and FedCNN, tailored for V2X networks. ImageFed was optimized for the CAN bus environment, whereas FedCNN employed a deep neural network to detect anomalies. A comparative analysis revealed that ImageFed demonstrated higher robustness, although its effectiveness was limited to IID datasets. Similarly, Roopak et al. [213] proposed an optimized feature selection (FS) approach for detecting Distributed Denial of Service (DDoS) attacks in IoT networks. Their method effectively reduced data dimensionality, improving IDS performance and classification accuracy.

Bovenzi et al. [214] proposed an intrusion detection framework that integrated IoT security solutions with machine learning algorithms, addressing key limitations of existing intrusion detection models while offering a scalable approach to network security.

Bangui et al. [215] applied a hybrid data methodology to detect malicious activities in V2X IDS systems. Their model integrated multiple data sources to enhance detection accuracy in diverse vehicular environments, including V2X and AAV networks. While their research highlighted key challenges in intrusion detection, such as adversarial attacks and concept drift, it lacked feasibility analysis. As emerging technologies such as beyond-5G continue to shape next-generation networks, Tang et al. [48] emphasized the importance of proactive security mechanisms to enhance network resilience.

Al-Khateeb et al. [216] proposed an anomaly detection strategy aimed at preventing cyberattacks on connected vehicles. The dynamic nature of evolving cyber threats limits the efficacy of traditional supervised ML models in real-time V2X scenarios, necessitating the use of reinforcement learning (RL) and imitation learning techniques. Xu et al. [217] and Liang et al. [53] demonstrated that RL-based detection mechanisms can dynamically improve IDS performance by learning from real-time interactions with network threats. Salehi and Rashidi [218] highlighted the challenges posed by high-dimensional V2X mobility data, which includes variables such as vehicle speed, geolocation, and movement trajectory. RL-based models can effectively manage such dynamic data distributions, even in scenarios with limited labeled training samples [219], [220]. More importantly, RL detection mechanisms can autonomously adapt to novel attack vectors, enhancing network security over time [221].

Sedar et al. [222] developed an RL-powered intrusion detection framework capable of identifying multiple variants of misbehavior attacks in the VeReMi dataset. Their model analyzed mobility patterns, including real-time vehicle positioning and speed vectors, to detect anomalous driving behavior. Jameel et al. [223] introduced a deep Q-learning-based power allocation strategy designed to optimize secrecy performance while ensuring compliance with quality of service (QoS) requirements, particularly in the presence of eavesdropping threats.

Alladi et al. [224] proposed two deep learning-based approaches for classifying misbehaving vehicles within edge computing environments. Their subsequent study [225] introduced a Mobile Edge Computing (MEC)-based intrusion detection framework integrating deep learning models for threat analysis in vehicular networks. However, MEC-based security solutions face challenges related to computation offloading in dynamic multi-vehicle scenarios, which can undermine system stability due to decision-making uncertainties [226]. A fundamental limitation of these approaches is the lack of a global view, which may result in the failure to detect specific attack types. To address this, hybrid detection strategies that combine local training with centralized anomaly detection have been proposed [227]. The deployment of anomaly detection mechanisms at various network layers—including edge, cloud, and MEC servers—further enhances detection accuracy and system scalability [228].

In dynamic V2X settings, where training datasets may be incomplete or infeasible due to resource constraints, online learning techniques provide a viable solution. Bosman et al. [229] employed an online learning strategy to incrementally train neural networks for anomaly detection on resource-constrained in-vehicle devices. Similarly, Zhang et al. [230] introduced an adaptive anomaly detection model that continuously updates itself in response to deviations from expected vehicular behavior. However, given that normal driving patterns often vary across geographical regions, V2X attack detection algorithms must remain adaptable to evolving threat landscapes. Active learning, a specialized online learning technique that prioritizes the selection of relevant training samples, has been shown to enhance IDS effectiveness in highly dynamic vehicular networks [231].

Karthiga et al. [232] designed an intelligent intrusion detection system (IDS) tailored for VANETs by integrating deep learning and machine learning techniques. Their framework utilized an adaptive neuro-fuzzy inference system (ANFIS) to detect known attack patterns and convolutional neural networks (CNNs) to identify emerging threats. By incorporating a modified LeNet (MLNET) architecture, their IDS demonstrated high precision and recall rates in detecting a wide range of cyberattacks, including botnet intrusions, distributed denial of service (DDoS) attacks, brute force attempts, and port scanning. Experimental evaluations using the CIC-IDS 2017 dataset confirmed the superior accuracy of their approach in comparison to conventional IDS methodologies.

Sinha et al. [233] applied deep learning techniques to develop a novel detection model for coronary artery disease, demonstrating the versatility of CNNs in diverse classification tasks, including medical diagnosis and network intrusion detection. These advancements underscore the potential of AI solutions in securing V2X networks, as well as their applicability across multiple domains requiring high-precision anomaly detection.

Wazid et al. [234] present a deep learning-based malware attack detection framework for IoT-enabled Intelligent Transportation Systems (ITS), termed SDLMA-IITS. The study highlights the increasing reliance on IoT devices for transportation network efficiency, safety, and monitoring while

emphasizing the security risks posed by malware attacks. The proposed framework incorporates Explainable AI (XAI) techniques to enhance malware detection, ensuring transparency in decision-making. The authors provide a comparative security analysis of SDLMA-IITS against existing schemes and discuss its practical implementation in real-world ITS environments. However, the study focuses primarily on malware detection and does not extensively cover adversarial AI attacks, dataset benchmarking, or real-time deployment challenges within vehicular networks.

Table XIX and Table XX summarize the key findings in deep learning based IDS in V2X communication.

XII. CHALLENGES IN REAL-WORLD DEPLOYMENT OF ML-IDS IN V2X SECURITY

ML-IDS have been widely explored for securing V2X networks. However, their real-world deployment remains challenging due to issues such as scalability, adversarial robustness, computational overhead, and compliance with V2X security standards. While many ML-IDS models achieve high detection accuracy in controlled simulation environments, their effectiveness in real-world vehicular networks remains an open challenge due to unpredictable environmental factors, heterogeneous network conditions, and the presence of sophisticated cyber threats.

This section categorizes and discusses existing ML-IDS validation methodologies, distinguishing between simulated and real-world implementations. In addition, it highlights deployment challenges and the necessity of real-world experimentation to bridge the gap between theoretical models and practical applications.

A. Simulation-Based Validation of ML-IDS

Numerous studies have relied on simulation environments to evaluate ML-based security models in vehicular networks due to the complexity and cost of real-world implementations. Simulations allow for controlled testing of security models in various traffic, mobility, and attack scenarios, providing essential information about their performance. However, they fail to capture the dynamics of the real-world network, latency constraints, and adversarial attack scenarios encountered in V2X communications.

Several studies have relied on simulation environments to evaluate ML-based security models in vehicular networks.

Iqbal et al. [235] explored cybersecurity threats in (VANETs), focusing on replay and bogus information attacks. They used the **Eclipse MOSAIC** simulation framework to model real-world attack scenarios, demonstrating that replay attacks—where attackers retransmit recorded emergency messages—can severely disrupt traffic flow. Similarly, bogus information attacks manipulate vehicle navigation by injecting false congestion warnings, leading to inefficient routing. Their study contributes an open-source dataset to support machine learning-based anomaly detection in VANET security. Rajak et al. [236] addressed security challenges in 5G-enabled V2X networks by developing a deep learning-based attack detection model (ADM) utilizing a stacked LSTM framework.

Their model, trained on the AIoT-SoL dataset, comprising 5,134,728 instances with 85 attributes, demonstrated exceptional performance:

- **Accuracy:** 99.8%
- **Precision:** 99.2%
- **F1-score:** 99.1%

The model effectively identifies a range of cyber threats, including:

- **Man-in-the-Middle (MitM) attacks:** Detection Rate: **98.8%**, Precision: **98.9%**.
- **Denial-of-Service (DoS) attacks:** Detection Rate: **99.1%**, Precision: **99.3%**.
- **Spoofing, malware, and Sybil attacks**, reinforcing the robustness of V2X security.

Jiménez et al. [237] introduced a deep learning framework to secure autonomous vehicle networks (AVN), focusing on detecting distributed denial of service (DDoS), black hole and selective forwarding attacks. Their study employed deep autoencoders (AEs) and LSTM-Recurrent Neural Networks (LSTM-RNNs), integrating sliding window techniques and dynamic feature extraction for enhanced detection. The model was validated in real-world conditions using synthetic attack scenarios, achieving:

- **Detection rate:** 95%
- **False alarm rate:** 3%
- **Detection probability:** 92%
- **Overall accuracy:** 94%

A gradient-boosted model was also incorporated to optimize network resilience, ensuring timely retraining of detection models.

Raja et al. [106] developed the Efficient Trajectory Anomaly Detection and Classification (ETADC) framework, a machine learning-based approach tailored to anomaly detection in a 6G-V2X environment. The ETADC framework employs the Deep Deterministic Policy Gradient (DDPG) algorithm to analyze vehicular movement parameters such as speed, distance, direction, and travel time, enabling the identification and classification of anomalous behaviors. The study utilized simulation data generated from SUMO and OMNET++, which incorporated dynamic vehicular characteristics and GPS information. Key performance metrics, including accuracy, precision, recall, loss, True Positive Rate (TPR), and False Positive Rate (FPR), were used for evaluation. The ETADC model demonstrated superior detection capabilities, achieving an accuracy of 97% and exhibiting the lowest false positive rate when compared to existing models such as Hybrid CNN-VAE and SpatioTemporal Autoencoders. The framework effectively detected a range of anomalous behaviors, including abrupt speed fluctuations, incorrect direction movements, inconsistencies in travel duration, and lane violations. Future work aims to expand the model's capability by incorporating long-distance trajectory data, refining anomaly classification, and exploring alternative distance metrics to improve detection precision.

Sousa et al. [238] investigated the deployment of an Intrusion Detection System (IDS) for 5G-enabled Internet of Vehicles (IoV), focusing on detecting Flooding attacks in vehicular networks. Given the increasing reliance on 5G

TABLE XIX
SUMMARIZING KEY FINDINGS OF IDS-BASED ADVANCED DEEP-LEARNING METHODS FOR V2X COMMUNICATION (PART 1)

Category	Study	Approach	Dataset/Results	Key Findings
DL Overview	General DL [199]	Feature extraction, classification	KDD Cup, image intrusion datasets	Improved adaptability and accuracy
GAN-Based Models	GANomaly, BiGAN [200], [201]	Dual encoders, reconstruction loss	KDD Cup and synthetic datasets	Enhanced anomaly detection using GANs
Survey	Thakkar et al. [205]	ML/DL survey for IDS	IoT intrusion datasets	DL is more suited for evolving threats
Simulated Dataset	Aloqaily et al. [206]	LAN traffic IDS dataset	9-week Air Force LAN, KDD99	Benchmark for evaluating multiple intrusions
Cybersecurity for CAVs	He et al. [207]	CAV-KDD with UML model	KDD-based IDS for AVs	New vulnerabilities for CAVs addressed
CNN+LSTM Hybrid	Thapa et al. [208]	Embedding + CNN + LSTM	CIC-IDS2017	Hybrid DL outperforms ML in V2X
Deep Transfer Learning	Vu et al. [209]	DTL with 2 AEs	Source + target domain IoT datasets	Generalization across unseen threats
Fog-based IDS	Samy et al. [210]	DL with fog node deployment	Distributed IDS architecture	Real-time detection, reduced latency
Feature Optimization	Roopak et al. [211]	Multi-objective feature selection	DDoS detection in IoT	Improved classification, reduced overhead
FL for CAN bus	Taslimasa et al. [212]	ImageFed, FedCNN models	CAN environment, FL-based	Privacy-preserving distributed detection
Hybrid IDS	Bangui et al. [215]	Multi-source hybrid detection	UAV, V2X scenarios	Improved detection via diverse data sources
RL-Based IDS	Al-Khateeb et al. [216]	Adaptive reinforcement learning	Dynamic real-time V2X threats	Learns from evolving cyber behaviors
RL for V2X	Salehi and Rashidi [218]	RL for high-dimensional V2X	Vehicle trajectory and speed	Effective with limited labeled data
RL for VeReMi	Sedar et al. [222]	RL on mobility traces (speed, pos.)	VeReMi dataset	Detects misbehavior, vehicle anomalies
Q-Learning + QoS	Jameel et al. [223]	Deep Q-Learning	Secure power allocation	Enhances secrecy in presence of eavesdroppers
MEC IDS	Alladi et al. [224], [225]	DL for edge devices	MEC + vehicular networks	High precision, but needs stable offloading
Online Learning	Bosman et al. [229]	Incremental DL models	On-device adaptive training	Good for constrained resources

technology in smart cities, IoT, and edge computing applications, the security of vehicular networks presents significant challenges due to decentralized management and high network mobility. The study employed the Network Simulator 3 (NS-3) to simulate vehicular scenarios and generated four datasets by varying the number of nodes, attackers, and mobility patterns extracted from the Simulation of Urban Mobility (SUMO). The IDS was designed using machine learning (ML) techniques, including decision trees, random forests, and multilayer perceptron models. The evaluation results demonstrated that decision trees and random forests

performed exceptionally well, achieving F1 scores of 1.00 and 0.98, respectively, effectively classifying the Flooding attack in the 5G vehicular environment. The findings emphasized the importance of dataset diversity in improving model robustness while also highlighting that more complex ML algorithms, such as neural networks, do not necessarily outperform simpler models like decision trees. This research underscores the applicability of ML-based IDS for securing next-generation vehicular networks, ensuring enhanced resilience against cyber threats in dynamic and high-mobility environments.

TABLE XX
SUMMARIZING KEY FINDINGS OF IDS-BASED ADVANCED DEEP-LEARNING METHODS FOR V2X COMMUNICATION (PART 2)

Category	Study	Approach	Dataset/Results	Key Findings
Adaptive Learning	Zhang et al. [230]	Continual self-updating models	Anomaly response adaptation	Real-time updates improve accuracy
ANFIS + CNN Hybrid	Karthiga et al. [232]	Hybrid DL for V2X attacks	CIC-IDS2017	Detects botnets, DDoS, brute-force, port scans
Medical Diagnosis (Transfer Study)	Sinha et al. [233]	CNN-based diagnosis model	Coronary disease dataset	DL transfer to non-network tasks validated
Malware Detection	Wazid et al. [234]	SDLMA-IITS with Explainable AI	IoT malware datasets	High performance, transparent threat analysis

Korba et al. [239] investigated the security and privacy vulnerabilities of Connected and Automated Vehicles (CAVs) operating within 5G and Beyond (5GB) networks. Traditional machine learning-based intrusion detection methods often struggle with detecting zero-day attacks and require extensive labeled datasets, posing privacy and scalability challenges. To overcome these limitations, the authors propose a federated learning-based intrusion detection system (IDS) that utilizes deep autoencoders to identify cyber threats based exclusively on benign network traffic patterns. Federated learning enables multiple CAVs to collaboratively train detection models without sharing raw data, thus preserving privacy while minimizing communication overhead. The study utilized the VDoS dataset, which encompasses benign and malicious traffic from diverse vehicular environments, including urban, rural, and highway settings. The dataset included three primary DoS attack types: UDP Flood, SYN Flood, and Slowloris. Performance evaluations, conducted using PyTorch in a cloud-based environment, demonstrated high detection accuracy with minimal false positives. The federated learning system, orchestrated through a Multi-access Edge Computing (MEC) server, required fewer training rounds compared to centralized models while achieving comparable performance. The proposed IDS exhibited a detection accuracy of 99.99%, an F1-score of 99.99%, and a false positive rate of only 0.01%, particularly excelling in identifying SYN Flood attacks due to its reliance on TCP-related features. Notably, training the IDS with data from only three CAVs achieved a performance level similar to centralized training while reducing overall training time by approximately 30%.

Magsi et al. [240] discuss the increasing relevance of Named Data Networking (NDN) in VANETs due to its capabilities such as in-network content caching, name-based routing, and mobility support. However, NDN faces several security, privacy, and routing challenges. Specific threats such as Content Poisoning Attacks (CPA) can introduce malicious content, leading to potential road accidents. To address these issues, the authors propose a threefold approach: (i) a machine learning-based reputation evaluation mechanism to detect CPA attackers, (ii) a decentralized blockchain system to ensure privacy, and (iii) a push-based content dissemination model using Publish-Subscribe (Pub-Sub) to improve routing

efficiency. The study utilizes the BurST-Australian dataset for Misbehavior Detection (BurST-ADMA), which contains 207,315 messages, including 179,126 legitimate and 28,189 malicious messages. The dataset includes multiple entities such as bicycles, buses, motorcycles, pedestrians, trucks, and passenger vehicles. Data preprocessing consolidates seven attack types into a binary classification of attackers (1) and legitimate (0) entities. The dataset is split into 70% for training and 30% for testing, with a 10-fold cross-validation employed to Explain performance. Five hybrid machine learning classifiers are evaluated: Logistic Regression (LR), Decision Tree (DT), K-Nearest Neighbors (KNN), Random Forest (RF), and Gaussian Naive Bayes (GNB). The Random Forest classifier achieves the highest accuracy rate of 100%, with an F1 score of 1.00, indicating perfect detection. The DT and KNN classifiers also show high accuracy, while LR and GNB provide satisfactory results.

Twardokus et al. [241] examine vulnerabilities in 5G Cellular Vehicle-to-Everything (5G C-V2X) technology, particularly focusing on the physical-layer attributes and decentralized Medium Access Control (MAC)-layer scheduling algorithm. The authors identify and exploit these vulnerabilities to develop two stealthy denial-of-service (DoS) attacks: (i) targeted sidelink jamming and (ii) sidelink resource exhaustion. The targeted sidelink jamming attack predicts and jams specific Basic Safety Messages (BSMs) by interfering with their control information, whereas the sidelink resource exhaustion attack disrupts the scheduling algorithm, reducing the overall packet delivery ratio (PDR). The study employs a comprehensive dataset from hardware testbeds using Universal Software Radio Peripherals (USRPs), commercial C-V2X devices, and extensive network and roadway simulations. Evaluation metrics include packet delivery rate, detection rate, and false positive rate. The attacks demonstrate a severe impact, with targeted jamming causing significant disruptions. The detection techniques developed achieve an 80% true positive rate with a 25% false positive rate for targeted jamming. The mitigation strategies, such as leveraging NR-V2X's aperiodic transmissions and shortening the semi-persistent scheduling listening period, successfully neutralize the attacks, reducing the impact of resource exhaustion by up to 80%.

Begum et al. [242] proposed the AI-based Sensor Attack Detection and Classification (SADC) framework for autonomous vehicles in 6G-V2X environments. Their approach integrates GPS and LiDAR-based anomaly detection with a Pattern-based Attack Classification (PAC) algorithm to identify malicious sensor activities based on four deviation types: instantaneous, constant, biased, and gradual drift anomalies. Their results indicate that SADC outperforms existing frameworks by 0.98% in attack detection and classification accuracy. The detection model comprises three Convolutional Neural Network (CNN) layers and two Long Short-Term Memory (LSTM) layers, optimized using L2 regularization and the Adam optimizer to mitigate overfitting. The evaluation was conducted on a dataset with a 75% training, 10% validation, and 15% testing split, ensuring robust performance.

Additionally, Begum et al. analyzed sensor spoofing attacks in which adversaries manipulate LiDAR data using laser interference. The SADC framework was tested against sophisticated cyber-attacks, including:

- **False Data Injection (FDI) Attack:** Alters sensor outputs to mislead the vehicle's control system.
- **Stealthy Attack:** Adjusts sensor readings to mimic expected motion patterns, making detection challenging.
- **Replay Attack:** Replaces real-time sensor observations with recorded data, deceiving the system with seemingly legitimate input.

I Prathiba et al. [243] proposed a Hybrid Deep Anomaly Detection (HDAD) framework for 6G-V2X cybersecurity, combining Multi-Agent Reinforcement Learning (MARL) with Maximum Entropy Inverse Reinforcement Learning (MaxEntIRL) to enhance anomaly detection. Their framework was validated using the **NS-3 simulator with the TeraSim extension**, testing **100 AVs, two 6G base stations, and ten Roadside Units (RSUs) over 6000 seconds**. Key results include:

- **Anomaly detection accuracy:** 98%
- **False prediction rate:** 4%
- **Processing delay:** Below 0.43s, outperforming traditional IDS approaches.

Table XXI provides a comparative analysis of ML-based intrusion detection systems (IDS) evaluated using simulation environments in V2X security. This table includes key performance metrics such as accuracy, precision, and F1-score for various methodologies, demonstrating their effectiveness under controlled conditions. While these results highlight the potential of ML-IDS models, they often fail to account for real-world network conditions, which significantly impact detection accuracy and adversarial robustness.

B. Experimental Validation: Real-World Deployments

Unlike simulation-based studies, real-world deployments provide a more realistic evaluation of ML-IDS models by considering factors such as environmental conditions, network delays, and actual vehicular traffic dynamics. However, real-world implementations are significantly more complex, requiring robust architectures, reliable hardware components, and seamless integration with existing vehicular systems.

While most research on intrusion detection systems (IDS) in vehicular networks relies on simulated data, Sousa et al. [244] conducted a real-world evaluation of flooding attacks using physical MK5 On-Board Unit (OBU) devices. This study represents one of the first publicly available datasets collected using IEEE 802.11p-based vehicular communication, providing valuable insights into real-world attack dynamics.

The experiment utilized three OBUs equipped with Dedicated Short-Range Communications (DSRC) radios to facilitate Vehicle-to-Vehicle (V2V) communication. Two main flooding attacks were tested:

- **SYN Flooding Attack:** Overwhelms a target node with TCP SYN packets to deplete system resources.
- **UDP Flooding Attack:** Generates excessive UDP traffic to degrade network performance.

To ensure realistic data collection, tcpdump was used to log all received packets and 200 experimental runs were conducted under varying transmission intervals. The researchers tested decision trees, random forests, and multilayer perceptron models to classify attack traffic, revealing the complexity of real vehicular network anomalies. The decision tree classifier was particularly effective, highlighting that traffic fluctuations in real-world conditions introduce additional detection challenges compared to controlled simulations.

A crucial aspect of this study was the comparison of real-world dataset performance against existing simulated datasets. The authors noted a significant gap between IDS performance in simulations and real vehicular environments. Previous studies using network simulators (e.g., NS-3) reported near-perfect classification metrics, often exceeding 99% F1-score. However, these datasets lacked real-world variability, making them potentially less reliable for practical deployment.

In contrast, the dataset from Sousa et al. demonstrated more realistic classification performance, with the Support Vector Machine (SVM) model achieving an F1-score of 0.9886, while the Random Forest and Decision Tree models attained 0.99989 and 0.99994, respectively. These results align closely with prior simulated dataset evaluations, raising key concerns about the generalizability of IDS models trained solely on synthetic data. These results emphasize the need for more real-world vehicular IDS datasets and highlight the importance of incorporating realistic attack scenarios to bridge the gap between theoretical ML models and practical cybersecurity solutions in connected vehicles.

Zhang et al. [245] conducted a real-world study on enhancing data transmission security and performance in Vehicular Ad-Hoc Networks (VANETs) by integrating machine learning techniques with 5G communication networks. Their research addressed the challenges of big data transfer in vehicular environments, particularly under varying traffic conditions. The study was conducted over three months in Berlin, covering highways, suburban roads, and urban roads, each representing different levels of traffic congestion, which directly impacted vehicle speed and inter-vehicle distance. The proposed approach utilized Support Vector Machine (SVM) algorithms and position-based routing protocols to detect non-line-of-sight (NLoS) conditions in real time, ensuring stable

TABLE XXI
SIMULATION-BASED ML METHODS FOR V2X INTRUSION DETECTION

Method	Model	Dataset	Acc (%)	Prec (%)	F1 (%)	Challenges	Ref
Replay Attack De- tection	RF	Eclipse MOSAIC (VANETs)	N/A	N/A	N/A	Dataset for anomaly detection	[235]
Stacked LSTM	Deep Learning	AIoT-SoL (5M+)	99.8	99.2	99.1	High GPU mem- ory; long training	[236]
Stacked (MitM)	LSTM	Deep Learning	AIoT-SoL (5M+)	98.9	98.8	High GPU mem- ory; long training	[236]
Stacked (DoS)	LSTM	Deep Learning	AIoT-SoL (5M+)	99.1	99.3	High GPU mem- ory; long training	[236]
Autoencoder LSTM-RNN	+	Deep Learning	AVN Attacks (Syn- thetic)	94.0	92.0	False alarm rate: 3%	[237]
DDPG	Reinforcement Learning	SUMO, OMNET++	97.0	96.0	96.5	Needs real-time tuning	[106]
Decision Tree	ML-Based IDS	NS-3 (SUMO)	100.0	100.0	100.0	Requires diverse datasets	[238]
Random Forest	ML-Based IDS	NS-3 (SUMO)	98.0	98.0	98.0	Simpler models work well	[238]
Deep Autoencoder (FL)	FL	VDoS Dataset (Ur- ban, Hwy)	99.99	99.99	99.99	Needs large labeled data	[239]
Random Forest	ML-Based IDS	BurST Dataset	100.0	100.0	100.0	Privacy, scalability issues	[240]
KNN	ML-Based IDS	BurST Dataset	98.0	98.0	98.0	Moderate accuracy	[240]
Logistic Regression	ML-Based IDS	BurST Dataset	92.0	92.0	92.0	High variance in performance	[240]
Gradient-Boosted Trees	ML-Based IDS	Custom AV Dataset	N/A	N/A	N/A	Ensures model re- training	[237]
Hybrid CNN-VAE	Hybrid Deep Learning	SUMO, OMNET++	95.0	N/A	N/A	High computa- tional demand	[106]
SVM	Supervised Learning	VANET Dataset	92.5	91.0	91.7	High computa- tional cost	[241]
CNN-LSTM	Deep Learning	LiDAR-GPS Data	98.0	97.5	97.0	Sensor reliability issue	[242]
HDAD	MARL + Max- EntIRL	NS-3 (TeraSim)	98.0	97.5	97.2	Processing delay in real-time	[243]

and secure data transmission without network performance degradation. The combination of 5G cellular networks with alternative data transmission channels allowed for optimized packet delivery, particularly in congested urban environments. The findings demonstrated that machine learning-based optimization significantly improved VANET performance, especially in areas with high traffic density. The proposed model effectively identified NLoS conditions, reducing packet loss and improving real-time communication reliability. The study also highlighted that high traffic density negatively affected data transmission, but the developed method successfully adapted to these challenges, ensuring continuous network performance. The results suggest that this AI- approach is suitable for real-world 5G-VANET applications, particularly in urban traffic management and autonomous driving systems, reinforcing the role of machine learning in securing and optimizing vehicular networks.

Salazar et al. [246] proposed the V2X Core Anomaly Detection System (VCADS), designed to enhance the security of V2X communications by identifying misbehavior through physics-based models. VCADS operates by constraining val-

ues within Basic Safety Messages (BSMs) transmitted by vehicles and detecting anomalies through comparative validation. The framework was evaluated against various attack scenarios, including false Forward Collision Warning (FCW), false Emergency Electronic Brake Light (EEBL), false Left Turn Assist (LTA), and false Intersection Collision Warning (ICW). The detection mechanism incorporates two primary components: Field Validation, which verifies the accuracy of individual BSM fields using predefined physical constraints, and Cross-Validation, which derives measurements from independent fields to Explain consistency. The results showed that VCADS effectively detected anomalies across multiple real-world datasets, including those collected from Ann Arbor, Arlington, Tampa, and Wyoming, achieving an attack detection rate between 85% and 95% with minimal false positives. This system presents a promising approach for mitigating adversarial threats in vehicular networks, ultimately contributing to enhanced road safety and improved traffic management. Minawi et al. [247] introduced a machine learning-based Intrusion Detection System (IDS) aimed at securing the Controller Area Network (CAN) bus in modern vehicles, which

remains vulnerable to various cyber threats due to its lack of authentication mechanisms and limited bandwidth. Their proposed IDS integrates multiple machine learning algorithms, including Random Tree (RT), Random Forest (RF), Stochastic Gradient Descent (SGD), and Naive Bayes (NB), to effectively identify and mitigate malicious activities. The system was evaluated using the “Car Hacking Dataset” from the Hacking and Countermeasure Research Lab (HCRL), which contains real CAN bus messages and diverse attack scenarios. The experimental results demonstrated remarkable accuracy levels: 100% detection accuracy for Denial of Service (DoS) and impersonation attacks (drive gear and RPM gauge) using the Random Tree algorithm, 99.99% accuracy in identifying fuzzy injection attacks with both Random Tree and Random Forest models, and 100% accuracy in detecting DoS attacks using Naive Bayes. Furthermore, the IDS maintained its high detection performance when tested on previously unseen datasets, underscoring its robustness and generalizability.

A real-world implementation of FL-IDS was presented by Bhavsar et al. [248], who deployed edge devices like Raspberry Pi 4 as clients and NVIDIA Jetson Xavier as the aggregation server. Their system utilized Logistic Regression and a PCC-CNN model to detect anomalies in vehicular networks using the NSL-KDD and Car-Hacking datasets. The FL-IDS achieved notable accuracy improvements, with 94% for NSL-KDD and 99% for Car-Hacking, demonstrating its effectiveness over centralized IDS approaches. However, the study highlighted critical deployment challenges, including computational limitations on edge devices, communication overhead, and security risks such as poisoning attacks. Addressing these concerns requires further research into lightweight FL models, secure aggregation methods, and dynamic learning strategies to enhance real-world applicability in autonomous vehicle security. Despite the success of FL-IDS, Bhavsar et al. [248] identified several challenges:

- Raspberry Pi 4 has limited processing power, leading to higher latency in intrusion detection.
- Jetson Xavier provides better performance but still has memory constraints when handling large-scale vehicular data.
- FL requires frequent model updates between clients and servers, increasing communication overhead.
- Bandwidth limitations in V2X networks may impact model synchronization and real-time effectiveness.
- Poisoning Attacks: Malicious clients could tamper with model updates.
- Privacy Leakage: Despite decentralized learning, gradient leakage attacks pose risks.

To overcome these challenges, future research should focus on:

- Lightweight FL models optimized for edge computing.
- Secure aggregation methods to prevent poisoning and privacy attacks.
- Efficient communication strategies to minimize latency and bandwidth usage.
- Dynamic federated learning to handle heterogeneous attack patterns across different vehicular networks.

Huang et al. [173] introduced FED-IoV, an FL-based intrusion detection system for the Internet of Vehicles (IoV), addressing high computational demands and privacy concerns. The framework transforms vehicular communication traffic into images and employs MobileNet-Tiny for efficient feature extraction, enabling deployment on resource-constrained devices. Evaluations on the CAN-Intrusion and CICIDS2017 datasets demonstrated 98.51% and 97.74% accuracy, respectively, highlighting the system’s ability to handle imbalanced datasets effectively. Additionally, FED-IoV maintained a prediction latency of under 10 milliseconds on Raspberry Pi 4 (8GB), achieving superior real-time performance compared to conventional methods. The study underscores the practicality of FL-based IDS in enhancing IoV security while preserving data privacy.

Monteuuis et al. [249] addressed the challenge of misbehavior detection in Connected and Automated Vehicles (CAVs) by examining anomalies in vehicular dimension data within V2X messages. Unlike conventional studies that primarily focus on mobility-based anomalies, their work introduces a misbehavior detection framework that classifies road users based on vehicle type and dimension characteristics using machine learning techniques. The study compares four classification models: MinMax, Adaboost, Multilayer Perceptron (MLP), and Random Forest. Given that CAVs rely on vehicular communication and sensor fusion to enhance traffic efficiency, incorrect information transmitted by compromised Intelligent Transportation System (ITS) stations presents a significant risk. The authors model two types of misbehavior: (i) the communication deceiver, which transmits fraudulent dimension data to gain spatial advantages on the road, and (ii) the fusion manipulator, which injects misleading sensor data to disrupt perception systems. The proposed framework comprises offline and online phases, where offline training involves data acquisition, preprocessing, training, and testing, while online deployment enables real-time classification of V2X messages. The study utilized datasets representing pedestrians, motorcycles, and passenger vehicles, applying preprocessing techniques such as data cleaning, transformation, and reduction. Performance evaluation was conducted using key metrics including Accuracy, Informedness, Cohen Kappa, and Matthews Correlation Coefficient (MCC). The Random Forest model outperformed other classifiers, achieving an MCC score of 93%, demonstrating its effectiveness in distinguishing between misbehaving and non-misbehaving entities.

Kong et al. [250] explored cybersecurity challenges in Vehicle-to-Vehicle (V2V) communication within smart city environments, employing advanced machine learning techniques in the context of 6G networks. The study emphasizes the vulnerabilities present in modern vehicular communication systems, highlighting the absence of comprehensive security frameworks. To address these issues, the authors introduce a Reinforcement Quantile Spatial Convolutional Neural Network (RQSCNN) model for identifying and mitigating cyber threats. The model was trained on a dataset consisting of 20,000 preprocessed images, evenly distributed between normal and abnormal conditions, with 70% allocated for training

and 30% for validation and testing. The study incorporated five-fold cross-validation to enhance model generalizability. Experimental results demonstrated notable improvements, with classification accuracy increasing from 84% to 86% over ten training epochs and training loss reducing from 0.52 to 0.40. Comparative analysis revealed that the proposed approach achieved superior performance in various V2V network scenarios, demonstrating higher throughput (up to 93%), improved quality of service (QoS) (up to 86%), and lower latency (reduced to 55%) relative to baseline models. The findings underscore the potential of machine learning in strengthening the cybersecurity of V2V networks, with implications for enhancing traffic safety, optimizing transportation efficiency, and reducing environmental impact through improved vehicular coordination.

Boddupalli et al. [251] developed **REDEM**, a real-time detection and mitigation framework for securing connected autonomous vehicle communications. Their system enhances security in Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) communications, particularly in cooperative platooning and adaptive cruise control applications. REDEM integrates anomaly detection models and real-time response mechanisms to mitigate cyber threats such as:

- **Man-in-the-Middle (MitM)**
- **Wormhole**
- **Sybil**
- **Denial-of-Service (DoS) attacks**

The framework was evaluated using a Multilayer Perceptron (MLP) model, trained on datasets reflecting:

- **Various road conditions** (highway, suburban, city).
- **Diverse weather scenarios** (rain, snow, clear, windy).
- **Different times of day** (day, night).

REDEM achieved a **95% prediction accuracy under adversarial conditions**, demonstrating low false positive and false negative rates.

Kihei et al. [252] examined the susceptibility of V2X communications to jamming attacks, demonstrating that existing vehicular networks lack effective mechanisms for detecting and mitigating these threats. Their study analyzed two primary types of jamming attacks, exposing critical security vulnerabilities in V2X networks. To address these issues, they introduced a supervised machine learning-based detection model deployed at the baseband chipset level. This approach significantly enhanced detection accuracy, achieving **99.84% precision**, emphasizing the need for AI security measures in V2X systems.

Unlike simulation-based methods, real-world ML-IDS deployments must consider factors such as environmental noise, mobility dynamics, and computational constraints. Table XXII presents an overview of ML-based IDS models validated in real-world vehicular networks, highlighting their strengths and challenges in practical implementations. These studies emphasize the need for robust architectures that ensure high detection rates while minimizing communication overhead and computational latency.

C. Challenges in Deploying ML-IDS in V2X Networks

The deployment of ML-IDS in V2X networks presents numerous challenges, both in simulated testing environments and real-world vehicular implementations. While simulations remain the primary approach for evaluating IDS performance, real-world deployment has exposed critical scalability, security, and integration barriers that hinder widespread adoption. This section outlines key challenges in both simulation-based and real-world IDS deployment, integrating observations from industry and experimental studies.

1) Challenges in Simulation-Based IDS Deployment:

Simulation-based evaluations have been widely used to test IDS solutions due to the high costs and difficulties associated with real-world experiments. However, simulated results often fail to accurately reflect real vehicular conditions, introducing several challenges:

- **Lack of Realistic Traffic Dynamics:** Many IDS models are trained on network simulators such as NS-3, OMNeT++, and Veins, which lack real-world mobility unpredictability, making it difficult to generalize findings.
- **Synthetic Datasets with Limited Variability:** Most IDS evaluations rely on synthetic datasets rather than real-world vehicular communication logs, leading to over-fitted models that struggle in dynamic environments.
- **Limited Testing in Adverse Conditions:** Simulators do not accurately account for non-line-of-sight (NLOS) issues, weather-related interference, or urban obstacles, which significantly impact V2X communication in real-world deployments.
- **Scalability and Computational Constraints:** Simulated IDS models may perform well with a small number of vehicles, but their effectiveness in large-scale, high-density urban traffic remains uncertain.

One recent study, CN+ Dataset from Bremen, Germany [253], highlighted these limitations by integrating real-world traffic data into simulation environments. By collecting data from over 25,000 vehicles at a signalized intersection, the authors demonstrated that real datasets inside a simulation can significantly improve accuracy and help achieve more realistic results. However, while real datasets enhance simulator realism, privacy concerns, permission issues, and high data collection costs remain significant barriers to obtaining large-scale vehicular network datasets.

2) Challenges in Real-World Deployment of ML-IDS:

In contrast to simulations, real-world ML-IDS deployment presents additional technical and operational challenges. Recent industry and academic efforts highlight several issues:

- **Scalability and Real-Time Processing:** ML-IDS must analyze vast amounts of V2X communication data in real-time, yet vehicular ECUs (Electronic Control Units) have limited processing power. Edge computing and federated learning offer potential solutions, but current automotive hardware is often inadequate for resource-intensive AI models.
- **False Positives and Model Generalization:** ML-based IDS solutions frequently misclassify normal variations in traffic as attacks, leading to high false positive rates. This creates alert fatigue and reduces system reliability.

TABLE XXII
EXPERIMENTAL VALIDATION OF ML-IDS IN REAL-WORLD V2X NETWORKS

Scenario	Method	Model	Acc (%)	Prec (%)	F1 (%)	Challenges	Ref
Flooding Attack Detection	Supervised ML	DT, RF, SVM	99.99	99.97	99.98	Traffic fluctuations affect detection	[244]
5G-VANET Security	Optimization	SVM + 5G	95.4	94.1	94.6	High congestion impacts efficiency	[245]
V2X Anomaly Detection	Physics-Based IDS	VCADS	85–95	84.2	90.0	Varies across datasets	[246]
CAN Bus IDS	Supervised ML	RT, RF, NB	100	100	100	Scalability in heterogeneous networks	[247]
Federated Learning IDS	FL	LR, PCC-CNN	94 (NSL), 99 (Car)	93.5	98.0	Overhead on edge devices	[248]
FL for IoV Security	FL + CNN	MobileNet-Tiny	98.51	97.2	97.74	Privacy risks in FL	[173]
Misbehavior Detection	Classification	RF	93.0	92.5	92.8	Needs cross-validation for unseen attacks	[249]
Cybersecurity in V2V	Deep Learning	RQSCNN	86.0	85.0	85.5	High computational cost	[250]
Real-Time IDS in CAVs	Supervised ML	MLP	95.0	94.7	94.8	Delay in high-speed networks	[251]
Jamming Attack Detection	Supervised ML	AI-Based Model	99.84	99.75	99.80	Needs dynamic interference adaptation	[252]

TABLE XXIII
COMPARISON OF SIMULATED AND REAL-WORLD ML-IDS STUDIES IN V2X SECURITY

Aspect	Simulated Studies	Real-World Studies
Environmental Conditions	Controlled, predefined scenarios [235, 236, 237]	Dynamic, unpredictable real-world factors [244, 245]
Network Delays	Assumed latency values in simulation models [106, 238]	Actual communication delays, interference from environmental factors [246, 247]
Scalability	Limited to the simulation environment (SUMO, NS-3, OMNET++) [239, 240]	Large-scale vehicular networks with real traffic dynamics [248, 173]
Attack Scenarios	Predefined attack models such as replay, DoS, and Sybil attacks [236, 241]	Realistic cyber threats, evolving attack tactics, and mixed attack patterns [245, 249]
Hardware Requirements	Minimal computing resources, often tested on virtual setups [237, 106]	Requires integration with physical OBUs, ECUs, and edge devices [244, 250]
Evaluation Metrics	Often show near-perfect detection rates (e.g., 99%+ accuracy) [236, 242]	More realistic results, often lower detection accuracy due to noise [246, 247]
Dataset Quality	Mostly synthetic and lacks real-world variability [239, 243]	Actual vehicle communication data, but harder to obtain and process [248, 173]
Deployment Complexity	Easier to implement and iterate, requires only software testing [238, 240]	Requires extensive testing, compliance with industry standards [245, 251]
Computational Cost	Low; models can be trained on local machines [235, 237]	High; requires powerful processing on edge/cloud servers [248, 252]
Adversarial Robustness	Assumes static attack patterns, limited exposure to evolving threats [241, 243]	Faces real adversarial attacks and adaptive cyber threats [249, 250]

- **Security of Over-the-Air (OTA) Updates:** While OTA updates enhance security by allowing manufacturers to update IDS models remotely, they introduce vulnerabilities, as attackers could

attempt to intercept and manipulate software updates.

- **Integration with Legacy Vehicles:** Many vehicles currently on the road lack the computational infrastructure

required to support real-time ML-IDS processing, making retrofitting older vehicles a complex challenge.

- **Adversarial Attacks on ML Models:** Attackers can manipulate sensor inputs or communication signals to fool ML-based IDS models, requiring continuous adaptation and model retraining using techniques such as adversarial machine learning.

3) *Key Observations and Lessons Learned:* From both simulated and real-world IDS deployments, several key lessons have emerged:

- High accuracy in controlled environments does not guarantee real-world effectiveness— Models trained on simulated data often fail when exposed to realistic mobility patterns and network interference.
- False positives remain a critical challenge – IDS models need to be optimized to reduce unnecessary alerts while maintaining high detection rates.
- Hardware-level integration improves IDS performance – Embedding ML-IDS at the chipset level (e.g., baseband processors) enhances real-time threat detection.
- Periodic retraining is essential for adapting to evolving cyber threats – Federated learning and incremental model updates can mitigate zero-day attack risks.
- Standardized benchmarking datasets are crucial— The lack of real-world IDS datasets makes it difficult to compare different models. Efforts such as the CN+ dataset and MK5 IDS testbeds provide valuable benchmarks but require broader adoption.

Table XXIII compares the key differences between simulated and real-world studies in ML-based for V2X security. Simulated studies allow controlled testing under predefined attack models and network conditions, leading to high detection rates. However, they lack the unpredictability and variability of real-world vehicular environments. On the other hand, real-world studies provide more realistic evaluations by considering actual environmental factors, network delays, and evolving attack scenarios, making deployment more complex but also ensuring better model generalizability and robustness.

D. Computational Complexity in Large-Scale V2X IDS Deployment

The computational complexity of Intrusion Detection Systems (IDS) plays a crucial role in their feasibility for real-time deployment in vehicular networks. Many deep learning-based IDS models, while achieving high detection accuracy in controlled environments, often face scalability and latency challenges when deployed in real-world V2X security applications. This subsection provides a comparative analysis of the computational complexity associated with different IDS methods, explores optimization strategies to enhance their real-world applicability, and discusses relevant studies addressing these challenges.

1) *Computational Complexity of IDS Models:* Machine learning-based IDS models exhibit varying levels of computational complexity, which directly impact their deployment feasibility. Theoretical complexity classifications for common IDS models are summarized as follows:

- **Polynomial-Time Complexity ($O(n^2)$, $O(n^3)$):** Traditional machine learning models such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) often have quadratic or cubic complexity, making them impractical for large-scale real-time applications in V2X environments [245].
- **Sub-Linear Approaches:** Techniques such as decision trees and random forests leverage hierarchical feature selection, reducing computational complexity while maintaining detection accuracy [236].
- **Deep Learning Methods:** Models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks require extensive computation due to their high-dimensional feature extraction processes, often demanding GPU acceleration for real-time operation [217].

A study by Sousa et al. [244] emphasized the trade-off between detection accuracy and computational feasibility, demonstrating that while deep learning models achieve superior accuracy, they incur high processing overhead, making real-time vehicular deployment difficult.

2) *Optimization Strategies for Real-World IDS Deployment:* To address the computational challenges associated with large-scale IDS implementation, several optimization techniques have been proposed:

- **Model Compression:** Techniques such as weight pruning, quantization, and knowledge distillation help reduce model size and computation requirements while preserving detection performance [169].
- **Edge Inference:** Deploying IDS models at the network edge (e.g., Roadside Units, On-Board Units) enables localized anomaly detection, minimizing cloud processing overhead and improving response time [173].
- **Federated Pruning:** By integrating federated learning with pruning techniques, IDS models can be trained collaboratively across multiple vehicular nodes while reducing unnecessary computational load and communication overhead [239].

The study by Korba et al. [239] explored federated learning with model pruning, demonstrating that it significantly reduces computational costs while maintaining high detection accuracy in vehicular networks.

3) *Trade-Offs in IDS Model Deployment:* The deployment of IDS in V2X networks involves several trade-offs between detection performance and computational efficiency:

- **Accuracy vs. Latency:** Centralized IDS models typically achieve higher accuracy due to global feature aggregation but suffer from increased latency. Decentralized approaches such as federated learning improve real-time detection but may face communication delays and training inconsistencies [248].
- **Energy Efficiency vs. Model Complexity:** While deep learning-based IDS provide high detection rates, they are often computationally expensive. Lightweight models optimized for edge computing offer lower energy consumption at the cost of slight accuracy degradation.

A recent evaluation by Bhavsar et al. [248] found that while federated learning reduces latency and improves scalability,

it introduces additional communication overhead, particularly in resource-constrained vehicular networks.

4) *Future Directions*: Future research should focus on balancing computational efficiency with detection accuracy by exploring hybrid approaches that integrate lightweight deep learning architectures with edge AI solutions. Additionally, the development of adaptive learning mechanisms capable of dynamically adjusting model complexity based on real-time traffic conditions and network constraints could significantly enhance IDS scalability in V2X security.

Recent advances in TinyML and lightweight federated learning show promise in making IDS models more efficient for vehicular environments by reducing model size without compromising detection accuracy. Additionally, dynamic model adaptation strategies could further enhance IDS scalability by adjusting computational complexity based on available resources in vehicular networks.

E. The Impact of Noise in ML-Based IDS for V2X Security

Noise in machine learning-based intrusion detection systems (ML-IDS) for V2X networks refers to any form of unstructured, misleading, or irrelevant data that affects the model's ability to accurately distinguish between benign and malicious traffic. In a highly dynamic vehicular network, noise can stem from sensor inaccuracies, fluctuating environmental conditions, channel interference, data transmission errors, and adversarial attacks designed to mimic normal behavior. The presence of noise in IDS datasets often leads to false alarms, increased computational complexity, and degraded model performance, particularly in real-time applications where rapid and precise detection is crucial.

In an ML-based IDS, the observed input signal X_{obs} consists of the true signal X_{true} and an added noise component N , where:

$$X_{\text{obs}} = X_{\text{true}} + N \quad (10)$$

The noise component can follow different distributions depending on its source. Gaussian noise ($N \sim N(\mu, \sigma^2)$) represents random fluctuations in sensor readings or communication interference, while adversarial noise is introduced through carefully crafted perturbations (N_{adv}) designed to mislead ML models into misclassifying inputs. Structural noise occurs due to incomplete or corrupted data in real-time transmission. The impact of noise can be quantified using the False Alarm Rate (FAR), defined as:

$$FAR = \frac{FP}{FP + TN} \quad (11)$$

where FP represents false positives and TN represents true negatives. A high FAR indicates excessive noise in the IDS system, leading to unnecessary security interventions and potential delays in real-time decision-making.

Addressing noise in ML-based IDS requires effective filtering techniques that balance computational efficiency with detection accuracy. One common approach is **feature selection and dimensionality reduction**, which eliminates irrelevant or redundant information contributing to noise. Techniques such as Principal Component Analysis (PCA) and autoencoders

improve IDS robustness. **Hybrid AI security mechanisms** have been proposed to dynamically allocate computational resources for filtering noisy features in V2X networks, as demonstrated by Bangui et al. [185].

Another effective method is **adaptive thresholding and dynamic sensitivity adjustment**, where detection thresholds adjust dynamically in response to fluctuating network traffic. This approach, exemplified by Deep-VFog, an AI-enhanced fog-assisted IDS introduced by Rihan et al. [186], significantly reduces the impact of noise while preserving detection accuracy.

Federated Learning (FL) for noise-resilient detection has gained prominence in vehicular networks, as it enables IDS models to learn collaboratively across multiple vehicles without sharing raw data, thereby mitigating inconsistencies and noise artifacts. Korba et al. [239] demonstrated that FL-based IDS achieved a 99.99% accuracy with an FPR of only 0.01%, making it a promising solution for reducing false alarms due to noisy data across distributed vehicular environments.

Additionally, **adversarial training and robust deep learning models** have been developed to counter adversarial noise. Williams et al. [187] proposed a hybrid AI security architecture combining edge computing with cloud intelligence, improving model resilience by training against perturbed data samples. Similarly, **Mobile Edge Computing (MEC)-based IDS**, as proposed by Alladi et al. [224], demonstrated that local anomaly filtering on RSUs and vehicle processors effectively reduces adversarial interference.

Lastly, **noise-tolerant learning algorithms** have been integrated into IDS frameworks to improve resilience against fluctuating network conditions. Samy et al. [210] developed a **fog-assisted attack detection framework** that integrates denoising autoencoders (DAEs) to filter out irrelevant network fluctuations before classification. This method significantly enhanced detection precision in resource-constrained vehicular environments by removing transient anomalies that do not correspond to actual threats. These advancements underscore the importance of noise mitigation in ensuring the reliability of ML-based IDS in V2X security, where real-time accuracy and computational efficiency remain critical challenges.

XIII. FUTURE DIRECTIONS AND LEARNED LESSONS

As V2X networks evolve, emerging technologies such as blockchain, quantum cryptography, and federated learning offer promising avenues for enhancing intrusion detection systems (IDS). This section outlines key research directions that can strengthen the security and resilience of ML-based IDS in V2X environments.

A. Blockchain-Enhanced V2X Security

The decentralized nature of blockchain makes it an ideal candidate for securing V2X communication networks. Future research should focus on:

- **Blockchain-Based Intrusion Detection**: Implementing blockchain-enabled IDS to ensure tamper-proof logging of malicious activities and secure data sharing across vehicles and roadside units (RSUs).

- **Decentralized Trust Mechanisms:** Leveraging blockchain to establish decentralized trust among vehicles, reducing reliance on centralized authentication authorities.
- **Efficient Consensus Algorithms:** Developing lightweight consensus protocols tailored for V2X environments to minimize computational overhead and energy consumption in resource-constrained vehicular devices.

Mishra et al. [254] propose a blockchain-based main key (MK) management system to enhance the security and credibility of sender identity and message authentication in vehicular networks. The study introduces a secure Vehicle-to-Vehicle (V2V) communication protocol where blockchain is leveraged for generating and validating MKs, ensuring improved security against unauthorized access and cyber threats. Additionally, an unsupervised deep learning model, specifically a long short-term memory (LSTM)-based anomaly detection approach, is integrated to identify malicious data in vehicular communication, addressing vulnerabilities in traditional V2V protocols. Experimental results demonstrate the protocol's resilience to cyber threats, and the blockchain fetch latency of 1.17 ms is considered acceptable for V2V communications. However, the study has several limitations, including increased computational overhead due to blockchain integration, which may impact real-time efficiency, and scalability concerns, as the performance in large-scale vehicular networks is not extensively analyzed. Furthermore, the combined use of blockchain and deep learning raises energy consumption issues, posing challenges for energy-efficient vehicular systems. Additionally, the study lacks extensive benchmarking against other state-of-the-art vehicular security frameworks, limiting direct comparative insights. These limitations highlight the need for further optimization in blockchain-based vehicular security solutions to balance security, performance, and scalability.

B. Quantum Cryptography for Secure V2X Communication

Quantum cryptography is emerging as a powerful tool to counter advanced cyber threats, particularly in 6G-enabled vehicular networks. Key research directions include:

- **Quantum Key Distribution (QKD):** Implementing QKD in V2X networks to enable secure communication channels resistant to quantum computing attacks.
- **Post-Quantum Cryptography:** Designing quantum-resistant cryptographic algorithms that can be integrated into ML-IDS frameworks to safeguard vehicular communication against future quantum threats.
- **Quantum Machine Learning for IDS:** Exploring quantum-enhanced ML techniques to accelerate intrusion detection and anomaly detection in V2X environments.

C. Federated Learning and Edge AI for Adaptive Security

Federated learning (FL) and Edge AI can improve privacy-preserving and real-time security mechanisms in V2X networks. Future work should explore:

- **Hierarchical Federated Learning:** Implementing multi-level FL architectures where vehicles, RSUs, and cloud servers collaborate to improve IDS model efficiency and reduce communication overhead.
- **Edge AI for Real-Time Threat Detection:** Deploying lightweight AI models at edge devices (e.g., RSUs, vehicle ECUs) to enable real-time anomaly detection with minimal latency.
- **Privacy-Preserving FL Techniques:** Enhancing FL security by integrating differential privacy, secure multi-party computation (SMPC), and homomorphic encryption to prevent model inversion attacks.

D. Explainable AI (XAI) for Interpretable IDS

A major challenge in deploying ML-based IDS is the lack of interpretability, making it difficult to understand why a model classifies certain behaviors as malicious. Future research should focus on:

- **XAI for IDS Decision-Making:** Incorporating explainability techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to enhance transparency in intrusion detection.
- **Human-Centric AI Models:** Designing AI-based IDS that provide actionable insights to human operators, improving trust and usability in real-world deployments.
- **Adversarial Robustness in XAI-Enabled IDS:** Developing methods to counter adversarial attacks that exploit model interpretability, ensuring robustness in dynamic V2X environments.

E. Standardized Benchmark Datasets and Real-World Deployment

A key limitation of current ML-IDS research is the reliance on synthetic datasets. To bridge the gap between theoretical models and practical applications, future research should explore:

- **Real-World IDS Dataset Collection:** Developing large-scale, publicly available datasets that capture real vehicular traffic conditions and attack scenarios.
- **Cross-Dataset Generalization:** Evaluating IDS models across multiple datasets to improve generalization and reduce overfitting to specific traffic patterns.
- **Experimental Testbeds for IDS Validation:** Establishing real-world V2X security testbeds to validate ML-IDS models in diverse environments, ensuring robustness against evolving cyber threats.

F. Towards 6G-Enabled Secure V2X Networks

As vehicular networks transition towards 6G, new security challenges and opportunities emerge. Future research should address:

- **6G-V2X Security Frameworks:** Developing next-generation IDS solutions optimized for ultra-low latency, terahertz (THz) communication, and AI-based network management.

- **AI-Based Network Slicing for Secure V2X:** Implementing AI-powered dynamic network slicing to isolate and mitigate cyber threats in real time.
- **Quantum-Resilient Cryptographic Mechanisms:** Designing cryptographic solutions that withstand quantum computing-based attacks, ensuring long-term security of vehicular communications.

1) Future Directions:

- **Robust AI Models for Adversarial Attack Resilience:** Future research should focus on designing more robust AI models capable of withstanding adversarial attacks in V2X networks. Techniques such as adversarial training, differential privacy, and robust deep learning architectures can be explored to improve model resilience.
- **Real-World Deployment and Dataset Expansion:** While adversarial attacks have been widely studied in simulated environments, real-world datasets capturing adversarial behavior in V2X networks remain limited. Future research should involve large-scale adversarial attack datasets from real vehicular environments to enhance IDS training and evaluation.
- **Federated Learning for Adversarial Defense:** Decentralized machine learning approaches, such as federated learning, should be explored to mitigate adversarial attacks. By training AI models across distributed edge nodes, federated learning can reduce centralized vulnerabilities and improve overall robustness.
- **Explainable AI (XAI) for Attack Detection:** The closed-box nature of deep learning models used in V2X IDS poses challenges in understanding attack decisions. Integrating explainable AI (XAI) techniques can improve trust and interpretability, allowing security analysts to better understand adversarial threats.
- **Blockchain-Enabled Adversarial Defense Mechanisms:** Blockchain technology can provide a decentralized authentication mechanism for V2X security. Future work should investigate blockchain-assisted defenses to ensure attack-resistant AI models in vehicular networks.
- **AI-Based Signal Authentication and Secure Modulation Techniques:** Signal manipulation attacks exploit vulnerabilities in AI-based spectrum classification. Future research should explore secure modulation techniques and signal authentication mechanisms to mitigate such threats.
- **Multi-Agent Learning for Distributed Security:** Multi-agent reinforcement learning (MAREL) approaches can be used to collaboratively detect and mitigate adversarial attacks across different V2X nodes. This can enhance the scalability of attack detection mechanisms.

2) Learned Lessons:

- Adversarial attacks significantly reduce the reliability of AI-Based V2X systems, leading to safety-critical failures such as misbehavior detection evasion and network congestion.
- Traditional ML-based security solutions struggle against advanced adversarial techniques such as PGD, PIA, and FGSM, necessitating the need for adaptive security mechanisms.

- Defensive techniques such as adversarial training and robust reinforcement learning models have shown promise in mitigating adversarial threats but require further validation in real-world vehicular environments.
- Decentralized approaches, including federated learning and blockchain authentication, present viable solutions for improving the resilience of ML-based V2X security frameworks.
- Future V2X security solutions must incorporate end-to-end encryption, zero-trust authentication, and multi-factor AI-based anomaly detection to enhance network robustness.

G. Future Directions and Learned Lessons in V2X IDS Datasets

1) Future Directions:

- **Development of Real-World V2X Datasets:** Existing datasets, such as ROAD [144] and VeReMi [146], offer valuable insights into vehicular misbehavior but lack the diversity of modern cyber threats. Future research should focus on acquiring large-scale, real-world vehicular datasets incorporating contemporary attack patterns, such as advanced GPS spoofing, AI-based evasion, and distributed coordinated attacks.
- **Hybrid Dataset Integration:** Combining synthetic and real-world data can enhance IDS training for better generalization. While datasets like VeReMi [146] simulate vehicular misbehavior, they do not capture broader cybersecurity threats. A hybrid approach that merges network-based datasets like CICIDS2017 [151], ToN-IoT [150], and UNSW-NB15 [152] with V2X-specific datasets can provide a comprehensive evaluation framework.
- **Benchmarking Frameworks for Standardized Evaluation:** The lack of standardization in dataset feature extraction and attack scenarios leads to inconsistencies in IDS evaluations. Establishing unified benchmarking methodologies, similar to those used in general network intrusion detection, can improve cross-dataset comparability and enhance model robustness.
- **Zero-Day Attack Detection Enhancement:** Current datasets primarily contain predefined attack signatures, limiting their effectiveness for detecting unknown threats. Future research should focus on adversarial dataset generation techniques that simulate adaptive attacks in V2X environments, ensuring IDS models can detect novel intrusion strategies.
- **Real-Time Data Collection for IDS Training:** Privacy concerns and infrastructure constraints hinder real-time vehicular data collection. Collaborations between academia, industry, and government agencies are necessary to establish privacy-preserving vehicular data-sharing frameworks while ensuring compliance with cybersecurity regulations.
- **Dataset Adaptability for 5G/6G-Enabled V2X Networks:** As V2X networks transition to 5G and 6G infrastructures, datasets must evolve to capture ultra-low

latency communication threats, network slicing vulnerabilities, and edge computing attack vectors. Integrating Multi-Access Edge Computing (MEC) traffic data into future datasets can enhance IDS training for next-generation vehicular security.

- **Incorporating Federated Learning for Distributed IDS Training:** Federated Learning (FL) offers a promising solution for dataset scarcity by enabling decentralized IDS model training across multiple vehicles without sharing raw data [106]. Future research should focus on implementing privacy-preserving FL approaches to enhance dataset scalability and resilience against adversarial model poisoning attacks.

2) *Learned Lessons:*

- The lack of real-world vehicular datasets limits IDS performance evaluation in practical deployment scenarios. Existing datasets rely heavily on simulated attacks, which may not fully capture dynamic adversarial strategies in V2X networks.
- No single dataset comprehensively addresses both vehicular misbehavior and broader cybersecurity threats. Hybrid dataset integration is essential for improving IDS robustness against diverse attack scenarios.
- Standardizing dataset structures, feature extraction methods, and evaluation metrics can improve the reliability and generalizability of IDS models across different vehicular environments.
- Traditional datasets, such as KDD Cup 99 and NSL-KDD, lack relevance for modern V2X networks, as they do not reflect real-world vehicular communication protocols or attack types.
- Zero-day attack detection remains a critical challenge due to the static nature of most datasets. Future research should emphasize dataset evolution strategies that dynamically incorporate emerging threats.
- Privacy concerns and security regulations restrict access to real vehicular data, necessitating privacy-preserving techniques, such as differential privacy and secure multi-party computation, to facilitate dataset collection and sharing.
- The emergence of 5G and 6G V2X communication introduces new security challenges that are not adequately represented in existing datasets. Future datasets must include scenarios involving network slicing attacks, edge computing threats, and AI-based adversarial strategies.
- Federated learning provides a viable alternative for overcoming dataset collection challenges by enabling collaborative IDS training across multiple V2X entities while preserving data privacy.

H. *Future Directions and Learned Lessons in ML-Based IDS for V2X Security*

1) *Future Directions:*

- **Integration of Federated Learning for IDS in V2X Networks:** Traditional ML-based IDS models rely on centralized data collection, raising concerns regarding

privacy and communication overhead. Future research should focus on integrating Federated Learning (FL) to enable decentralized IDS training across V2X nodes while preserving data privacy [9].

- **Enhancing Real-World Deployment of ML-Based IDS:** Most existing ML-based IDS models are tested on benchmark datasets like VeReMi and CICIDS2017, but their real-world deployment remains limited. Future studies should emphasize practical implementation in real V2X environments to Explain the impact of latency, communication constraints, and real-time adaptability.
- **Explainable AI (XAI) for ML-Based Intrusion Detection:** While ML-based IDS models have achieved high accuracy, their decision-making process remains a “closed-box” [125]. Future research should focus on integrating Explainable AI (XAI) techniques such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) to enhance interpretability and trustworthiness.
- **Hybrid IDS Approaches Combining Signature and Anomaly Detection:** Future IDS solutions should incorporate both signature-based and anomaly-based detection for robust security. Studies such as Yang et al. highlight the need for multi-layered IDS frameworks that integrate rule-based Snort detection with ML-based anomaly classifiers.
- **Leveraging Reinforcement Learning for Adaptive Security Mechanisms:** Reinforcement Learning (RL) provides an adaptive security mechanism capable of dynamically responding to evolving threats. Future IDS frameworks should explore Deep Q-Networks (DQN) and Actor-Critic RL models for optimizing misbehavior detection [92].
- **Adapting ML Models for 5G/6G-Enabled V2X Networks:** The transition to 5G and beyond introduces new security challenges such as network slicing vulnerabilities and ultra-low latency attack surfaces. Future IDS research should focus on optimizing ML models to address these evolving threats.
- **Mitigating Model Poisoning Attacks in ML-Based IDS:** Attackers can manipulate training data to mislead IDS models (e.g., data poisoning attacks). Future research should explore adversarial training, differential privacy, and blockchain-enhanced verification to improve model resilience.
- **Reducing False Positives in Anomaly-Based IDS Models:** Anomaly-based IDS models often suffer from high false positive rates. Future research should focus on optimizing feature selection, employing semi-supervised learning, and leveraging ensemble models to improve detection accuracy while minimizing false alarms.

2) *Learned Lessons:*

- **ML-Based IDS Achieves High Detection Accuracy but Faces Deployment Challenges:** While ML techniques such as Random Forest, SVM, and Neural Networks exhibit high accuracy in simulated environments,

TABLE XXIV
REAL TESTING AND VALIDATION FOR IDS IMPLEMENTATION USING MACHINE LEARNING ALGORITHMS FOR V2X NETWORKS(PART 1)

Study	Focus	Key Contributions	ML Methods & Metrics	Dataset & Results
Billy Kihei et al. [252]	Vulnerability of V2X to jamming attacks	Highlighted V2X vulnerabilities to jamming; implemented high-accuracy detection model	Supervised ML at baseband chipset level; detection accuracy: 99.84%	Lab experiments demonstrated V2X networks' susceptibility to jamming and lack of standards
Begum et al. [242]	Sensor attack detection for AVs in 6G-V2X	Improved AV security; developed SADC framework with higher accuracy	CNN and LSTM layers; 0.98% higher accuracy than existing solutions	AV sensor data; accuracy: 98%; 75% training, 15% testing, 10% validation; attacks: False Data Injection, Stealthy Attack
Iqbal et al. [235]	Cybersecurity threats in VANETs	Investigated replay and bogus information attacks; provided public dataset for anomaly detection solutions	Simulations with Eclipse MOSAIC framework	Public dataset: significant disruptions and delays due to replay attacks; useful for ML-based solutions
Rajak et al. [236]	Security challenges in 5G V2X	Developed deep learning-based attack detection model	Stacked LSTM; accuracy: 99.8%, precision: 99.2%, F1-score: 99.1%	AIoT-SoL dataset; 5,134,728 instances; practical for DoS, MitM, spoofing
Carlos Jiménez [237]	Security of AV networks	Enhanced detection of cyber-attacks in AVNs	Deep Autoencoders, LSTM-RNNs; detection rate: 95%, false alarm rate: 3%, overall accuracy: 94%	Simulated AVN environment; synthetic attack scenarios; achieved 94% accuracy
Boddupalli et al. [251]	Real-time detection & mitigation framework for CAVs	Developed REDEM for real-time anomaly detection	Multilayer Perceptron (MLP) model; accuracy: 95%	Physical automotive simulator; low mean absolute error, high accuracy
Prathiba et al. [243]	Anomaly detection and cyber-attack mitigation in AVs	Proposed HDAD framework; high accuracy, low computation cost	HDAD: MARL + MaxEntIRL; accuracy: 98%, recall: 95%	NS-3 with TeraSim extension, SUMO; 100 AVs; high anomaly detection accuracy
Minawi et al. [247]	Intrusion Detection System for CAN bus	Secured CAN bus with ML-based IDS	ML: Random Tree, RF, SGD, Naive Bayes; 100% accuracy for DoS, impersonation attacks	"Car Hacking Dataset" from HCRL; effective across attack types

real-world deployment introduces latency, computational, and scalability constraints.

- **Supervised Learning is Effective for Known Attacks but Limited for Zero-Day Threats:** Traditional supervised learning models rely on labeled datasets and struggle to detect previously unseen attacks. Hybrid approaches incorporating unsupervised learning and reinforcement learning are necessary for adaptive security.
- **Dataset Limitations Impact the Generalization of ML Models:** The lack of diverse real-world datasets affects the robustness of IDS models. Future research should focus on creating datasets that capture a wider range of attack scenarios and real vehicular network conditions.
- **Federated Learning Enhances Privacy but Introduces Communication Costs:** FL enables privacy-preserving IDS training across distributed V2X nodes but requires optimization to reduce communication overhead and ensure model convergence.

- **Ensemble and Hybrid Learning Techniques Improve IDS Performance:** Studies demonstrate that ensemble methods, such as XGBoost, AdaBoost, and stacking, outperform individual classifiers by reducing false positives and improving detection rates [129].
- **Adaptive IDS Solutions are Needed for 5G and 6G V2X Environments:** Next-generation vehicular networks introduce new security threats that require IDS models capable of real-time adaptation, cross-layer security analysis, and edge-assisted intrusion detection.

I. Future Directions and Learned Lessons in Federated Learning and Edge AI for V2X Security

1) Future Directions:

- **Enhancing Privacy-Preserving FL Mechanisms:** While FL minimizes the need for centralized data collection, privacy risks such as model inversion and gradient leakage remain significant. Future research should integrate Homomorphic Encryption (HE) and Differential Privacy

TABLE XXV
REAL TESTING AND VALIDATION FOR IDS IMPLEMENTATION USING MACHINE LEARNING ALGORITHMS FOR V2X NETWORKS (PART 2)

Study	Focus	Key Contributions	ML Methods & Metrics	Dataset & Results
Salazar et al. [246]	Anomaly detection in V2X communications	Enhanced V2X security by detecting misbehavior	Physics-based models; attack detection: 85–95%	Ann Arbor, Arlington, Tampa, Wyoming datasets; improved road safety
Raja et al. [106]	Trajectory anomaly detection in 6G-V2X	Improved trajectory anomaly detection	Deep Deterministic Policy Gradient (DDPG) algorithm; accuracy: 97%	SUMO, OMNET++; detected speed variations, wrong movements
Monteuuis et al. [249]	Misbehavior detection in CAVs	Framework for detecting misbehaving ITS-Stations	ML: MinMax, Adaboost, MLP, RF; MCC: 93%	Pedestrians, motorcycles, cars datasets; high detection accuracy
Kong et al. [250]	Cybersecurity in V2V communication	Addressed V2V communication vulnerabilities	Reinforcement Quantile Spatial CNN; accuracy: 86%	20,000 images dataset; improved throughput, reduced latency
Korba et al. [239]	IDS for CAVs in 5G+ networks	Proposed federated learning-based IDS	Deep autoencoder; accuracy: 99.99%, FPR: 0.01%	VDoS dataset; detected SYN Flood attacks effectively
Magsi et al. [240]	NDN in VANETs	Enhanced VANET security using ML and blockchain	Hybrid ML: LR, DT, KNN, RF, GNB; RF: 100% accuracy	BurST-Australian dataset; 207,315 messages; RF outperformed others
Twardokus et al. [241]	Vulnerabilities in 5G C-V2X	Developed detection and mitigation for DoS attacks	Unsupervised clustering (DBSCAN) for jamming; accuracy: 80% TPR	Hardware testbeds and simulations; targeted jamming reduced PDR by 90%

(DP) into FL-based IDS to prevent unauthorized access to sensitive vehicular data.

- **Adaptive Federated Learning for Dynamic V2X Environments:** V2X networks exhibit high mobility, leading to frequent disconnections and varying data distributions. Future studies should explore adaptive FL models that dynamically adjust aggregation frequency and client participation rates based on network conditions.
- **Hybrid FL-Edge AI Models for Low-Latency IDS:** Combining FL with Edge AI enables real-time intrusion detection in vehicular networks. Future work should investigate Hierarchical FL (HFL) models that distribute learning across vehicles, RSUs, and cloud servers to optimize computational efficiency and reduce latency.
- **Blockchain-Enabled Federated Learning for Secure Aggregation:** Integrating blockchain with FL can enhance the security of model aggregation by ensuring decentralized trust and tamper-proof updates. Future research should examine lightweight blockchain architectures optimized for V2X environments.
- **Quantum-Resilient FL for 6G-V2X Networks:** As quantum computing evolves, traditional cryptographic mechanisms for FL aggregation may become vulnerable. Future research should explore Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC) to secure FL-based intrusion detection models.
- **Optimized FL for Energy-Efficient V2X Security:** Training FL models on edge devices consumes significant

energy. Future studies should develop lightweight FL architectures and introduce AI-based model pruning techniques to enhance energy efficiency while maintaining security performance.

- **Mitigating FL Model Poisoning Attacks:** Attackers can manipulate model updates to introduce vulnerabilities in FL-based IDS. Future work should explore anomaly detection techniques for FL training data validation, as well as robust aggregation mechanisms like Byzantine-robust FL.
- **Federated Learning for Next-Generation Autonomous Vehicles:** With the growing adoption of autonomous vehicles (AVs), FL can enable collaborative threat detection across AV fleets. Future research should focus on Vehicle-to-Cloud (V2C) FL models that integrate autonomous driving security with intrusion detection.

2) Learned Lessons:

- **FL Enhances Privacy but Requires Secure Aggregation:** While FL eliminates the need for raw data sharing, adversarial threats such as model inversion and membership inference attacks necessitate secure aggregation mechanisms like HE and DP.
- **Hierarchical FL Improves Scalability in V2X Networks:** Multi-tier FL architectures, incorporating vehicles, RSUs, and cloud servers, can reduce bandwidth consumption and optimize IDS performance in large-scale V2X environments.

- **FL-Based IDS Achieves High Accuracy but Faces Deployment Challenges:** Studies show that FL-based IDS can achieve up to 98.5% accuracy in detecting DoS and Sybil attacks. However, real-world deployment requires addressing issues related to network instability, heterogeneous data distributions, and computational limitations.
- **Edge AI Reduces Latency and Enhances Real-Time Security:** Deploying AI-based IDS at edge nodes significantly improves response times and reduces reliance on cloud-based solutions. However, Edge AI models must be optimized for resource-constrained environments.
- **FL and Blockchain Synergy Strengthens Trust and Security:** Studies demonstrate that blockchain-enhanced FL improves the integrity of IDS model aggregation while mitigating adversarial manipulations. Future implementations should explore efficient consensus mechanisms for lightweight blockchain integration.
- **Real-World Testing of FL-Based IDS is Limited:** Most existing FL-based IDS models are evaluated using simulated datasets such as VeReMi and CICIDS2017. Future research should focus on real-world implementations to explain the impact of network congestion, mobility, and adversarial conditions.
- **Energy Constraints Must be Addressed for Sustainable FL Deployment:** FL training consumes substantial energy, particularly on resource-limited vehicular devices. Efficient training techniques such as Adaptive FL and Federated Transfer Learning (FTL) should be explored to minimize energy consumption while maintaining security effectiveness.
- **FL for 6G-V2X Networks Requires Quantum-Resilient Security Mechanisms:** The advent of 6G and quantum computing introduces new security challenges for FL-based intrusion detection. Research into Quantum-Resilient FL and Quantum Cryptography is necessary to future-proof V2X security frameworks.

XIV. CONCLUSION

Ensuring the security and resilience of Vehicle-to-Everything (V2X) communication networks is critical for the widespread adoption of intelligent transportation systems. This study provides a comprehensive analysis of Intrusion Detection Systems (IDS) in V2X environments, exploring both traditional security mechanisms and advanced AI-based approaches, including machine learning-based intrusion detection and privacy-preserving solutions such as Federated Learning (FL) and Edge AI. By systematically evaluating existing methodologies, datasets, and real-world applicability, this study highlights the challenges and opportunities in securing V2X communication networks.

The findings reveal that machine learning-based IDS offer significant improvements in detecting cyber threats in vehicular networks. Hybrid deep learning architectures, particularly CNN-LSTM models, demonstrate superior performance, achieving detection accuracy of up to 99.2%. These models also exhibit lower false alarm rates, with false positive rates below 2%, ensuring greater reliability in real-time V2X

security applications. Moreover, generative models such as GAN-based anomaly detection contribute to enhanced robustness against adversarial attacks, a crucial aspect in ensuring the adaptability of IDS solutions to evolving cyber threats.

Federated Learning presents a promising approach for implementing decentralized IDS while preserving data privacy. Experimental results indicate that FL-based IDS reduce communication overhead by 5–10% compared to traditional centralized IDS architectures. Although slight accuracy degradation of approximately 1–2% is observed in FL models, fine-tuning and optimized model aggregation mitigate this issue, making FL a viable security solution for large-scale V2X deployments. However, challenges remain in optimizing communication efficiency, addressing aggregation security risks, and ensuring seamless coordination among participating nodes.

The study also highlights the increasing threat posed by adversarial attacks, which exploit vulnerabilities in AI-based IDS models. The incorporation of adversarial training significantly improves detection robustness, with an observed enhancement of up to 30% against evasion attacks. The potential integration of blockchain with FL-based IDS offers an additional layer of security, particularly in securing data exchange and model updates among connected vehicles. However, existing datasets used for IDS research still present limitations, particularly in representing real-world adversarial attack scenarios, emphasizing the need for more diverse and standardized security datasets.

Despite the advancements in AI-based IDS, real-world deployment poses several challenges. Scalability and computational complexity remain significant concerns, especially when deploying deep learning-based IDS in resource-constrained vehicular environments. Communication latency and bandwidth limitations also impact IDS efficiency, necessitating the adoption of edge computing and optimized model aggregation techniques. Furthermore, the lack of high-quality V2X-specific datasets limits the generalizability of IDS solutions, underscoring the need for more representative real-world datasets that capture diverse attack scenarios.

To address these challenges, future research should explore the integration of post-quantum cryptographic techniques to enhance the resilience of IDS frameworks against emerging cyber threats. The adoption of Explainable AI (XAI) will further improve the interpretability and trustworthiness of AI-based IDS, facilitating their real-world deployment. Additionally, hybrid security frameworks combining AI, blockchain, and FL offer a promising direction toward achieving a scalable, privacy-preserving IDS for V2X networks. The development of 6G-enabled IDS architectures leveraging ultra-low latency and distributed intelligence will also play a vital role in securing the next generation of autonomous and connected vehicles.

By bridging the gap between theoretical IDS frameworks and practical V2X security implementations, this study provides valuable insights for researchers and industry practitioners. The comparative evaluations, dataset analyses, and experimental findings contribute to a deeper understanding of IDS methodologies and their real-world applicability. The

results emphasize the necessity of developing adaptive, scalable, and privacy-preserving IDS solutions capable of securing connected vehicles against increasingly sophisticated cyber threats.

REFERENCES

- [1] M. Hasan, S. Mohan, T. Shimizu, and H. Lu, "Securing vehicle-to-everything (V2X) communication platforms," *IEEE Trans. Intell. Vehicles*, vol. 5, no. 4, pp. 693–713, Dec. 2020.
- [2] A. Alnasser, H. Sun, and J. Jiang, "Cyber security challenges and solutions for V2X communications: A survey," *Comput. Netw.*, vol. 151, pp. 52–67, Mar. 2019.
- [3] J. Huang, D. Fang, Y. Qian, and R. Q. Hu, "Recent advances and challenges in security and privacy for V2X communications," *IEEE Open J. Veh. Technol.*, vol. 1, pp. 244–266, 2020.
- [4] S. A. A. Hakeem, A. A. Hady, and H. Kim, "Current and future developments to improve 5G-NewRadio performance in vehicle-to-everything communications," *Telecommun. Syst.*, vol. 75, no. 3, pp. 331–353, Nov. 2020.
- [5] H. M. Furqan, M. Sohaib J. Solaija, J. M. Hamamreh, and H. Arslan, "Intelligent physical layer security approach for V2X communication," 2019, *arXiv:1905.05075*.
- [6] A. Ghosal and M. Conti, "Security issues and challenges in V2X: A survey," *Comput. Netw.*, vol. 169, Mar. 2020, Art. no. 107093.
- [7] D. Ulybyshev, A. O. Alsalem, B. Bhargava, S. Savvides, G. Mani, and L. B. Othmane, "Secure data communication in autonomous V2X systems," in *Proc. IEEE Int. Congr. Internet Things (ICIoT)*, Jul. 2018, pp. 156–163.
- [8] P. K. Mvula, P. Branco, G. V. Jourdan, and H. L. Viktor, "A survey on the applications of semi-supervised learning to cyber-security," *ACM Comput. Surv.*, vol. 56, no. 10, pp. 1–41, 2024.
- [9] A. Selamnia, B. Brik, S. M. Senouci, A. Boualouache, and S. Hossain, "Edge computing-enabled intrusion detection for C-V2X networks using federated learning," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2022, pp. 2080–2085, doi: [10.1109/GLOBECOM48099.2022.10001675](https://doi.org/10.1109/GLOBECOM48099.2022.10001675).
- [10] H. Navidan et al., "Generative adversarial networks (GANs) in networking: A comprehensive survey & evaluation," *Comput. Netw.*, vol. 194, Jul. 2021, Art. no. 108149.
- [11] F. W. Alsaade and M. H. Al-Adhaileh, "Cyber attack detection for self-driving vehicle networks using deep autoencoder algorithms," *Sensors*, vol. 23, no. 8, p. 4086, Apr. 2023.
- [12] A. Boualouache, T. E. T. Djaidja, S. Senouci, and T. Engel, "Deep learning-based intra-slice attack detection for 5G-V2X sliced networks," in *Proc. IEEE 95th Vehicular Technol. Conf. (VTC2022-Spring)*, Helsinki, Finland, 2022, pp. 1–5, doi: [10.1109/VTC2022-Spring54318.2022.9860373](https://doi.org/10.1109/VTC2022-Spring54318.2022.9860373).
- [13] T. Sowmya and E. A. Mary Anita, "A comprehensive review of AI based intrusion detection system," *Meas., Sensors*, vol. 28, Aug. 2023, Art. no. 100827.
- [14] S. So, J. Petit, and D. Starobinski, "Physical layer plausibility checks for misbehavior detection in V2X networks," in *Proc. 12th Conf. Secur. Privacy Wireless Mobile Netw.*, Miami, FL, USA, May 2019, pp. 84–93.
- [15] N. Bißmeyer, J. Njeukam, J. Petit, and K. M. Bayarou, "Central misbehavior evaluation for VANETs based on mobility data plausibility," in *Proc. 9th ACM Int. workshop Veh. Inter-Netw., Syst., Appl.*, Jun. 2012, pp. 73–82.
- [16] M. Sun, M. Li, and R. Gerdes, "A data trust framework for VANETs enabling false data detection and secure vehicle tracking," in *Proc. IEEE Conf. Commun. Netw. Secur. (CNS)*, Oct. 2017, pp. 1–9.
- [17] Y. Yao et al., "Voiceprint: A novel Sybil attack detection method based on RSSI for VANETs," in *Proc. 47th Annu. IEEE/IFIP Int. Conf. Dependable Syst. Netw. (DSN)*, Jun. 2017, pp. 591–602.
- [18] Q. A. Chen, Y. Yin, Y. Feng, Z. M. Mao, and H. X. Liu, "Exposing congestion attack on emerging connected vehicle based traffic signal control," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2018, pp. 1–5.
- [19] T. Zhang and Q. Zhu, "Distributed privacy-preserving collaborative intrusion detection systems for VANETs," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 4, no. 1, pp. 148–161, Mar. 2018.
- [20] J. Mahmood, Z. Duan, Y. Yang, Q. Wang, J. Nebhen, and M. N. M. Bhutta, "Security in vehicular ad hoc networks: Challenges and countermeasures," *Secur. Commun. Netw.*, vol. 2021, Jun. 2021, Art. no. 9997771.
- [21] A. Irshad, M. Usman, S. A. Chaudhry, H. Naqvi, and M. Shafiq, "A provably secure and efficient authenticated key agreement scheme for energy internet-based vehicle-to-grid technology framework," *IEEE Trans. Ind. Appl.*, vol. 56, no. 4, pp. 4425–4435, Jul. 2020.
- [22] S. M. Faisal and T. Zaidi, "Timestamp-based detection of Sybil attack in VANET," *Int. J. Netw. Secur.*, vol. 22, no. 3, pp. 399–410, 2020.
- [23] K. Mahmood, J. Arshad, S. A. Chaudhry, and S. Kumari, "An enhanced anonymous identity-based key agreement protocol for smart grid advanced metering infrastructure," *Int. J. Commun. Syst.*, vol. 32, no. 16, Nov. 2019, Art. no. e4137.
- [24] Y. Mirsky, T. Doitsman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," 2018, *arXiv:1802.09089*.
- [25] S. Sharma, H. Sharma, and J. B. Sharma, "An adaptive color image watermarking using RDWT-SVD and artificial bee colony based quality metric strength factor optimization," *Appl. Soft Comput.*, vol. 84, Nov. 2019, Art. no. 105696.
- [26] A. Nayyar, "Flying adhoc network (FANETs): Simulation based performance comparison of routing protocols: AODV, DSDV, DSR, OLSR, AOMDV and HWMP," in *Proc. Int. Conf. Adv. Big Data, Comput. Data Commun. Syst. (icABCD)*, Durban, South Africa, Aug. 2018, pp. 1–9.
- [27] I. Naqvi, A. Chaudhary, and A. Rana, "Intrusion detection in VANETs," in *Proc. 9th Int. Conf. Rel., INFOCOM Technol. Optim. (Trends Future Directions) (ICRITO)*, Noida, India, Sep. 2021, pp. 1–5.
- [28] M. Yao, X. Wang, Q. Gan, Y. Lin, and C. Huang, "An improved and privacy-preserving mutual authentication scheme with forward secrecy in VANETs," *Secur. Commun. Netw.*, vol. 2021, Apr. 2021, Art. no. 6698099.
- [29] J. Firl, H. Stübging, S. A. Huss, and C. Stiller, "MARV-X: Applying maneuver assessment for reliable verification of car-to-X mobility data," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 3, pp. 1301–1312, Sep. 2013.
- [30] F. Guo et al., "Detecting vehicle anomaly in the edge via sensor consistency and frequency characteristic," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 5618–5628, Jun. 2019.
- [31] R. W. van der Heijden, S. Dietzel, T. Leinmüller, and F. Kargl, "Survey on misbehavior detection in cooperative intelligent transportation systems," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 779–811, 1st Quart., 2019.
- [32] H. Zhu, K.-V. Yuen, L. Mihaylova, and H. Leung, "Overview of environment perception for intelligent vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 10, pp. 2584–2601, Oct. 2017.
- [33] T. Zhou, R. R. Choudhury, P. Ning, and K. Chakrabarty, "P2DAP—Sybil attacks detection in vehicular ad hoc networks," *IEEE J. Sel. Areas Commun.*, vol. 29, no. 3, pp. 582–594, Mar. 2011.
- [34] A. Abdelaziz, R. Burton, F. Barickman, J. Martin, J. Weston, and C. E. Koksall, "Enhanced authentication based on angle of signal arrivals," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4602–4614, May 2019.
- [35] V.-L. Nguyen, P.-C. Lin, and R.-H. Hwang, "Multi-array relative positioning for verifying the truthfulness of V2X messages," *IEEE Commun. Lett.*, vol. 23, no. 10, pp. 1704–1707, Oct. 2019.
- [36] M. Naserian and A. Lewis, "Detecting misbehavior in vehicle-to-vehicle communications," U.S. Patent 9 865 168, Jan. 9, 2018.
- [37] A. Talpur and M. Gurusamy, "Machine learning for security in vehicular networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 1, pp. 346–379, 1st Quart., 2022.
- [38] X. Sun, F. R. Yu, and P. Zhang, "A survey on cyber-security of connected and autonomous vehicles (CAVs)," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6240–6259, Jul. 2022.
- [39] M. Dibaei et al., "Attacks and defences on intelligent connected vehicles: A survey," *Digit. Commun. Netw.*, vol. 6, no. 4, pp. 399–421, 2020.
- [40] Z. Ying, K. Wang, J. Xiong, and M. Ma, "A literature review on V2X communications security: Foundation, solutions, status, and future," *IET Commun.*, vol. 18, no. 20, pp. 1683–1715, Dec. 2024.
- [41] K. H. M. Gularte et al., "Safeguarding the V2X pathways: Exploring the cybersecurity landscape through systematic review," *IEEE Access*, vol. 12, pp. 72871–72895, 2024.
- [42] J. R. V. Solaas, N. Tuptuk, and E. Mariconti, "Systematic review: Anomaly detection in connected and autonomous vehicles," 2024, *arXiv:2405.02731*.
- [43] A. Boualouache and T. Engel, "A survey on machine learning-based misbehavior detection systems for 5G and beyond vehicular networks," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1128–1172, 2nd Quart., 2023.

- [44] F. Sakiz and S. Sen, "A survey of attacks and detection mechanisms on intelligent transportation systems: VANETs and IoV," *Ad Hoc Netw.*, vol. 61, pp. 33–50, Jun. 2017.
- [45] J. Nagarajan et al., "Machine learning based intrusion detection systems for connected autonomous vehicles: A survey," *Peer-Peer Netw. Appl.*, vol. 16, no. 5, pp. 2153–2185, Sep. 2023.
- [46] R. Sedar, C. Kalalas, F. Vázquez-Gallego, L. Alonso, and J. Alonso-Zarate, "A comprehensive survey of V2X cybersecurity mechanisms and future research paths," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 325–391, 2023.
- [47] S. Sharma and A. Kaul, "A survey on intrusion detection systems and honeypot based proactive security mechanisms in VANETs and VANET cloud," *Veh. Commun.*, vol. 12, pp. 138–164, Apr. 2018.
- [48] F. Tang, Y. Kawamoto, N. Kato, and J. Liu, "Future intelligent and secure vehicular network toward 6G: Machine-learning approaches," *Proc. IEEE*, vol. 108, no. 2, pp. 292–307, Feb. 2020.
- [49] G. K. Rajbahadur, A. J. Malton, A. Walenstein, and A. E. Hassan, "A survey of anomaly detection for connected vehicle cybersecurity and safety," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 421–426.
- [50] A. M. Alrehan and F. A. Alhaidari, "Machine learning techniques to detect DDoS attacks on VANET system: A survey," in *Proc. 2nd Int. Conf. Comput. Appl. Inf. Secur. (ICCAIS)*, May 2019, pp. 1–6.
- [51] F. Gonçalves et al., "A systematic review on intelligent intrusion detection systems for VANETs," in *Proc. 11th Int. Congr. Ultra Modern Telecommun. Control Syst. Workshops (ICUMT)*, Oct. 2019, pp. 1–10.
- [52] P. Asuquo et al., "Security and privacy in location-based services for vehicular and mobile communications: An overview, challenges, and countermeasures," *IEEE Internet Things J.*, vol. 5, no. 6, pp. 4778–4802, Dec. 2018.
- [53] L. Liang, H. Ye, and G. Y. Li, "Toward intelligent vehicular networks: A machine learning framework," *IEEE Internet Things J.*, vol. 6, no. 1, pp. 124–135, Feb. 2019.
- [54] S. Kuutti, R. Bowden, Y. Jin, P. Barber, and S. Fallah, "A survey of deep learning applications to autonomous vehicle control," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 2, pp. 712–733, Feb. 2021.
- [55] M. A. Hossain, R. M. Noor, K. A. Yau, S. R. Azzuhri, M. R. Z'aba, and I. Ahmedy, "Comprehensive survey of machine learning approaches in cognitive radio-based vehicular ad hoc networks," *IEEE Access*, vol. 8, pp. 78054–78108, 2020.
- [56] Z. Lu, G. Qu, and Z. Liu, "A survey on recent advances in vehicular network security, trust, and privacy," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 2, pp. 760–776, Feb. 2019.
- [57] M. S. Sheikh and J. Liang, "A comprehensive survey on VANET security services in traffic management system," *Wireless Commun. Mobile Comput.*, vol. 2019, Sep. 2019, Art. no. 2423915.
- [58] W. Tong, A. Hussain, W. X. Bo, and S. Maharjan, "Artificial intelligence for Vehicle-to-everything: A survey," *IEEE Access*, vol. 7, pp. 10823–10843, 2019.
- [59] E. Farsimadan, L. Moradi, and F. Palmieri, "A review on security challenges in V2X communications technology for VANETs," *IEEE Access*, vol. 13, pp. 31069–31094, 2025, doi: [10.1109/ACCESS.2025.3541035](https://doi.org/10.1109/ACCESS.2025.3541035).
- [60] X. Zhang et al., "Vehicle-to-everything communication in intelligent connected vehicles: A survey and taxonomy," *Automot. Innov.*, vol. 8, no. 1, pp. 13–45, Feb. 2025.
- [61] A. Maria, M. Biagi, and R. Cusani, "Smart vehicles, technologies and main applications in vehicular ad hoc networks," *Veh. Technol.-Deployment Appl.*, vol. 2013, pp. 3–20, Feb. 2013.
- [62] S. Chen et al., "Vehicle-to-Everything (V2X) services supported by LTE-based systems and 5G," *IEEE Commun. Standards Mag.*, vol. 1, no. 2, pp. 70–76, Feb. 2017.
- [63] N. Xia and C.-S. Yang, "Vehicular communications: Standards and challenges," in *Proc. 4th Int. Conf. Internet Vehicles. Technol. Services Smart Cities (IOV)*, Kanazawa, Japan, Cham, Switzerland: Springer, Jan. 2017, pp. 1–12.
- [64] J. Kolleda et al., "National security credential management system (SCMS) deployment support: SCMS baseline summary report," U.S. Dept. Transp. (USDOT), Intell. Transp. Syst. Joint Program Office (ITS JPO), Washington, DC, USA, Tech. Rep. FHWA-JPO-18-686, 2018.
- [65] R. D. Murrall, M. D. Furtado, and H. Liu, "EmuLab of security credential management system (SCMS) for vehicular communications," in *Proc. IEEE 88th Veh. Technol. Conf. (VTC-Fall)*, Aug. 2018, pp. 1–5.
- [66] S. A. Abdel Hakeem and H. Kim, "Centralized threshold key generation protocol based on Shamir secret sharing and HMAC authentication," *Sensors*, vol. 22, no. 1, p. 331, Jan. 2022, doi: [10.3390/s22010331](https://doi.org/10.3390/s22010331).
- [67] M. Houmer, M. Ouaisa, and M. Ouaisa, "Secure authentication scheme for 5G-based V2X communications," *Proc. Comput. Sci.*, vol. 198, pp. 276–281, Sep. 2022.
- [68] S. A. Abdel Hakeem and H. Kim, "Authentication and encryption protocol with revocation and reputation management for enhancing 5G-V2X security," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 35, no. 7, Jul. 2023, Art. no. 101638, doi: [10.1016/j.jksuci.2023.101638](https://doi.org/10.1016/j.jksuci.2023.101638).
- [69] Y. Yang, Z. Wei, Y. Zhang, H. Lu, K.-K.-R. Choo, and H. Cai, "V2X security: A case study of anonymous authentication," *Pervas. Mobile Comput.*, vol. 41, pp. 259–269, Oct. 2017.
- [70] J. Camenisch, M. Drijvers, A. Lehmann, G. Neven, and P. Towa, "Zone encryption with anonymous authentication for V2V communication," in *Proc. IEEE Eur. Symp. Secur. Privacy (EuroS&P)*, Sep. 2020, pp. 405–424.
- [71] M. Wu et al., "Research on certificate management and key management of C-V2X security authentication technology in intelligent network vehicle," *Proc. SPIE*, vol. 12641, pp. 115–123, May 2023.
- [72] G. Rigazzi, A. Tassi, R. J. Piechocki, T. Tryfonas, and A. Nix, "Optimized certificate revocation list distribution for secure V2X communications," in *Proc. IEEE 86th Veh. Technol. Conf. (VTC-Fall)*, Sep. 2017, pp. 1–7.
- [73] W. Whyte, A. Weimerskirch, V. Kumar, and T. Hehn, "A security credential management system for V2V communications," in *Proc. IEEE Veh. Netw. Conf.*, Dec. 2013, pp. 1–8.
- [74] T. Yoshizawa and B. Preneel, "A new approach to pseudonym certificate management in V2X communication," in *Proc. IEEE Veh. Netw. Conf. (VNC)*, Apr. 2023, pp. 25–32.
- [75] T. Yoshizawa and B. Preneel, "On handling of certificate digest in V2X communication," in *Proc. 18th Int. Conf. Wireless Mobile Comput., Netw. Commun. (WiMob)*, Oct. 2022, pp. 160–165.
- [76] H. Aliev, H. Kim, and S. Choi, "A scalable and secure group key management method for secure V2V communication," *Sensors*, vol. 20, no. 21, p. 6137, Oct. 2020.
- [77] V. Sharma, I. You, and N. Guizani, "Security of 5G-V2X: Technologies, standardization, and research directions," *IEEE Netw.*, vol. 34, no. 5, pp. 306–314, Sep. 2020.
- [78] S. A. Abdel Hakeem, M. A. Abd El-Gawad, and H. Kim, "A decentralized lightweight authentication and privacy protocol for vehicular networks," *IEEE Access*, vol. 7, pp. 119689–119705, 2019.
- [79] S. Anbalagan, G. Raja, S. Gurumoorthy, R. D. Suresh, and K. Dev, "IIDS: Intelligent intrusion detection system for sustainable development in autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15866–15875, Dec. 2023.
- [80] E. S. Ali et al., "Machine learning technologies for secure vehicular communication in Internet of Vehicles: Recent advances and applications," *Secur. Commun. Netw.*, vol. 2021, Mar. 2021, Art. no. 8868355.
- [81] N. Trkulja, D. Starobinski, and R. A. Berry, "Denial-of-Service attacks on C-V2X networks," 2020, *arXiv:2010.13725*.
- [82] SolarMainframe. *IDS Intrusion CSV*. Accessed: Apr. 25, 2025. [Online]. Available: <https://www.kaggle.com/datasets/solarmainframe/ids-intrusion-csv>
- [83] M. N. Saulaiman, M. Kozlovsky, and Á. Csilling, "A survey on vulnerabilities and classification of cyber-attacks on 5G-V2X," in *Proc. IEEE 21st Int. Symp. Comput. Intell. Informat. (CINTI)*, Nov. 2021, pp. 000235–000240.
- [84] N. Mazher, M. Alhadaad, and O. Shagdar, "A brief summary of cybersecurity attacks in V2X communication," Nat. Renew. Energy Lab. (NREL), Golden, CO, USA, Tech. Rep., 2022.
- [85] S. A. Abdel Hakeem, A. A. Hady, and H. Kim, "5G-V2X: Standardization, architecture, use cases, network-slicing, and edge-computing," *Wireless Netw.*, vol. 26, no. 8, pp. 6015–6041, Nov. 2020.
- [86] M. A. Mohiuddin, K. Nirosha, D. Anusha, M. Nazeer, G. R. Nv, and S. Lakhanpal, "AI to V2X privacy and security issues in autonomous vehicles: Survey," *MATEC Web Erences*, vol. 392, Jan. 2024, Art. no. 01097.
- [87] S. Ullah et al., "TNN-IDS: Transformer neural network-based intrusion detection system for MQTT-enabled IoT networks," *Comput. Netw.*, vol. 237, Dec. 2023, Art. no. 110072.
- [88] M. Muhammad and G. A. Safdar, "Survey on existing authentication issues for cellular-assisted V2X communication," *Veh. Commun.*, vol. 12, pp. 50–65, Apr. 2018.
- [89] S. A. A. Hakeem, S. M. A. El-Kader, and H. Kim, "A key management protocol based on the hash chain key generation for securing LoRaWAN networks," *Sensors*, vol. 21, no. 17, p. 5838, Aug. 2021, doi: [10.3390/s21175838](https://doi.org/10.3390/s21175838).

- [90] S. A. A. Hakeem, H. H. Hussein, and H. Kim, "Security requirements and challenges of 6G technologies and applications," *Sensors*, vol. 22, no. 5, p. 1969, Mar. 2022, doi: [10.3390/s22051969](#).
- [91] S. Taha and X. Shen, "Lightweight group authentication with dynamic vehicle-clustering for 5G-based V2X communications," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2018, pp. 1–6.
- [92] P. Sharma, D. Austin, and H. Liu, "Attacks on machine learning: Adversarial examples in connected and autonomous vehicles," in *Proc. IEEE Int. Symp. Technol. Homeland Secur. (HST)*, Woburn, MA, USA, Nov. 2019, pp. 1–7, doi: [10.1109/HST47167.2019.9032989](#).
- [93] R. Sedar, C. Kalalas, F. Vázquez-Gallego, and J. Alonso-Zárate, "Deep reinforcement learning-based adversarial defense in vehicular communication systems," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2024, pp. 5250–5256, doi: [10.1109/icc51166.2024.10622762](#).
- [94] Z. H. Khattak, B. L. Smith, and M. D. Fontaine, "Cyberattack monitoring architectures for resilient operation of connected and automated vehicles," *IEEE Open J. Intell. Transp. Syst.*, vol. 5, pp. 322–341, 2024, doi: [10.1109/OJITS.2024.3391830](#).
- [95] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *Proc. 5th Int. Conf. Learn. Represent.*, Toulon, France, 2017, pp. 1–14.
- [96] A. Lad, R. Bhale, and S. Belgamwar, "Fast gradient sign method (FGSM) variants in white box settings: A comparative study," in *Proc. Int. Conf. Inventive Comput. Technol. (ICICT)*, Lalitpur, Nepal, Apr. 2024, pp. 382–386, doi: [10.1109/icict60155.2024.10544606](#).
- [97] J. Zhang, W. Qian, R. Nie, J. Cao, and D. Xu, "Generate adversarial examples by adaptive moment iterative fast gradient sign method," *Int. J. Speech Technol.*, vol. 53, no. 1, pp. 1101–1114, Jan. 2023, doi: [10.1007/s10489-022-03437-z](#).
- [98] L. He, Z. Wang, S. Yang, T. Liu, and Y. Huang, "Generalizing projected gradient descent for deep-learning-aided massive MIMO detection," *IEEE Trans. Wireless Commun.*, vol. 23, no. 3, pp. 1827–1839, Mar. 2024.
- [99] Y. A. Ergu et al., "Efficient adversarial attacks against DRL-based resource allocation in intelligent O-RAN for V2X," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, pp. 1674–1686, Jan. 2025.
- [100] S. Soltani, M. Shojafar, A. Brighente, M. Conti, and R. Tafazolli, "Poisoning bearer context migration in O-RAN 5G network," *IEEE Wireless Commun. Lett.*, vol. 12, no. 3, pp. 401–405, Mar. 2023.
- [101] S. A. A. Hakeem, H. H. Hussein, and H. Kim, "Vision and research directions of 6G technologies and applications," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 6, pp. 2419–2442, Jun. 2022, doi: [10.1016/j.jksuci.2022.03.019](#).
- [102] A. Alshammari, M. A. Zohdy, D. Debnath, and G. Corser, "Classification approach for intrusion detection in vehicle systems," *Wireless Eng. Technol.*, vol. 9, no. 4, pp. 79–94, 2018.
- [103] G. Karopoulos, G. Kambourakis, E. Chatzoglou, J. L. Hernández-Ramos, and V. Kouliaridis, "Demystifying in-vehicle intrusion detection systems: A survey of surveys and a meta-taxonomy," *Electronics*, vol. 11, no. 7, p. 1072, Mar. 2022.
- [104] F. Gonçalves, J. Macedo, and A. Santos, "Intelligent hierarchical intrusion detection system for VANETs," in *Proc. 13th Int. Congr. Ultra Modern Telecommun. Control Syst. Workshops (ICUMT)*, Oct. 2021, pp. 50–59.
- [105] B. Lampe and W. Meng, "Intrusion detection in the automotive domain: A comprehensive review," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2356–2426, 4th Quart., 2023.
- [106] G. Raja et al., "AI-empowered trajectory anomaly detection and classification in 6G-V2X," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 4599–4607, Apr. 2023.
- [107] D. Man, F. Zeng, J. Lv, S. Xuan, W. Yang, and M. Guizani, "AI-based intrusion detection for intelligence Internet of Vehicles," *IEEE Consum. Electron. Mag.*, vol. 12, no. 1, pp. 109–116, Jan. 2023.
- [108] S. Rajapaksha, H. Kalutarage, M. O. Al-Kadri, A. Petrovski, G. Madzudzo, and M. Cheah, "AI-based intrusion detection systems for in-vehicle networks: A survey," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–40, Nov. 2023.
- [109] W. Gou, H. Zhang, and R. Zhang, "Multi-classification and tree-based ensemble network for the intrusion detection system in the Internet of Vehicles," *Sensors*, vol. 23, no. 21, p. 8788, Oct. 2023.
- [110] T. Schlegel, P. Seeböck, S. M. Waldstein, U. Schmidt-Erfurth, and G. Langs, *Unsupervised Anomaly Detection With Generative Adversarial Networks To Guide Marker Discovery*. Cham, Switzerland: Springer, Jan. 2017, pp. 146–157.
- [111] W. Xu, J. Jang-Jaccard, A. Singh, Y. Wei, and F. Sabrina, "Improving performance of autoencoder-based network anomaly detection on NSL-KDD dataset," *IEEE Access*, vol. 9, pp. 140136–140146, 2021.
- [112] M. S. Amir et al., "Efficient & sustainable intrusion detection system using machine learning & deep learning for IoT," in *Proc. 4th Int. Conf. Comput., Math. Eng. Technol.*, May 2023, pp. 1–6.
- [113] F. A. Khan, A. Gumaei, A. Derhab, and A. Hussain, "A novel two-stage deep learning model for efficient network intrusion detection," *IEEE Access*, vol. 7, pp. 30373–30385, 2019.
- [114] K. Kim, M. E. Aminanto, and H. C. Tanuwidjaja, *Network Intrusion Detection Using Deep Learning: A Feature Learning Approach*. Cham, Switzerland: Springer, 2018.
- [115] C. Zhang, D. Jia, L. Wang, W. Wang, F. Liu, and A. Yang, "Comparative research on network intrusion detection methods based on machine learning," *Comput. Secur.*, vol. 121, Oct. 2022, Art. no. 102861.
- [116] J. Grover, N. K. Prajapati, V. Laxmi, and M. S. Gaur, "Machine learning approach for multiple misbehavior detection in VANET," in *Advanced Computational and Communication Paradigms*. Berlin, Germany: Springer, 2011, pp. 644–653.
- [117] S. Y. Wang and C. L. Chou, "NCTUns 5.0 network simulator for advanced wireless vehicular network researches," in *Proc. 10th Int. Conf. Mobile Data Manage., Syst., Services Middleware*, May 2009, pp. 375–376.
- [118] W. Li, A. Joshi, and T. Finin, "SVM-CASE: An SVM-based context aware security framework for vehicular ad-hoc networks," in *Proc. IEEE 82nd Veh. Technol. Conf. (VTC-Fall)*, Sep. 2015, pp. 1–5.
- [119] M. Sarhan, S. Layeghy, N. Moustafa, M. Gallagher, and M. Portmann, "Feature extraction for machine learning-based intrusion detection in IoT networks," 2021, *arXiv:2108.12722*.
- [120] L. Yang, A. Moubayed, and A. Shami, "MTH-IDS: A multitiered hybrid intrusion detection system for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 616–632, Jan. 2022.
- [121] H. Sedjelmaci and S. M. Senouci, "An accurate and efficient collaborative intrusion detection framework to secure vehicular networks," *Comput. Elect. Eng.*, vol. 43, pp. 33–47, Apr. 2015.
- [122] Y. Li, F. Li, and J. Song, "The research of random forest intrusion detection model based on optimization in Internet of Vehicles," *J. Phys., Conf. Ser.*, vol. 1757, no. 1, Jan. 2021, Art. no. 012149.
- [123] E. A. Shams, A. Rizaner, and A. H. Ulusoy, "Trust aware support vector machine intrusion detection and prevention system in vehicular ad hoc networks," *Comput. Secur.*, vol. 78, pp. 245–254, Sep. 2018.
- [124] A. Shenfield, D. Day, and A. Ayesh, "Intelligent intrusion detection systems using artificial neural networks," *ICT Exp.*, vol. 4, no. 2, pp. 95–99, 2018.
- [125] S. Mane and D. Rao, "Explaining network intrusion detection system using explainable AI framework," 2021, *arXiv:2103.07110*.
- [126] G. Zhao, C. Zhang, and L. Zheng, "Intrusion detection using deep belief network and probabilistic neural network," in *Proc. IEEE Int. Conf. Comput. Sci. Eng. (CSE) IEEE Int. Conf. Embedded Ubiquitous Comput. (EUC)*, vol. 1, Jul. 2017, pp. 639–642.
- [127] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON-IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.
- [128] A. R. Gad, A. A. Nashat, and T. M. Barkat, "Intrusion detection system using machine learning for vehicular ad hoc networks based on ToN-IoT dataset," *IEEE Access*, vol. 9, pp. 142206–142217, 2021.
- [129] J. Grover, V. Laxmi, and M. S. Gaur, "Misbehavior detection based on ensemble learning in VANET," in *Proc. Adv. Comput. Commun. Secur.*, Jan. 2012, pp. 602–611.
- [130] P. K. Singh, S. Gupta, R. Vashistha, S. K. Nandi, and S. Nandi, "Machine learning-based approach to detect position falsification attack in VANETs," in *Security Privacy*. Singapore: Springer, 2019, pp. 166–178.
- [131] R. W. van der Heijden, T. Lukaseder, and F. Kargl, "VeReMi: A dataset for comparable evaluation of misbehavior detection in VANETs," in *Security Privacy Communication Networks*. Cham, Switzerland: Springer, 2018, pp. 318–337.
- [132] S. So, P. Sharma, and J. Petit, "Integrating plausibility checks and machine learning for misbehavior detection in VANET," in *Proc. 17th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2018, pp. 564–571.
- [133] O. A. Wahab, A. Mourad, H. Otrouk, and J. Bentahar, "CEAP: SVM-based intelligent detection model for clustered vehicular ad hoc networks," *Exp. Syst. Appl.*, vol. 50, pp. 40–54, May 2016.

- [134] S. Gyawali, Y. Qian, and R. Q. Hu, "Machine learning and reputation based misbehavior detection in vehicular communication networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 8871–8885, Aug. 2020.
- [135] F. Hawlader, A. Boualouache, S. Faye, and T. Engel, "Intelligent misbehavior detection system for detecting false position attacks in vehicular networks," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC)*, Aug. 2021, pp. 1–6.
- [136] A. Sharma and A. Jaekel, "Machine learning based misbehaviour detection in VANET using consecutive BSM approach," *IEEE Open J. Veh. Technol.*, vol. 3, pp. 1–14, 2022.
- [137] S. Ercan, M. Ayaida, and N. Messai, "Misbehavior detection for position falsification attacks in VANETs using machine learning," *IEEE Access*, vol. 10, pp. 1893–1904, 2022.
- [138] Z. Yang, K. Zhang, L. Lei, and K. Zheng, "A novel classifier exploiting mobility behaviors for Sybil detection in connected vehicle systems," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2626–2636, Apr. 2019.
- [139] S. Tariq, S. Lee, and S. S. Woo, "CANTransfer: Transfer learning based intrusion detection on a controller area network using convolutional LSTM network," in *Proc. 35th Annu. ACM Symp. Appl. Comput.*, Mar. 2020, pp. 1048–1055.
- [140] K. A. Tait et al., "Intrusion detection using machine learning techniques: An experimental comparison," 2021, *arXiv:2105.13435*.
- [141] A. Rosay, E. Cheval, F. Carlier, and P. Leroux, "CIC-IDS2017 dataset report," Canadian Inst. Cybersecurity (CIC), Univ. New Brunswick, Fredericton, NB, Canada, Tech. Rep., 2022.
- [142] M. Almi'ani, A. A. Ghazleh, A. Al-Rahayfeh, and A. Razaque, "Intelligent intrusion detection system using clustered self organized map," in *Proc. 5th Int. Conf. Softw. Defined Syst. (SDS)*, Barcelona, Spain, Apr. 2018, pp. 138–144.
- [143] L. A. Maglaras, "A novel distributed intrusion detection system for vehicular ad hoc networks," *Int. J. Adv. Comput. Sci. Appl.*, vol. 6, no. 4, pp. 101–106, 2015.
- [144] M. E. Verma et al., "A comprehensive guide to CAN IDS data and introduction of the ROAD dataset," 2020, *arXiv:2012.14600*.
- [145] F. A. Alhaidari and A. M. Alrehan, "A simulation work for generating a novel dataset to detect distributed denial of service attacks on vehicular ad hoc NETWORK systems," *Int. J. Distrib. Sensor Netw.*, vol. 17, no. 3, Mar. 2021, Art. no. 155014772110002, doi: [10.1177/15501477211000287](https://doi.org/10.1177/15501477211000287).
- [146] *Veremi Dataset*. Accessed: Apr. 25, 2025. [Online]. Available: <https://veremi-dataset.github.io/>
- [147] R. Rahal, A. A. Korba, and N. Ghoualmi-Zine, "Towards the development of realistic DoS dataset for intelligent transportation systems," *Wireless Pers. Commun.*, vol. 115, no. 2, pp. 1415–1444, Nov. 2020, doi: [10.1007/s11277-020-07635-1](https://doi.org/10.1007/s11277-020-07635-1).
- [148] S. Kumar, Sunanda, and S. Arora, "A statistical analysis on KDD cup'99 dataset for the network intrusion detection system," in *Applied Soft Computing and Communication Networks*, vol. 125, S. M. Thampi, Ed., Singapore: Springer, 2020, pp. 83–92.
- [149] H. P. Vinutha and B. Poornima, "Analysis of NSL-KDD dataset using K-means and canopy clustering algorithms based on distance metrics," in *Integrated Intelligent Computing, Communication and Security*, vol. 771. Singapore: Springer, 2019, pp. 239–248.
- [150] N. Moustafa, "ToN_IoT datasets," IEEE Dataport, Piscataway, NJ, USA, Tech. Rep., Oct. 2019, doi: [10.21227/fesz-dm97](https://doi.org/10.21227/fesz-dm97).
- [151] Can. Inst. Cybersecurity. (2028). *Intrusion Detection Evaluation Dataset (CSE-CIC-IDS2018)*. [Online]. Available: <https://www.unb.ca/cic/datasets/ids-2018.html>
- [152] *The UNSW-NB15 Dataset*. Accessed: Jul. 10, 2024. [Online]. Available: <https://research.unsw.edu.au/projects/unswnb15-dataset>
- [153] N. Moustafa and J. Slay, "The evaluation of network anomaly detection systems: Statistical analysis of the UNSW-NB15 data set and the comparison with the KDD99 data set," *Inf. Secur. J., A Global Perspective*, vol. 25, nos. 1–3, pp. 18–31, Apr. 2016.
- [154] M. Al-Hawawreh, E. Simikova, and N. Aboutorab, "X-IIoTID: A connectivity- and device-agnostic intrusion dataset for industrial Internet of Things," IEEE Dataport, Piscataway, NJ, USA, Tech. Rep., Jul. 2021, doi: [10.21227/mpb6-py55](https://doi.org/10.21227/mpb6-py55).
- [155] S. T. Banafshehvaragh and A. M. Rahmani, "Intrusion, anomaly, and attack detection in smart vehicles," *Microprocessors Microsyst.*, vol. 96, Feb. 2023, Art. no. 104726.
- [156] *CAN Dataset for intrusion detection (OTIDS)*. Accessed: Jan. 18, 2024. [Online]. Available: <https://ocslab.hksecurity.net/Dataset/CAN-intrusion-dataset>
- [157] *Car-Hacking Dataset*. Accessed: Oct. 18, 2024. [Online]. Available: <https://ocslab.hksecurity.net/Datasets/car-hacking-dataset>
- [158] S. Burschka and B. Dupasquier, "Tranalyzer: Versatile high performance network traffic analyzer," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2016, pp. 1–8.
- [159] A. Azab, M. Khasawneh, S. Alrabaa, K.-K.-R. Choo, and M. Sarsour, "Network traffic classification: Techniques, datasets, and challenges," *Digit. Commun. Netw.*, vol. 10, no. 3, pp. 676–692, Jun. 2024.
- [160] E. Moradi-Pari, D. Tian, M. Bahramgiri, S. Rajab, and S. Bai, "DSRC versus LTE-V2X: Empirical performance analysis of direct vehicular communication technologies," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 4889–4903, May 2023.
- [161] N. T. Tangirala, A. Abraham, A. Choudhury, P. Vyas, R. Zhang, and J. Dauwels, "Analysis of packet drops and channel crowding in vehicle platooning using V2X communication," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Nov. 2018, pp. 281–286.
- [162] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th AISTATS*, Fort Lauderdale, FL, USA, Jan. 2016, pp. 1–8.
- [163] T. Li et al., "Federated optimization in heterogeneous networks," in *Proc. 3rd MLSys Conf.*, Austin, TX, USA, 2020, pp. 1–13.
- [164] R. Song, L. Zhou, V. Lakshminarasimhan, A. Festag, and A. Knoll, "Federated learning framework coping with hierarchical heterogeneity in cooperative ITS," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Macau, China, Oct. 2022, pp. 3502–3508.
- [165] S. Narkedimilli et al., "FL-DECO-BC: A privacy-preserving, secure, and provenance-preserving FL framework with blockchain for VANETs," 2024, *arXiv:2407.21141*.
- [166] L. Alekszejenko et al., "A V2X-based privacy-preserving federated learning system," 2024, *arXiv:2401.13848*.
- [167] H. Korba, "Federated learning-based intrusion detection in vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2456–2472, Feb. 2024.
- [168] Y. Zhou, "Privacy-preserving federated learning for autonomous vehicles," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 6783–6797, May 2024.
- [169] X. Li, "Federated intrusion detection for V2X networks: Challenges and solutions," *Comput. Netw.*, vol. 225, Jul. 2024, Art. no. 109835.
- [170] J. Huang, "An execution & evaluation dual-network FL framework for edge security in IoT," *IEEE Trans. Dependable Secure Comput.*, vol. 21, no. 1, pp. 89–102, Jan. 2024.
- [171] M. Raza, "SDN-enabled federated learning for secure V2X communication," *IEEE Trans. Mobile Comput.*, vol. 23, no. 3, pp. 1347–1361, Mar. 2024.
- [172] X. Chen, W. Qiu, L. Chen, Y. Ma, and J. Ma, "Fast and practical intrusion detection system based on federated learning for VANET," *Comput. Secur.*, vol. 142, Jul. 2024, Art. no. 103881, doi: [10.1016/j.cose.2024.103881](https://doi.org/10.1016/j.cose.2024.103881).
- [173] K. Huang, R. Xian, M. Xian, H. Wang, and L. Ni, "A comprehensive intrusion detection method for the Internet of Vehicles based on federated learning architecture," *Comput. Secur.*, vol. 147, Dec. 2024, Art. no. 104067, doi: [10.1016/j.cose.2024.104067](https://doi.org/10.1016/j.cose.2024.104067).
- [174] A. Renda et al., "Federated learning of explainable AI models in 6G systems: Towards secure and automated vehicle networking," *Information*, vol. 13, no. 8, p. 395, Aug. 2022, doi: [10.3390/info13080395](https://doi.org/10.3390/info13080395).
- [175] M. Doostmohammadian, H. Zarrabi, H. R. Rabiee, U. A. Khan, and T. Charalambous, "Distributed detection and mitigation of biasing attacks over multi-agent networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 4, pp. 3465–3477, Oct. 2021, doi: [10.1109/TNSE.2021.3115032](https://doi.org/10.1109/TNSE.2021.3115032).
- [176] P. H. Mirzaee, M. Shojafar, H. Bagheri, T. H. Chan, H. Cruickshank, and R. Tafazolli, "A two-layer collaborative vehicle-edge intrusion detection system for vehicular communications," in *Proc. IEEE 94th Veh. Technol. Conf. (VTC-Fall)*, Norman, OK, USA, Sep. 2021, pp. 1–6, doi: [10.1109/VTC2021-Fall52928.2021.9625388](https://doi.org/10.1109/VTC2021-Fall52928.2021.9625388).
- [177] D. J. Yeong, K. Panduru, and J. Walsh, "Exploring the unseen: A survey of multi-sensor fusion and the role of explainable AI (XAI) in autonomous vehicles," *Sensors*, vol. 25, no. 3, p. 856, Jan. 2025, doi: [10.3390/s25030856](https://doi.org/10.3390/s25030856).
- [178] M. Doostmohammadian and T. Charalambous, "Distributed anomaly detection and estimation over sensor networks: Observational-equivalence and Q-redundant observer design," in *Proc. Eur. Control Conf. (ECC)*, London, U.K., Jul. 2022, pp. 460–465, doi: [10.23919/ECC55457.2022.9838396](https://doi.org/10.23919/ECC55457.2022.9838396).

- [179] M. Zabihi, R. V. Mehrizi, A. Kasaiezadeh, M. Pirani, and A. Khajepour, "A hybrid model-data vehicle sensor and actuator fault detection and diagnosis system," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 8121–8133, Jul. 2024.
- [180] X. He, X. Ren, H. Sandberg, and K. H. Johansson, "How to secure distributed filters under sensor attacks," *IEEE Trans. Autom. Control*, vol. 67, no. 6, pp. 2843–2856, Jun. 2022, doi: [10.1109/TAC.2021.3092603](https://doi.org/10.1109/TAC.2021.3092603).
- [181] M. Doostmohammadian and H. R. Rabiee, "Distributed observer design for tracking platoon of connected and autonomous vehicles," 2025, *arXiv:2501.18890*.
- [182] K. Samy, "Edge AI-based real-time intrusion detection for smart vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 4, pp. 3401–3416, May 2024.
- [183] T. Alladi, "MEC-enabled intrusion detection systems for secure V2X networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 2, pp. 211–226, Feb. 2024.
- [184] M. Hasan, "Lightweight federated learning for energy-efficient IoT security," *IEEE Access*, vol. 12, pp. 7456–7472, 2024.
- [185] H. Bangui, "Hybrid AI-driven security mechanism for V2X networks," *ACM Trans. Cyber-Phys. Syst.*, vol. 8, no. 1, pp. 56–72, 2024.
- [186] M. Rihan, M. Elwekeil, Y. Yang, L. Huang, C. Xu, and M. M. Selim, "Deep-VFog: When artificial intelligence meets fog computing in V2X," *IEEE Syst. J.*, vol. 15, no. 3, pp. 3492–3505, Sep. 2021.
- [187] K. L. Williams, Y. D. Prasanth, and M. Jeyaselvi, "Hybrid AI architecture using edge-cloud computing for secure V2X communication," in *Proc. 9th Int. Conf. Commun. Electron. Syst. (ICCES)*, Dec. 2024, pp. 913–920, doi: [10.1109/icc63552.2024.10859430](https://doi.org/10.1109/icc63552.2024.10859430).
- [188] C. Zhang, X. Liu, X. Zheng, R. Li, and H. Liu, "FengHuoLun: A federated learning based edge computing platform for cyber-physical systems," in *Proc. Int. Conf. Pervasive Comput. Commun. (PerCom) Workshops*, 2020, pp. 1–4, doi: [10.1109/PerComWorkshops48775.2020.9156259](https://doi.org/10.1109/PerComWorkshops48775.2020.9156259).
- [189] M. M. Moussa and L. Alazzawi, "A hybrid deep learning cyber-attacks intrusion detection system for CAV path planning," in *Proc. IEEE Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Aug. 2021, pp. 607–610, doi: [10.1109/MWSCAS47672.2021.9531858](https://doi.org/10.1109/MWSCAS47672.2021.9531858).
- [190] A. Patel, M. Bhavsar, and K. Roy, "Enhanced CAV security using machine learning," in *Proc. Int. Conf. Artif. Intell., Big Data, Comput. Data Commun. Syst. (icABCD)*, Aug. 2024, pp. 1–6, doi: [10.1109/icabcd62167.2024.10645287](https://doi.org/10.1109/icabcd62167.2024.10645287).
- [191] M. Kim, I. Oh, K. Yim, M. Sahlabadi, and Z. Shukur, "Security of 6G-enabled vehicle-to-everything communication in emerging federated learning and blockchain technologies," *IEEE Access*, vol. 12, pp. 33972–34001, 2024, doi: [10.1109/ACCESS.2023.3348409](https://doi.org/10.1109/ACCESS.2023.3348409).
- [192] M. Yoshida, K. Mori, T. Inoue, and H. Tanaka, "EdgeRE: An edge computing-enhanced network redundancy elimination service for connected cars," in *Proc. 6th Int. Conf. Fog Mobile Edge Comput. (FMEC)*, Dec. 2021, pp. 1–6, doi: [10.1109/FMEC54266.2021.9732548](https://doi.org/10.1109/FMEC54266.2021.9732548).
- [193] M. A. Khan et al., "Robust, resilient and reliable architecture for V2X communications," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4414–4430, Jul. 2021.
- [194] V. Rishiwal, U. Agarwal, A. Alotaibi, S. Tanwar, P. Yadav, and M. Yadav, "Exploring secure V2X communication networks for human-centric security and privacy in smart cities," *IEEE Access*, vol. 12, pp. 138763–138788, 2024, doi: [10.1109/ACCESS.2024.3467002](https://doi.org/10.1109/ACCESS.2024.3467002).
- [195] H. Ibn-Khedher, M. Laroui, H. Mounsla, H. Afifi, and E. Abd-Elrahman, "Next-generation edge computing assisted autonomous driving based artificial intelligence algorithms," *IEEE Access*, vol. 10, pp. 53987–54001, 2022, doi: [10.1109/ACCESS.2022.3174548](https://doi.org/10.1109/ACCESS.2022.3174548).
- [196] N. Albogami, "Intelligent deep federated learning model for enhancing security in Internet of Things enabled edge computing environment," *Sci. Rep.*, vol. 15, no. 1, p. 4041, Feb. 2025, doi: [10.1038/s41598-025-88163-5](https://doi.org/10.1038/s41598-025-88163-5).
- [197] R. Taslimasa, "ImageFed: Federated learning for cyberattack detection in autonomous vehicles," *IEEE Trans. Artif. Intell.*, vol. 5, no. 3, pp. 312–328, Mar. 2024.
- [198] D. Bhavsar, "Blockchain-enhanced federated learning for secure vehicular networks," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1458–1473, 2024.
- [199] D. Swessi and H. Idoudi, "A comparative review of security threats datasets for vehicular networks," in *Proc. Int. Conf. Innov. Intell. Informat., Comput., Technol. (3ICT)*, Manama, Bahrain, Sep. 2021, pp. 746–751, doi: [10.1109/3ICT53449.2021.9581683](https://doi.org/10.1109/3ICT53449.2021.9581683).
- [200] K. Geeta and K. Gulshan, "Machine learning and deep learning methods for intrusion detection systems: Recent developments and challenges," *Soft Comput.*, vol. 25, no. 15, pp. 9731–9763, 2021.
- [201] P. Vijayakumar, M. Azees, A. Kannan, and L. J. Deborah, "Dual authentication and key management techniques for secure data transmission in vehicular ad hoc networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 4, pp. 1015–1028, Apr. 2016.
- [202] G. Loukas, T. Vuong, R. Heartfield, G. Sakellari, Y. Yoon, and D. Gan, "Cloud-based cyber-physical intrusion detection for vehicles using deep learning," *IEEE Access*, vol. 6, pp. 3491–3508, 2018.
- [203] A. Rosay, F. Carlier, and P. Leroux, "Feed-forward neural network for network intrusion detection," in *Proc. IEEE 91st Veh. Technol. Conf. (VTC-Spring)*, May 2020, pp. 1–6.
- [204] O. Y. Al-Jarrah, C. Maple, M. Dianati, D. Oxtoby, and A. Mouzakitis, "Intrusion detection systems for intra-vehicle networks: A review," *IEEE Access*, vol. 7, pp. 21266–21289, 2019.
- [205] F. A. Ghaleb et al., "Misbehavior-aware on-demand collaborative intrusion detection system using distributed ensemble learning for VANET," *Electronics*, vol. 9, no. 9, p. 1411, 2020.
- [206] A. Thakkar and R. Lohiya, "A review on machine learning and deep learning perspectives of IDS for IoT: Recent updates, security issues, and challenges," *Arch. Comput. Methods Eng.*, vol. 28, no. 4, pp. 3211–3243, Jun. 2021.
- [207] M. Aloqaily, S. Otoum, I. A. Ridhawi, and Y. Jararweh, "An intrusion detection system for connected vehicles in smart cities," *Ad Hoc Netw.*, vol. 90, Jul. 2019, Art. no. 101842.
- [208] Q. He, X. Meng, R. Qu, and R. Xi, "Machine learning-based detection for cybersecurity attacks on connected and autonomous vehicles," *Mathematics*, vol. 8, no. 8, p. 1311, 2020.
- [209] N. Thapa, Z. Liu, D. B. Kc, B. Gokaraju, and K. Roy, "Comparison of machine learning and deep learning models for network intrusion detection systems," *Future Internet*, vol. 12, no. 10, p. 167, Sep. 2020.
- [210] L. Vu, Q. U. Nguyen, D. N. Nguyen, D. T. Hoang, and E. Dutkiewicz, "Deep transfer learning for IoT attack detection," *IEEE Access*, vol. 8, pp. 107335–107344, 2020.
- [211] A. Samy, H. Yu, and H. Zhang, "Fog-based attack detection framework for Internet of Things using deep learning," *IEEE Access*, vol. 8, pp. 74571–74585, 2020.
- [212] M. Roopak, G. Y. Tian, and J. Chambers, "Multi-objective-based feature selection for DDoS attack detection in IoT networks," *IET Netw.*, vol. 9, no. 3, pp. 120–127, May 2020.
- [213] H. Taslimasa et al., "ImageFed: Practical privacy preserving intrusion detection system for in-vehicle CAN bus protocol," in *Proc. IEEE IEEE 9th Intl. Conf. Big Data Secur. Cloud (BigDataSecurity) Intl. Conf. High Perform. Smart Comput., (HPSC) IEEE Intl. Conf. Intell. Data Secur. (IDS)*, May 2023, pp. 122–129.
- [214] G. Bovenzi, G. Aceto, D. Ciunzio, V. Persico, and A. Pescapé, "A hierarchical hybrid intrusion detection approach in IoT scenarios," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Taipei, Taiwan, Dec. 2020, pp. 1–7.
- [215] H. Bangui, M. Ge, and B. Buhnova, "A hybrid data-driven model for intrusion detection in VANET," *Proc. Comput. Sci.*, vol. 184, pp. 516–523, Jun. 2021.
- [216] H. Al-Khateeb, G. Epiphaniou, A. Reviczky, P. Karadimas, and H. Heidari, "Proactive threat detection for connected cars using recursive Bayesian estimation," *IEEE Sensors J.*, vol. 18, no. 12, pp. 4822–4831, Jun. 2018.
- [217] Y. Xu, J. Xia, H. Wu, and L. Fan, "Q-learning based physical-layer security game against multiagent attacks," *IEEE Access*, vol. 7, pp. 49212–49222, 2019.
- [218] M. Salehi and L. Rashidi, "A survey on anomaly detection in evolving data: [With application to forest fire risk prediction]," *ACM SIGKDD Explorations Newslett.*, vol. 20, no. 1, pp. 13–23, May 2018.
- [219] G. Pang, A. van den Hengel, C. Shen, and L. Cao, "Toward deep supervised anomaly detection: Reinforcement learning from partially labeled anomaly data," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discov. Data Min.*, 2021, pp. 1298–1308.

- [220] C. Huang, Y. Wu, Y. Zuo, K. Pei, and G. Min, "Towards experienced anomaly detector through reinforcement learning," in *Proc. 32nd AAAI Conf. Artif. Intell. 13th Innovat. Appl. Artif. Intell. Conf. 8th AAAI Symp. Educ. Adv. Artif. Intell.*, 2018, pp. 8087–8088.
- [221] Z. Li, Y. Kong, C. Wang, and C. Jiang, "DDoS mitigation based on space-time flow regularities in IoV: A feature adaption reinforcement learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2262–2278, Mar. 2022.
- [222] R. Sedar, C. Kalalas, F. Vázquez-Gallego, and J. Alonso-Zarate, "Reinforcement learning based misbehavior detection in vehicular networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2022, pp. 3550–3555.
- [223] F. Jameel, M. A. Javed, S. Zeadally, and R. Jäntti, "Secure transmission in cellular V2X communications using deep Q-learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17167–17176, Oct. 2022.
- [224] T. Alladi, V. Kohli, V. Chamola, and F. R. Yu, "Securing the Internet of Vehicles: A deep learning-based classification framework," *IEEE Netw. Lett.*, vol. 3, no. 2, pp. 94–97, Jun. 2021.
- [225] T. Alladi, V. Kohli, V. Chamola, F. R. Yu, and M. Guizani, "Artificial intelligence (AI)-empowered intrusion detection architecture for the Internet of Vehicles," *IEEE Wireless Commun.*, vol. 28, no. 3, pp. 144–149, Jun. 2021.
- [226] X. Zhu, Y. Luo, A. Liu, M. Z. A. Bhuiyan, and S. Zhang, "Multi-agent deep reinforcement learning for vehicular computation offloading in IoT," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9763–9773, Jun. 2021.
- [227] T. D. Nguyen, S. Marchal, M. Miettinen, H. Fereidooni, N. Asokan, and A.-R. Sadeghi, "DfIoT: A federated self-learning anomaly detection system for IoT," in *Proc. IEEE 39th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jul. 2019, pp. 756–767.
- [228] W. Y. B. Lim et al., "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 2031–2063, 3rd Quart., 2020.
- [229] H. H. W. J. Bosman, G. Iacca, A. Tejada, H. J. Wörtche, and A. Liotta, "Ensembles of incremental learners to detect anomalies in ad hoc sensor networks," *Ad Hoc Netw.*, vol. 35, pp. 14–36, Dec. 2015.
- [230] Y. Zhang, N. Meratnia, and P. Havinga, "Adaptive and online one-class support vector machine-based outlier detection techniques for wireless sensor networks," in *Proc. Int. Conf. Adv. Inf. Netw. Appl. Workshops*, May 2009, pp. 990–995.
- [231] M. Almgren and E. Jonsson, "Using active learning in intrusion detection," in *Proc. 17th IEEE Comput. Secur. Found. Workshop*, May 2004, pp. 88–98.
- [232] B. Karthiga, D. Durairaj, N. Nawaz, T. K. Venkatasamy, G. Ramasamy, and A. Hariharasudan, "Intelligent intrusion detection system for VANET using machine learning and deep learning approaches," *Wireless Commun. Mobile Comput.*, vol. 2022, pp. 1–13, Oct. 2022.
- [233] D. Sinha, A. Sharma, and S. Sharma, "Automated detection of coronary artery disease comparing arterial fat accumulation using CNN," *J. Electron. Imag.*, vol. 31, no. 5, Jan. 2022, Art. no. 051405.
- [234] M. Wazid et al., "Explainable deep learning-enabled malware attack detection for IoT-enabled intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, early access, Jan. 16, 2025, doi: 10.1109/TITS.2025.3525505.
- [235] S. Iqbal, P. Ball, M. H. Kamarudin, and A. Bradley, "Simulating malicious attacks on VANETs for connected and autonomous vehicle cybersecurity: A machine learning dataset," in *Proc. 13th Int. Symp. Commun. Syst., Netw. Digit. Signal Process. (CSNDSP)*, Jul. 2022, pp. 332–337.
- [236] P. Rajak and P. K. Mishra, "An attack detection model for enhancing the security of 5G-enabled vehicle-to-everything (V2X) communication for smart vehicles," in *Proc. FedCSIS (Commun. Papers)*, 2023, pp. 229–236.
- [237] C. Jiménez, "Deep learning-based cyber attack detection in autonomous vehicle networks," *J. AI-Assist. Sci. Discovery*, vol. 2, no. 2, pp. 1–25, 2022.
- [238] B. Sousa, N. Magaia, and S. Silva, "An intelligent intrusion detection system for 5G-enabled Internet of Vehicles," *Electronics*, vol. 12, no. 8, p. 1757, Apr. 2023, doi: 10.3390/electronics12081757.
- [239] A. A. Korba, A. Boualouache, B. Briki, R. Rahal, Y. Ghamri-Doudane, and S. M. Senouci, "Federated learning for zero-day attack detection in 5G and beyond V2X networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2023, pp. 1137–1142.
- [240] A. H. Magsi, A. Ghulam, S. Memon, K. Javeed, M. Alhussein, and I. Rida, "A machine learning-based attack detection and prevention system in vehicular named data networking," *Comput., Mater. Continua*, vol. 77, no. 2, pp. 1445–1465, 2023.
- [241] G. Twardokus and H. Rahbari, "Towards protecting 5G sidelink scheduling in C-V2X against intelligent DoS attacks," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7273–7286, Nov. 2023.
- [242] M. Begum, G. Raja, and M. Guizani, "AI-based sensor attack detection and classification for autonomous vehicles in 6G-V2X environment," *IEEE Trans. Veh. Technol.*, vol. 73, no. 4, pp. 5054–5063, Apr. 2024.
- [243] S. B. Prathiba, G. Raja, S. Anbalagan, K. S. Arikumar, S. Gurumoorthy, and K. Dev, "A hybrid deep sensor anomaly detection for autonomous vehicles in 6G-V2X environment," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 3, pp. 1246–1255, May 2023.
- [244] B. Sousa, N. Magaia, S. Silva, N. T. Hieu, and Y. L. Guan, "Vehicle-to-vehicle flooding datasets using MK5 on-board unit devices," *Sci. Data*, vol. 11, no. 1, p. 1363, Dec. 2024. [Online]. Available: <https://www.nature.com/articles/s41597-024-03663-5>
- [245] S. Zhang, M. Lagutkina, K. O. Akpinar, and M. Akpinar, "Improving performance and data transmission security in VANETs," *Comput. Commun.*, vol. 180, pp. 126–133, Dec. 2021.
- [246] A. A. Andrade Salazar, P. D. McDaniel, R. Sheatsley, and J. Petit, "Physics-based misbehavior detection system for V2X communications," *SAE Int. J. Connected Automated Vehicles*, vol. 5, no. 3, pp. 237–258, Mar. 2022.
- [247] O. Minawi, J. Whelan, A. Almeahmadi, and K. El-Khatib, "Machine learning-based intrusion detection system for controller area networks," in *Proc. 10th ACM Symp. Design Anal. Intell. Veh. Netw. Appl.*, Nov. 2020, pp. 41–47.
- [248] M. H. Bhavsar, Y. B. Bekele, K. Roy, J. C. Kelly, and D. Limbrick, "FL-IDS: Federated learning-based intrusion detection system using edge devices for transportation IoT," *IEEE Access*, vol. 12, pp. 52215–52226, 2024, doi: 10.1109/ACCESS.2024.3386631.
- [249] J.-P. Monteuis, J. Petit, J. Zhang, H. Labiod, S. Mafrika, and A. Serval, "'My autonomous car is an elephant': A machine learning-based detector for implausible dimension," in *Proc. 3rd Int. Conf. Secur. Smart Cities, Ind. Control Syst. Commun. (SSIC)*, Oct. 2018, pp. 1–8.
- [250] S. Kong, K. Wang, C. Feng, and J. Wang, "Smart cities and transportation based vehicle-to-vehicle communication and cyber security analysis using machine learning model in 6G network," *Wireless Pers. Commun.*, vol. 2024, pp. 1–19, Jun. 2024.
- [251] S. Boddupalli and S. Ray, "REDEM: Real-time detection and mitigation of communication attacks in connected autonomous vehicle applications," in *Proc. IFIP Int. Internet Things Conf.*, Jan. 2020, pp. 105–122.
- [252] B. Kihei, H. Wilson, and M. Fall, "Experimental results of detecting primitive jamming attacks using machine learning in vehicle-to-everything communication networks," in *Proc. IEEE 7th World Forum Internet Things (WF-IoT)*, Jun. 2021, pp. 530–535.
- [253] T. Karunathilake, M. Zongo, D. Amarawardana, and A. Förster, "CN+: Vehicular dataset at traffic light regulated intersection in bremen, Germany," *Sci. Data*, vol. 11, no. 1, p. 1363, Jun. 2024.
- [254] S. Mishra, N. Sengar, and D. Har, "A secure, blockchain-enabled vehicular sensor communication protocol with deep learning-assisted anomaly detection," *IEEE Intell. Transp. Syst. Mag.*, early access, Jan. 7, 2025, doi: 10.1109/MITS.2024.3519620.



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